Real asymptotic expansions and inverse functions

Power–logarithmic series, real branches, and error transport

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Abstract

Real powers and logarithms lead naturally beyond Taylor and Puiseux series. For example, the positive inverse of $x + x^2(1 + \log x)$ begins $y - y^2(1 + \log y) + y^3(2\log^2 y + 5\log y + 3)$, whereas the inverse of $x + x^{\sqrt{2}}$ begins $y - y^{\sqrt{2}} + \sqrt{2}\, y^{2\sqrt{2}-1}$. We develop a calculus of complete power–logarithmic blocks with real exponents, valid at finite endpoints and at infinity.

Marker Lagrange–Bürmann inversion and an Euler-operator formula give the coefficients of arbitrary real powers of an inverse. Real branch selection, weighted truncation, and error transport are treated explicitly. Convergence is proved for the analytic finite-parameter class; finite smoothness yields separate asymptotic theorems. The calculus extends to logarithmic coordinates, exact inverse cores, finite flat sectors, and Fourier-valued coefficients. Special-function tails and exponential normalization illustrate the distinction between convergent local expansions and Poincaré expansions. Absolute error envelopes transport exponential and oscillatory special-function approximations through finite compositions. A common moment theorem gives integral and Lerch tails, while the zeta defining sum supplies a convergent expansion with logarithmic exponent weights. Logarithmic normalization of Gamma products supplies an exact growth factor and a relative correction series. Inverting Gamma around a retained Lambert core gives ordered corrections with polynomial reciprocal-logarithmic coefficients and finite-order Stirling error bounds. The increasing Barnes inverse has a quadratic logarithmic Lambert balance, with a shifted reciprocal-logarithmic coefficient coordinate and finite-order error transport from its logarithmic asymptotic expansion. Rational interval bounds convert a residual estimate into a rigorous real root enclosure. Each extension states its own hypotheses, ordering, and remainder meaning.

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1 Introduction

1.1 Two questions

The first question [1] asks for an asymptotic expansion, as $y \to 0^+$, of the inverse of

$$f_1(x) = x + x^2(1 + \log x), \qquad x > 0, \tag{1.1}$$

on the branch that maps $(0, \infty)$ to itself. Choosing this branch is essential: near a nonzero complex preimage of 0, the same expression has an analytic inverse germ of the form

$$c + \frac{y}{c-1} - \frac{3 + 2\mathrm{W}_{-1}(-e)}{2(c-1)^3}\, y^2 + \mathcal{O}(y^3), \qquad c = e^{\mathrm{W}_{-1}(-e)-1} \approx -0.1178 - 0.5332\, i, \tag{1.2}$$

Here c is a nonzero complex root of $1 + x(1 + \log x) = 0$, where f_1 is analytic with a nonzero derivative (Section 4). The real branch has no Taylor expansion at all. Its expansion is

$$\begin{aligned} g_1(y) = y - y^2(1+L) + y^3(2L^2 + 5L + 3) - y^4\Big(5L^3 + \tfrac{41}{2}L^2 + 27L + \tfrac{23}{2}\Big) \\ + y^5\Big(14L^4 + \tfrac{241}{3}L^3 + \tfrac{335}{2}L^2 + 151L + \tfrac{299}{6}\Big) + \mathcal{O}(y^6(1+|L|)^5), \end{aligned} \tag{1.3}$$

with $L = \log y$: every power of y carries a polynomial in $\log y$, of degree growing with the power, so the scale of the expansion is neither a Taylor nor a Puiseux scale. The question also asks, as a second example, for

$$g_2(y) = y - y^{\sqrt{2}} + \sqrt{2}\, y^{2\sqrt{2}-1} - \frac{6-\sqrt{2}}{2}\, y^{3\sqrt{2}-2} + \Big(\frac{17\sqrt{2}}{3} - 4\Big) y^{4\sqrt{2}-3} + \mathcal{O}(y^{5\sqrt{2}-4}), \tag{1.4}$$

the inverse of $f_2(x) = x + x^{\sqrt{2}}$: no logarithms this time, but exponents in the semigroup $1 + (\sqrt{2}-1)\mathbb{N}$, which no rational grid contains.

The second question [2] concerns the forward asymptotics of $(1 + x + x^{\sqrt{2}})^{\sqrt{2}}$ as $x \to \infty$. Its leading term is x^2, but the subsequent terms require ordering two incommensurable positive gaps. The expansion is

$$\begin{aligned} (1 + x + x^{\sqrt{2}})^{\sqrt{2}} = x^2 + \sqrt{2}\, x^{3-\sqrt{2}} + \Big(1 - \tfrac{1}{\sqrt{2}}\Big)x^{4-2\sqrt{2}} + \Big(\tfrac{2\sqrt{2}}{3} - 1\Big)x^{5-3\sqrt{2}} \\ + \sqrt{2}\, x^{2-\sqrt{2}} + \tfrac{13-9\sqrt{2}}{12}\, x^{6-4\sqrt{2}} + (2-\sqrt{2})\, x^{3-2\sqrt{2}} + \mathcal{O}(x^{7-5\sqrt{2}}), \end{aligned} \tag{1.5}$$

and its inverse, as $y \to \infty$, begins $\sqrt{y} - \frac{1}{\sqrt{2}} y^{(2-\sqrt{2})/2} + \frac{3-\sqrt{2}}{4} y^{(3-2\sqrt{2})/2} + \mathcal{O}(y^{(4-3\sqrt{2})/2})$.

1.2 What is common to all of this

All these expansions are finite or infinite sums of *power–log blocks* $w^\beta P(\log w)$, where $w \to 0^+$ is a local variable ($w = y$, $w = 1/x, \ldots$), β is a real exponent, and P is a polynomial. Their supports lie in translates of semigroups generated by finitely many positive gaps. These semigroups are locally finite, even when the groups generated by the same gaps are dense in $\mathbb{R}$, so every truncation is a finite computation. The logarithmic polynomials are handled by the Euler identity $\frac{\mathrm{d}}{\mathrm{d}w}[w^\beta P(\log w)] = w^{\beta-1}(\beta + \frac{\mathrm{d}}{\mathrm{d}L})P(L)$; and remainders must keep their logarithmic factor, because $y^3(2\log^2 y + 5\log y + 3)$ is not $\mathcal{O}(y^3)$.

This article develops the algebra and analysis needed to construct and justify these expansions. Its starting points are the questions [1, 2] and the phase-coordinate approach of [11]. The treatment gives coefficient formulas, branch theorems, convergence proofs and remainder estimates, then extends the construction to finite endpoints from either side, infinity, observables of an inverse, and scales involving iterated logarithms or exponentially small terms.

1.3 Summary of results

- **Scale** (Section 2). Power–log blocks are totally ordered by size, the finitely generated exponent semigroups are locally finite, and the truncated arithmetic (products, real powers, logarithms, exponentials, composition with polynomials in the logarithm) is exact and terminates. The single primitive is the composition recurrence $Q_{k+1} = ((a-k)Q_k + Q_k')/(k+1)$ for $(1+t)^a P(L + \log(1+t))$.
- **Forward expansions** (Section 3). Expressions built from constants, the variable, arithmetic, real powers, logarithms, bounded exponentials and analytic functions of expressions with a limit have convergent power–log expansions; structural error propagation supplies a rigorous remainder class, which answers the second question.
- **Branch** (Section 4). For $f(x) = ax^p(1 + \sum_i x^{\delta_i} B_i(\log x))$ with $a, p \neq 0$, $\delta_i > 0$, the real branch is the one with $x > 0$ and $x/(y/a)^{1/p} \to 1$; it is unique, and the complex germ (1.2) is explained.
- **Lagrange–Bürmann with a marker** (Section 5). A residue proof of $\Phi(u) = \Phi(q) + \sum_N \frac{(-\eta)^N}{N!}\partial_q^{N-1}[\Phi'(q)h(q)^N]$ at a positive point, valid analytically and formally, with the transport of observables and the general-core form.
- **Coefficients** (Section 6). For every multi-index $\boldsymbol{k}$ of weight σ, the block of $g(y)^r$ at $z^{r+\sigma}$, $z = (y/a)^{1/p}$, is $\frac{(-1)^{|\boldsymbol{k}|}r}{p^{|\boldsymbol{k}|}\boldsymbol{k}!}\prod_{j=1}^{|\boldsymbol{k}|-1}(\mathcal{D} + r + \sigma + pj)\prod_i B_i(L)^{k_i}$, $\mathcal{D} = \mathrm{d}/\mathrm{d}L$; equal weights are added. Degrees, leading logarithms (Catalan numbers), closed forms for constant coefficients (Fuss–Catalan), $\log g$, and the formal uniqueness through the triangular equation, Picard iteration and Newton iteration.
- **Convergence** (Section 7). The expansion converges absolutely for small positive y after substituting finitely many small parameters $z^{\delta_i}(\log z)^j$, with coefficient bounds $|C_{\boldsymbol{k}}(L)| \leq MK^{|\boldsymbol{k}|}(1 + |L|)^{D(\boldsymbol{k})}$; an explicit conditional tail majorant follows from Rouché's theorem.
- **Remainders** (Section 8). Truncation below any exponent cutoff leaves $z^{1+\sigma_*}C_{\sigma_*}(L) + o(z^{1+\sigma_*})$ where σ_* is the first omitted weight and C_{σ_*} the complete coefficient there, computed from the one-step boundary of the retained multi-index set; depth truncation has its own bound.
- **Transport** (Section 9). Forward remainders and residuals are transported to the inverse with the correct scaling z^{1-p}; a residual bracket gives existence and a two-sided error bound; and a theorem with a complete proof shows that finite smoothness of the perturbation, with derivative bounds in the power–log scale, suffices for the marker formula with an explicit remainder, which covers perturbations such as $x^2 \sin(\log x)$.
- **Endpoints** (Section 10). Finite points from either side and $\pm\infty$ reduce to the same local inversion problem through a local variable and a signed leading power $p \neq 0$; the limit y_0 may be finite or infinite; powers $(x - x_0)^r$ and x^r of the inverse are obtained directly. Worked examples include the Wright omega function and the inverse of (1.5).

- **Lambert coordinates and boundaries** (Section 12). Leading logarithmic and exponential cores reduce to a convergent Lambert expansion in a reciprocal-logarithm coordinate, with a polynomial in its logarithm at every order. This covers $x \log x$ near 0 and xe^x at infinity. Zero gaps, flat perturbations and oscillatory coefficients require further scale choices; symbolic exponents can use depth truncation when their branch assumptions are proved.
- **Coordinates and extended scales**. Exact changes of source and target coordinate transport inverse expansions and their errors. Weighted-support algorithms organize coefficient calculation; interval residual bounds turn local approximations into enclosures of unique roots. Further sections develop exact-core perturbations, iterated logarithmic scales, Fourier coefficients and exponentially small sectors.

1.4 Conventions

$\mathbb{N} = \{0, 1, 2, \dots\}$. For a local variable $w \to 0^+$ we write $L = \log w$ and $\mathrm{M}(w) = 1 + |\log w|$. $\mathcal{O}$ and o refer to $w \to 0^+$ with all other data fixed; no uniformity in parameters is asserted unless stated. $\mathcal{D} = \mathrm{d}/\mathrm{d}L$ acts on polynomials in L. For a multi-index $\boldsymbol{k} = (k_1, \dots, k_m) \in \mathbb{N}^m$ we write $|\boldsymbol{k}| = \sum k_i$, $\boldsymbol{k}! = \prod k_i!$, and $\boldsymbol{k} \cdot \boldsymbol{\delta} = \sum k_i \delta_i$. Real powers of positive quantities are $w^\beta = e^{\beta \log w}$. Real observables use the selected real branches; auxiliary complex powers may occur in analytic proofs or Fourier coefficient representations. Power and weight cutoffs are *exclusive*: a cutoff q retains the blocks of exponent $< q$. An exponential sector depth N is inclusive and retains degrees through N.

2 The power–log scale and its exact arithmetic

2.1 Blocks and their order

Throughout, w denotes a positive real variable tending to 0^+, $L = \log w \to -\infty$ and $\mathrm{M}(w) = 1 + |\log w|$.

Definition 2.1 (Blocks)**.** A *power–log block* is an expression $w^\beta P(L)$ with $\beta \in \mathbb{R}$ and $P \in \mathbb{R}[L]$, $P \neq 0$. Its *exponent* is β and its *logarithmic degree* is $\deg P$.

Lemma 2.2 (Power beats logarithm)**.** *For every $\varepsilon > 0$ and every real b, $w^\varepsilon \mathrm{M}(w)^b \to 0$ as $w \to 0^+$. Consequently, if $\beta < \gamma$ and $P, Q \neq 0$, then $w^\gamma Q(L) = o(w^\beta P(L))$; and within a fixed exponent, a higher logarithmic degree dominates: $w^\beta L^j = o(w^\beta L^k)$ for $j < k$.*

Proof. Put $w = e^{-t}$, $t \to +\infty$: $w^\varepsilon \mathrm{M}(w)^b = e^{-\varepsilon t}(1+t)^b \to 0$. For the second claim the ratio is $w^{\gamma-\beta} Q(L)/P(L)$, and $|Q(L)/P(L)| \le C\mathrm{M}(w)^{\deg Q}$ for w small because $P(L)$ has no zeros for L sufficiently negative and $|P(L)| \ge c|L|^{\deg P} \ge c$ there; the first claim finishes. The third claim is clear. □

The growth classes of nonzero blocks are totally ordered by increasing exponent and, at equal exponents, by decreasing logarithmic degree; blocks of the same exponent and degree have the same growth class. A finite sum of blocks with distinct exponents is asymptotic to its block of least exponent (whole polynomial included), and vanishes identically only if every block does. This is why truncations are taken by *complete blocks*: cutting a block inside its polynomial would drop terms larger than the retained terms of the next exponent.

Definition 2.3 (Remainder classes)**.** For $\beta \in \mathbb{R}$ and an integer $k \ge 0$ we write $R(w) = \mathcal{O}_{\beta,k}$, for $R(w) = \mathcal{O}(w^\beta \mathrm{M}(w)^k)$ as $w \to 0^+$.

By Lemma 2.2, $\mathcal{O}_{\beta,k} \subseteq \mathcal{O}_{\beta',k'}$ whenever $\beta > \beta'$, or $\beta = \beta'$ and $k \le k'$; the class with the smaller exponent, and at equal exponents the larger degree, is the coarser one. A block $w^\beta P(L)$ with $\deg P = k$ belongs to $\mathcal{O}_{\beta,k}$ and to no $\mathcal{O}_{\beta,k'}$ with $k' < k$. Sums and products of remainder classes obey

$$\mathcal{O}_{\beta,k} + \mathcal{O}_{\beta',k'} = \mathcal{O}_{\min(\beta,\beta'),\,k''}, \qquad \mathcal{O}_{\beta,k} \cdot \mathcal{O}_{\beta',k'} = \mathcal{O}_{\beta+\beta',\,k+k'}, \tag{2.1}$$

where k'' is the degree attached to the smaller exponent, and the larger of the two when the exponents coincide.

2.2 Exponent semigroups

Lemma 2.4 (Local finiteness)**.** *Let $\delta_1, \ldots, \delta_m > 0$ and $S = \{\boldsymbol{k} \cdot \boldsymbol{\delta} : \boldsymbol{k} \in \mathbb{N}^m\}$. For every real H the set $\{\boldsymbol{k} \in \mathbb{N}^m : \boldsymbol{k} \cdot \boldsymbol{\delta} \le H\}$ is finite, hence $S \cap (-\infty, H]$ is finite, and every $\sigma \in S$ has only finitely many representations $\sigma = \boldsymbol{k} \cdot \boldsymbol{\delta}$.*

Proof. If $\boldsymbol{k} \cdot \boldsymbol{\delta} \le H$ then $k_i \le H/\delta_i$ for each i. □

When the gaps are incommensurable the *group* generated by them is dense in $\mathbb{R}$ (for instance $\mathbb{Z} + \sqrt{2}\mathbb{Z}$), but the semigroup of nonnegative combinations has only finitely many elements below any bound. This is the whole reason why expansions with irrational exponents can be computed term by term: the ordinary power–log expansions developed below have support in a translate $\beta_0 + S$ of such a semigroup. Following [11], a subset of $\mathbb{R}$ that is finite below every bound is called *left-finite*; the semigroups S are left-finite, and so are their translates and finite unions.

2.3 Jets and their arithmetic

Definition 2.5 (Jets)**.** Fix gaps as above and $\beta_0 \in \mathbb{R}$. A *jet* (of cutoff H) is a finite sum $U = \sum_{\sigma \in T} w^\sigma C_\sigma(L)$ with $T \subset \beta_0 + S$ finite, $C_\sigma \in \mathbb{R}[L]$, all $\sigma < H$; equal exponents are merged and zero polynomials are discarded. Its *valuation* $\operatorname{val} U$ is its least exponent ($+\infty$ for the empty jet). For nonnegative exponents and $H > 0$, the jet algebra $\mathcal{P}_{<H}$ uses addition and multiplication followed by truncation below H.

In that nonnegative-exponent ring, blocks of exponent $\ge H$ form an ideal and $\mathcal{P}_{<H}$ is the corresponding quotient. This quotient description does not apply to arbitrary real exponents: multiplication by a negative power can move a discarded term below the cutoff. General shifted jets therefore require precision tracking; a factor of valuation $\gamma < 0$ lowers an unknown exponent cutoff by $|\gamma|$ (Section 3.2). Equal exponents must be identified before their coefficient polynomials are added. In particular, exact identities among algebraic exponents determine which terms belong to the same block.

This formal exponent filtration must be distinguished from an analytic remainder class. For example, $w^H \log w$ is discarded at cutoff H, but is not $\mathcal{O}(w^H)$: its ratio to w^H is unbounded. Thus a formal cutoff alone cannot replace the pair (H, k) in an analytic error bound. The logarithmic degrees of retained blocks likewise do not determine those of the tail. A complete-tail estimate requires the hypotheses of Section 3.2 or another applicable asymptotic theorem.

Differentiation

Lemma 2.6 (Euler identity)**.** *For $\beta \in \mathbb{R}$, $P \in \mathbb{R}[L]$ and $r \in \mathbb{N}$,*

$$\frac{\mathrm{d}}{\mathrm{d}w}[w^\beta P(L)] = w^{\beta-1}(\beta + \mathcal{D})P(L), \qquad \frac{\mathrm{d}^r}{\mathrm{d}w^r}[w^\beta P(L)] = w^{\beta-r} \prod_{j=0}^{r-1} (\beta - j + \mathcal{D})P(L). \tag{2.2}$$

The operators $\beta - j + \mathcal{D}$ commute, and $(\beta + \mathcal{D})$ preserves $\deg P$ when $\beta \neq 0$ and lowers it by one when $\beta = 0$.

Proof. $\frac{\mathrm{d}}{\mathrm{d}w}[w^\beta P(\log w)] = \beta w^{\beta-1} P + w^\beta P'(\log w)/w$. Iterate. $\square$

The identity says that differentiation acts on a block as the scalar β plus $\mathcal{D}$ on the polynomial; the exponent bookkeeping and the polynomial arithmetic separate completely. In the coefficient formulas below operators $\mathcal{D} + c$ appear with real c (not necessarily positive), and their action on the coefficient vector $(z_0, \ldots, z_d)$ of $P = \sum z_s L^s$ is bidiagonal: $(c + \mathcal{D})P = \sum_s (c z_s + (s+1) z_{s+1}) L^s$.

Unit series

Let U be a jet with $\operatorname{val} U > 0$. Then U^k has valuation $\geq k \operatorname{val} U$, so in $\mathcal{P}_{<H}$ only finitely many powers of U are nonzero, and every power series $\Phi(t) = \sum_k \phi_k t^k$ can be substituted:

$$\Phi(U) = \sum_{k < H/\operatorname{val} U} \phi_k U^k \in \mathcal{P}_{<H}. \tag{2.3}$$

In particular $(1+U)^a = \sum_k \binom{a}{k} U^k$ for every real a (with $\binom{a}{k} = a(a-1)\cdots(a-k+1)/k!$), $\log(1+U) = \sum_{k\geq 1} (-1)^{k+1} U^k/k$, $e^U = \sum_k U^k/k!$, and $\phi(c+U) = \sum_k \phi^{(k)}(c) U^k/k!$ for any ϕ analytic at c. These are identities between finite sums of blocks modulo exponents $\geq H$; their analytic content is discussed in Section 7.

Composition with the logarithm

The one operation that mixes the exponent bookkeeping with the polynomial coefficients is the substitution of a jet into the logarithm: if w is replaced by $w(1+U)$ then L becomes $L + \log(1+U)$, and a block $w^a P(L)$ becomes $w^a (1+U)^a P(L + \log(1+U))$. The following recurrence computes this without expanding $\log(1+U)$ separately.

Lemma 2.7 (Composition recurrence)**.** *Let $P \in \mathbb{R}[L]$ and $a \in \mathbb{R}$. For $|t| < 1$,*

$$(1+t)^a P(L + \log(1+t)) = \sum_{k\geq 0} Q_k(L)\, t^k, \qquad Q_0 = P, \qquad Q_{k+1} = \frac{(a-k)Q_k + Q_k'}{k+1}, \tag{2.4}$$

with $Q_k \in \mathbb{R}[L]$, $\deg Q_k \leq \deg P$, and, when $a \in \mathbb{N}$ and P is constant, $Q_k = 0$ for $k > a$. More generally, if Q_m is identically zero, then $Q_k = 0$ for every $k \geq m$. Consequently, for a jet U of positive valuation, $(1+U)^a P(L + \log(1+U)) = \sum_k Q_k(L) U^k$ in $\mathcal{P}_{<H}$, and an identically zero Q_m ends this sum before any further powers of U are needed.

Proof. Put $F(t) = (1+t)^a P(L + \log(1+t))$; it is analytic in $|t| < 1$ with values in $\mathbb{R}[L]$, so $F = \sum Q_k t^k$ with $Q_k \in \mathbb{R}[L]$ and $Q_0 = P$. Differentiating, $(1+t)\partial_t F = aF + \partial_L F$, because ∂_t acting on the two factors gives $a(1+t)^{a-1} P + (1+t)^{a-1} P'$. Comparing coefficients of t^k in $(1+t) \sum (k+1) Q_{k+1} t^k = \sum (a Q_k + Q_k') t^k$ gives $(k+1) Q_{k+1} + k Q_k = a Q_k + Q_k'$. The degree does not increase, and for $a \in \mathbb{N}$ and constant P the factor $(a-k)$ annihilates Q_{a+1} and hence all later Q_k (for nonconstant P the derivative keeps the sequence alive: $\log(1+t)$ is not a polynomial). In general, $Q_m = 0$ implies $Q_m' = 0$, so the homogeneous recurrence gives $Q_{m+1} = 0$ and induction proves permanent annihilation. The jet statement follows because substitution of U for t is a ring homomorphism on finite sums. $\square$

This stopping criterion uses the homogeneous recurrence, not merely an observed zero coefficient of an arbitrary power series. For example, $\sin t$ has zero even coefficients and nonzero odd coefficients. Likewise, $Q_m(L_0) = 0$ at a single value of L does not imply $Q'_m = 0$ and is insufficient; the zero must be a polynomial identity under the fixed parameter hypotheses.

Remark 2.8. The recurrence is the whole composition calculus needed here: the forward model $ax^p(1 + \sum x^{\delta_i} B_i(\log x))$ evaluated at $x = z(1 + U)$ is $az^p(1 + U)^p(1 + \sum_i z^{\delta_i}(1 + U)^{\delta_i} B_i(L + \log(1 + U)))$, each factor of which is an instance of (2.4) (with $P = 1$ for the first). Its derivative is obtained from the same lemma with a replaced by $a - 1$ and P by $aP + P'$, since $\frac{\mathrm{d}}{\mathrm{d}x}[x^a P(\log x)] = x^{a-1}(aP + P')(\log x)$.

2.4 Blocks with nonconstant leading coefficient

Everything above allows the polynomial in the leading block to be nonconstant. In general, a real non-integer power or a logarithm of such a block produces $P(L)^a$ or $\log P(L)$ outside $\mathbb{R}[L]$. Special algebraic simplifications may stay in the class, but generic expressions require larger scales (rational functions of L, or iterated logarithms, Section 12). Nonnegative integer powers of any jet stay in the class.

3 Forward expansions

Before inverting anything we need the forward function in the normal form $ax^p(1 + \sum_i x^{\delta_i} B_i(\log x))$, and the second question [2] is a forward problem in its own right. This section shows that a large class of expressions has convergent power–log expansions, and establishes rules for propagating a rigorous remainder class through arithmetic and composition.

3.1 Convergent power–log expansions

Definition 3.1. A function F on $(0, w_0)$ has a *convergent power–log expansion* with support in $\beta_0 + S$ (S a finitely generated positive semigroup with gaps $\delta_1, \dots, \delta_m$) if there are polynomials $C_\sigma \in \mathbb{R}[L]$, constants $M, K \geq 1$, and integers $d_0, d \geq 0$ such that

$$F(w) = \sum_{\sigma \in S} w^{\beta_0+\sigma} C_\sigma(L) \quad \text{for } 0 < w < w_1, \qquad \|C_\sigma\|_1 \leq MK^{n(\sigma)}, \quad \deg C_\sigma \leq d_0 + d\, n(\sigma), \tag{3.1}$$

where $n(\sigma)$ is the least $|\boldsymbol{k}|$ with $\boldsymbol{k} \cdot \boldsymbol{\delta} = \sigma$, $\|P\|_1$ is the sum of the absolute values of the coefficients of P, and the series converges absolutely. Thus $|C_\sigma(L)| \leq MK^{n(\sigma)} \mathrm{M}(w)^{d_0+dn(\sigma)}$. Zero coefficients satisfy the degree bound by convention. We write $F \in \mathcal{E}$. The fixed allowance d_0 is essential: it admits a polynomial in $\log w$ at the leading exponent, including $\log w$ itself.

The coefficient bound makes the tail after any cutoff controllable (Lemma 3.2 below), and it is stable under all the operations we need.

Lemma 3.2 (Tail of a convergent expansion)**.** *Let $F \in \mathcal{E}$ as in* (3.1)*, $\delta_* = \min_i \delta_i$, and $H > 0$. Then*

$$F(w) - \sum_{\sigma < H} w^{\beta_0+\sigma} C_\sigma(L) = \mathcal{O}(w^{\beta_0+H} \mathrm{M}(w)^{d_0+Nd}), \qquad N = \lceil H/\delta_* \rceil. \tag{3.2}$$

More precisely, the omitted part equals $w^{\beta_0+\sigma_*} C_{\sigma_*}(L) + o(w^{\beta_0+\sigma_*})$ where $\sigma_* = \min\{\sigma \in S : \sigma \geq H\}$.

Proof. Split the omitted terms into those with $n(\sigma) \leq N - 1$ and the rest. The first group is finite (Lemma 2.4) and each term is $\mathcal{O}(w^{\beta_0+H} \mathrm{M}^{d_0+(N-1)d})$. For the rest, the number of weights whose shortest representation has length n is at most m^n, and $\sigma \ge n\delta_*$, so their sum is bounded by

$$Mw^{\beta_0}\mathrm{M}^{d_0}\sum_{n\ge N}(mKw^{\delta_*}\mathrm{M}^d)^n = \mathcal{O}(w^{\beta_0+N\delta_*}\mathrm{M}^{d_0+Nd}).$$

Since $N\delta_* \ge H$, this proves the bound. For the refined statement let σ_{**} be the next element of S after σ_*, which exists by local finiteness when S has a generator, and choose $\sigma_* < H' < \sigma_{**}$. Applying the bound at H' shows that everything beyond σ_* is $\mathcal{O}(w^{\beta_0+H'}\mathrm{M}^{d_0+N'd}) = o(w^{\beta_0+\sigma_*})$. The difference $\sigma_{**}-\sigma_*$ need not be at least δ_*. Finite support with no further terms has zero tail. □

Theorem 3.3 (Closure). *The class $\mathcal{E}$ contains the constants and w and is closed under sums, products, and the following operations, with the supports and gaps indicated.*

1. *If $F \in \mathcal{E}$ has leading block cw^β with a* constant $c > 0$*, then $F^a \in \mathcal{E}$ for every real a and $\log F \in \mathcal{E}$; their supports are contained in $a\beta + S'$ and $\{0\} \cup S'$ respectively, with block $\log c + \beta L$ at exponent 0 in the logarithm. Here S' is a finitely generated positive semigroup containing the exponents of $F/(cw^\beta) - 1$; the proof establishes that such a semigroup exists.*
2. *If $F \in \mathcal{E}$ has no negative-exponent blocks and its exponent-0 block is $c + kL$ with $k \in \mathbb{R}$, then $e^F \in \mathcal{E}$ with leading block $e^c w^k$. The argument F may grow logarithmically.*
3. *If $F \in \mathcal{E}$ has $\beta_0 \ge 0$ and constant exponent-0 block c, and ϕ is analytic at c, then $\phi \circ F \in \mathcal{E}$.*

Proof. First note that a positive-valuation expansion U can be represented with support in a positive semigroup based at 0: if necessary, add its positive leading exponent as a generator and enlarge the constants in (3.1). Choose one shortest multi-index $\boldsymbol{k}$ for each resulting weight σ, put $D = d_0 + d$, and introduce

$$T_{ij} = w^{\delta_i}L^j, \qquad 1 \le i \le m, \quad 0 \le j \le D.$$

For $n = |\boldsymbol{k}| \ge 1$, every $w^\sigma L^s$ with $s \le d_0 + dn \le Dn$ is a product of n such parameters: distribute the s logarithmic factors among the n generator factors, each receiving at most D. Fix one such distribution for every monomial. The coefficient-norm bound makes the resulting series analytic in a sufficiently small polydisc, since its terms of degree n have total absolute coefficient at most $M(mK)^n$. Analytic composition with $(1+U)^a$, $\log(1+U)$, e^U, or $\phi(c+U)$ is therefore legitimate. Cauchy estimates give exponential coefficient bounds in the new parameters, and their logarithmic degree grows at most linearly in parameter degree.

Regrouping by weight preserves bounds of the form (3.1): a representation of weight σ has length at most σ/δ_*, whereas its shortest length is at least $\sigma/\delta_{\max}$; their ratio is bounded, and the number of parameter monomials of a given total degree is at most exponential. Enlarge M, K, d to absorb both factors.

For sums and products, use the union of the generator sets and, for a sum of differently shifted supports, the positive difference between their shifts as an additional generator. Coefficient norms are subadditive and submultiplicative, so the same bounds apply. If leading blocks cancel, rebasing at the next nonzero exponent still admits finitely many positive generators: the multi-indices above that exponent form an upward-closed subset of $\mathbb{N}^m$, with finitely many coordinatewise minimal elements. Subtracting the new leading exponent from the weights of those elements supplies finitely many nonnegative shifts, together with the original generators. This also justifies normalizing F by its actual leading monomial in (1). Separate the finite leading polynomial before applying the small-parameter construction; in (2) separate $c + kL$ to obtain the factor $e^c w^k$. □

The theorem supplies sufficient closure conditions, and every truncation is a genuine asymptotic expansion by Lemma 3.2. They are not necessary conditions for every individual expression: for example $\sqrt{L^2} = -L$ for $L < 0$, and polynomials may be applied to an unbounded logarithm. In general, non-integer powers of a nonconstant leading polynomial, its logarithm, exponentials of negative-power arguments, and oscillatory functions of an unbounded logarithm require larger scales (Section 12).

Corollary 3.4 (Laurent and Puiseux composition)**.** *If $U \in \mathcal{E}$ has a positive constant leading coefficient and positive valuation, and a selected real branch of $\phi(c+v)$ has a convergent Puiseux expansion $v^{n_0/d}A(v^{1/d})$ for $v \to 0^+$, with $d \geq 1$ an integer and A analytic at 0, then $\phi(c+U) \in \mathcal{E}$. The same holds from below using $\phi(c-v)$.*

Proof. Apply the closure theorem to $U^{1/d}$, the analytic function A, and $U^{n_0/d}$. Negative n_0 includes poles. The side of approach fixes the real branch of a ramified function. □

Variable powers also reduce to the closure theorem through $F^G = \exp(G \log F)$ when $F > 0$ and the resulting exponential argument satisfies its hypotheses; this includes $w^w = \exp(w \log w)$. For a nonzero leading block $w^\beta P(L)$, the eventual sign is the sign of $(-1)^{\deg P}[L^{\deg P}]P$, so absolute values reduce to multiplication by that sign whenever it is provable.

3.2 Propagation of finite-order errors

A finite-order expansion of a function F consists of a jet T and a remainder assertion

$$F = T + \mathcal{O}(w^P \mathrm{M}(w)^D), \qquad P \in \mathbb{R} \cup \{+\infty\}, \tag{3.3}$$

where $P = +\infty$ denotes an exact finite expression. The following rules construct expansions of compound functions from those of their constituents.

- For a sum, add the jets and use the coarser of the two remainder classes. If a block is discarded at the boundary exponent, include its whole logarithmic degree in that remainder.
- For a product $(T_1 + R_1)(T_2 + R_2)$, the unknown part is $R_1T_2 + R_2T_1 + R_1R_2$. Apply (2.1) to all three contributions, and to every explicit product block removed by truncation. A negative leading exponent in either factor can reduce the absolute precision.
- Suppose $F = cw^\beta(1 + U)$ with $c > 0$ and $U \to 0$. An absolute error $\mathcal{O}(w^P \mathrm{M}^D)$ in F is a relative error $\mathcal{O}(w^{P-\beta}\mathrm{M}^D)$ in U. For fixed real a, expand $(1 + U)^a$ and multiply by $c^a w^{a\beta}$; the propagated input error is $\mathcal{O}(w^{P+(a-1)\beta}\mathrm{M}^D)$. Integer powers also follow by finite multiplication, without a positivity assumption.
- With the same positive leading monomial, $\log F = \log c + \beta L + \log(1 + U)$. The propagated error is measured at relative exponent $P - \beta$.
- If $F = c + kL + U$, where $k \in \mathbb{R}$, $U \to 0$ and the input uncertainty tends to zero, then $e^F = e^c w^k e^U$. The input error is multiplied by the exact factor $e^c w^k$.
- For a function ϕ analytic at c and a vanishing increment U, use $\phi(c+U) = \sum_{j=0}^{n} \phi^{(j)}(c)U^j/j! + \mathcal{O}(U^{n+1})$. Laurent and Puiseux expansions on a fixed side reduce to powers of the signed increment and analytic composition.

These analytic unit substitutions require their propagated relative uncertainties to tend to zero. All unit-series truncations are finite by positive valuation. The error from their explicit truncation is combined with the propagated uncertainty of their arguments. If cancellation conceals the leading increment needed for a pole or a ramified branch, more terms of that increment must first be determined.

Lemma 3.5 (Nonlinear logarithmic bound at the precision boundary)**.** *Let* $U = \sum_{\beta>0} w^\beta P_\beta(L)$ *be a nonzero finite jet,* $\nu = \operatorname{val} U > 0$*, and* $d = \max_\beta \deg P_\beta$*. Suppose* F *is real-valued and* $F = U + \mathcal{O}(w^P \mathrm{M}^D)$ *with* $P > 0$ *and* $D \in \mathbb{N}$*. An exact argument* $F = U$ *is also allowed, with* $P = +\infty$*. Let* Φ *be real analytic near* 0*, and let* $H > 0$ *be finite. Put*

$$c = \min(P, H), \qquad N = \lceil c/\nu \rceil, \qquad J = \left[\sum_{j=0}^{N-1} \frac{\Phi^{(j)}(0)}{j!} U^j \right]_{<c},$$

where the brackets retain complete blocks of exponent less than c*. Then*

$$\Phi(F) = J + \mathcal{O}(w^c \mathrm{M}^K), \qquad K = \begin{cases} Nd, & H < P, \\ \max(D, Nd), & P \le H. \end{cases} \tag{3.4}$$

Proof. The finite jet satisfies $U = \mathcal{O}(w^\nu \mathrm{M}^d)$ and tends to zero. Taylor's theorem gives a tail $\mathcal{O}(U^N)$, which is $\mathcal{O}(w^{N\nu} \mathrm{M}^{Nd}) \subseteq \mathcal{O}(w^c \mathrm{M}^{Nd})$. Every block discarded from the finite Taylor polynomial has exponent at least c and logarithmic degree at most $(N-1)d$, so it obeys the same bound. Both F and U tend to zero, and Φ' is bounded on a neighborhood containing them. The mean value theorem therefore gives $\Phi(F) - \Phi(U) = \mathcal{O}(w^P \mathrm{M}^D)$. If $H < P$, Lemma 2.2 absorbs this into $\mathcal{O}(w^H)$; if $P \le H$, its degree must be combined with the nonlinear truncation degree at the same power. For $F = U$ this last contribution is absent. □

In particular, retaining more powers than the input precision permits does not remove the logarithms generated by nonlinear truncation. For $F = wL + \mathcal{O}(w^3)$ and $H \ge 3$,

$$\log(1 + F) = wL - \tfrac{1}{2} w^2 L^2 + \mathcal{O}(w^3 \mathrm{M}^3).$$

The cubic Taylor term has degree three although the inherited degree is zero. The bound uses the largest degree of *all* retained blocks: $U = w + w^2 L^7$, $P = 3$, $D = 0$ gives the valid conservative degree 21, without asserting it is minimal. If $U = 0$, boundedness of Φ' instead directly gives $\Phi(F) = \Phi(0) + \mathcal{O}(w^P \mathrm{M}^D)$; for $F = 0$ this is exact.

Proposition 3.6 (Soundness of compositional error propagation)**.** *Suppose the initial expansions satisfy their stated branch and remainder hypotheses. Every finite composition of the rules above satisfies its resulting assertion* (3.3)*.*

Proof. Induct on the number of operations. For sums and products, use (2.1) and include the explicit discarded blocks. Lemma 3.5 combines the Taylor truncation error with the argument error of an analytic unit function; Lemma 3.2 supplies finite-order input bounds for convergent expansions. Factoring the leading monomial gives the stated shifts for powers and logarithms; the same factorization reduces a Laurent or Puiseux composition to the preceding rules. □

Each initial analytic remainder estimate is a hypothesis of this proposition. Knowing the degree of one omitted coefficient does not by itself bound an arbitrary unknown tail. For a convergent expansion in the class $\mathcal{E}$, Lemma 3.2 supplies the needed complete-tail assertion. For other functions, an independent finite-order asymptotic theorem is required.

3.3 The second question

For $F(x) = (1 + x + x^{\sqrt{2}})^{\sqrt{2}}$ at $x \to +\infty$ the local variable is $w = 1/x$ and

$$F = w^{-2}(1 + w^{\sqrt{2}-1} + w^{\sqrt{2}})^{\sqrt{2}} = w^{-2} \sum_{k \ge 0} \binom{\sqrt{2}}{k} (w^{\sqrt{2}-1} + w^{\sqrt{2}})^k,$$

an instance of item (1) of Theorem 3.3 with gaps $\sqrt{2}-1$ and $\sqrt{2}$. The exponents of x are $2-(\sqrt{2}-1)k_1-\sqrt{2}k_2$; ordering the semigroup elements gives (1.5), and the first omitted exponent is $7-5\sqrt{2}$ (multi-index $(5,0)$). The tail estimate therefore gives the remainder $\mathcal{O}(x^{7-5\sqrt{2}})$ in (1.5). Every larger finite cutoff is treated by the same multinomial expansion and ordering of the positive-gap semigroup.

4 The model and the real branch

4.1 The normal form

Let $u \to 0^+$ be the local variable (later $u = x - x_0$, $x_0 - x$, $1/x$ or $-1/x$). The central object is a finite *power–log germ*

$$f(u) = y_0 + a\,u^p\Big(1 + \sum_{i=1}^{m} u^{\delta_i} B_i(\log u)\Big), \qquad a \neq 0,\ p \neq 0,\ \delta_i > 0,\ B_i \in \mathbb{R}[L] \setminus \{0\}, \tag{4.1}$$

with $y_0 \in \mathbb{R}$ when $p > 0$ and $y_0 = 0$ when $p < 0$ (a constant term is then a correction with gap $-p$). Equal gaps are merged. The *leading block* au^p has a constant coefficient; a logarithmic factor in the leading block ($u \log u$, $u(\log u)^{-1}$) changes the problem (Section 12). Everything below is stated for this exact finite germ; functions that are only known through a truncated expansion are handled in Section 9.

The *uniformizer* is

$$z = \Big(\frac{v}{a}\Big)^{1/p}, \qquad v = y - y_0 \ (p > 0), \quad v = y \ (p < 0), \qquad L = \log z = \frac{\log(v/a)}{p}, \tag{4.2}$$

so that $z \to 0^+$ in every case: for $p > 0$ because $v \to 0$ with $v/a > 0$, for $p < 0$ because $v \to \pm\infty$ with the sign of a. The role of z is that the inverse satisfies $u \sim z$.

4.2 Existence and uniqueness of the local inverse

Theorem 4.1 (Positive local branch)**.** *For the germ (4.1) there is $\varepsilon > 0$ such that f is strictly monotone on $(0, \varepsilon)$, with*

$$f(u) - y_0 \sim a\,u^p, \qquad f'(u) \sim a\,p\,u^{p-1} \qquad (u \to 0^+). \tag{4.3}$$

Its inverse g, defined on the image of $(0,\varepsilon)$, is the unique function with $g > 0$, $g \to 0$ and $f(g(y)) = y$ there, and it satisfies

$$g(y) \sim z, \qquad g'(y) \sim \frac{z}{p\,v} \qquad (z \to 0^+). \tag{4.4}$$

Proof. Write $R(u) = \sum_i u^{\delta_i} B_i(\log u)$. By Lemma 2.2, $R(u) \to 0$ and, since $uR'(u) = \sum_i u^{\delta_i}(\delta_i B_i + B_i')(\log u)$ by Lemma 2.6, also $uR'(u) \to 0$. Hence

$$f'(u) = apu^{p-1}\Big(1 + R(u) + \frac{uR'(u)}{p}\Big) = apu^{p-1}(1 + o(1)),$$

which has the sign of ap on some $(0,\varepsilon)$; this gives monotonicity and (4.3). The inverse exists on the image, tends to 0 because f is a homeomorphism of $(0,\varepsilon)$ onto its image, and is the only solution of $f(u) = y$ in $(0,\varepsilon)$. Substituting $u = g(y)$ into $f(u) - y_0 \sim au^p$ gives $v \sim ag(y)^p$, i.e. $g \sim z$; and $g' = 1/f'(g) \sim 1/(apg^{p-1}) = g/(pag^p) \sim z/(pv)$. □

The theorem is local. Whether the local branch extends to a global inverse is a separate question, answered for the two examples of [1] as follows.

Proposition 4.2 (Global invertibility of the examples)**.** *$f_1(x) = x + x^2(1+\log x)$ and $f_2(x) = x + x^{\sqrt{2}}$ are strictly increasing bijections of $(0,\infty)$ onto itself, with*

$$f_1'(x) = 1 + x(3+2\log x) \ \geq\ m_1 := 1 - 2e^{-5/2} > 0.835, \qquad f_2'(x) = 1 + \sqrt{2}\,x^{\sqrt{2}-1} > 1. \tag{4.5}$$

Their inverses g_1, g_2 are C^1 up to $y = 0$ with $g_i(0) = 0$, $g_i'(0) = 1$, but g_1 is not C^2 at 0: $g_1''(y) = -f_1''(g_1)/f_1'(g_1)^3 \sim -(5+2\log y) \to +\infty$.

Proof. $x(3+2\log x)$ has derivative $5+2\log x$, minimum at $x = e^{-5/2}$ with value $-2e^{-5/2}$; the limits of f_1 at 0 and ∞ are 0 and ∞. The rest is immediate. □

So for both examples the branch of Theorem 4.1 is *the* real inverse, and the expansions (1.3)–(1.4) describe it near 0; the absence of a second-order Taylor expansion of g_1 at 0 is exactly why $\log y$ must appear at y^2.

4.3 Competing complex inverse germs

The equation $f_1(x) = 0$, $x \neq 0$, reads $1 + x(1+\log x) = 0$. With $t = 1 + \log x$ it becomes $te^t = -e$, so $t = \mathrm{W}_k(-e)$ and

$$c_k = e^{\mathrm{W}_k(-e)-1} = -\frac{1}{\mathrm{W}_k(-e)}, \qquad f_1'(c_k) = c_k - 1, \qquad f_1''(c_k) = 3 + 2\mathrm{W}_k(-e), \tag{4.6}$$

where the derivative identities follow from $c_k\mathrm{W}_k(-e) = -1$, provided the logarithm near c_k is chosen to have value $\mathrm{W}_k(-e) - 1$ there. This branch qualification matters: with the fixed principal logarithm, the relevant roots are the conjugate pair $k = 0, -1$; other k require other local logarithm branches. For each compatible choice the derivative is nonzero, and the analytic inverse function theorem gives an inverse germ through c_k. For the choice $k = -1$, the Taylor expansion is

$$c + \frac{y}{c-1} - \frac{f_1''(c)}{2f_1'(c)^3}y^2 + \cdots = c + \frac{y}{c-1} - \frac{3 + 2\mathrm{W}_{-1}(-e)}{2(c-1)^3}y^2 + \mathcal{O}(y^3),$$

which is (1.2). This germ is the inverse through a complex root. Restricting the target to real positive y does not select a real preimage. The positive real branch is instead characterized by $g(y) > 0$ and $g(y) \to 0$, as in Theorem 4.1; this limiting condition distinguishes it from every germ through a nonzero root.

5 Lagrange–Bürmann inversion with a marker

The classical Lagrange inversion theorem expands the inverse of an analytic map at a point where its derivative does not vanish. Our maps are not analytic at 0, and the inverse has no Taylor series there; but at every *positive* point the map is analytic, and the trick is to invert a perturbation $u + \eta h(u) = q$ around the positive point q in the perturbation parameter η. The logarithms and irrational powers in h are then analytic functions of u near $q > 0$, and the derivatives that appear are computed by the Euler identity without ever evaluating anything at 0.

5.1 The marker theorem

Theorem 5.1 (Lagrange–Bürmann with a marker)**.** *Let $q > 0$ and let h and Φ be analytic in a complex neighbourhood of q. For $|\eta|$ sufficiently small the equation*

$$u + \eta\, h(u) = q \tag{5.1}$$

has a unique solution $u(\eta)$ near q with $u(0) = q$, it is analytic in η, and

$$\Phi(u(\eta)) = \Phi(q) + \sum_{N\geq 1} \frac{(-\eta)^N}{N!} \frac{\mathrm{d}^{N-1}}{\mathrm{d}q^{N-1}} \Big[\Phi'(q)\, h(q)^N\Big]. \tag{5.2}$$

The same identity holds coefficientwise in η for arbitrary formal power series h, Φ with coefficients in any commutative $\mathbb{Q}$-algebra, i.e. as an identity of formal power series in η whose coefficients are polynomials in finitely many derivatives of h and Φ at q.

Proof. Choose $\rho > 0$ so small that h and Φ are analytic on the closed disc $|s - q| \leq \rho$, and let $M = \max_{|s-q|=\rho} |h(s)|$. For $|\eta| < \rho/M$ we have $|\eta h(s)| < |s - q|$ on the circle $C = \{|s - q| = \rho\}$, so by Rouché's theorem $F_\eta(s) = s - q + \eta h(s)$ has exactly one zero $u(\eta)$ in the disc, and by the argument principle

$$\Phi(u(\eta)) = \frac{1}{2\pi i} \oint_C \Phi(s)\, \frac{F_\eta'(s)}{F_\eta(s)}\, \mathrm{d}s = \frac{1}{2\pi i} \oint_C \Phi(s)\, \frac{1 + \eta h'(s)}{s - q + \eta h(s)}\, \mathrm{d}s.$$

On C the geometric series $\frac{1}{s-q+\eta h(s)} = \sum_{N\geq 0} \frac{(-\eta)^N h(s)^N}{(s-q)^{N+1}}$ converges uniformly, so

$$\Phi(u(\eta)) = \sum_{N\geq 0} (-\eta)^N \Big[\frac{1}{2\pi i} \oint_C \frac{\Phi(s) h(s)^N}{(s-q)^{N+1}}\, \mathrm{d}s + \frac{1}{2\pi i} \oint_C \frac{\eta\, \Phi(s) h'(s) h(s)^N}{(s-q)^{N+1}}\, \mathrm{d}s\Big].$$

By Cauchy's formula the first integral is $\frac{1}{N!} \frac{\mathrm{d}^N}{\mathrm{d}q^N} [\Phi h^N](q)$ and the second, after shifting the index $N \to N - 1$, contributes $-\frac{1}{(N-1)!} \frac{\mathrm{d}^{N-1}}{\mathrm{d}q^{N-1}} [\Phi h' h^{N-1}](q)$ to the coefficient of $(-\eta)^N$ for $N \geq 1$. Since $\frac{\mathrm{d}}{\mathrm{d}q}[\Phi h^N] = \Phi' h^N + N \Phi h' h^{N-1}$, the coefficient of $(-\eta)^N$ is

$$\frac{1}{N!} \frac{\mathrm{d}^{N-1}}{\mathrm{d}q^{N-1}} [\Phi' h^N + N\Phi h' h^{N-1}] - \frac{1}{(N-1)!} \frac{\mathrm{d}^{N-1}}{\mathrm{d}q^{N-1}} [\Phi h' h^{N-1}] = \frac{1}{N!} \frac{\mathrm{d}^{N-1}}{\mathrm{d}q^{N-1}} [\Phi' h^N],$$

which is (5.2); the constant term is $\Phi(q)$. Analyticity of $u(\eta)$ follows from the implicit function theorem at $(q, 0)$, where $\partial_u F_0 = 1$. For the formal statement, both sides of (5.2) have coefficients that are universal polynomials in the Taylor coefficients of h and Φ at q; two polynomials that agree for all analytic data agree identically (they agree on a Zariski-dense set), so the identity holds for arbitrary formal coefficients. □

Remark 5.2. Other proofs exist [4, 5, 3]; the one above is chosen because it exhibits the analytic content: the solution is the unique root in a disc around q, a fact reused in Section 7 to obtain explicit constants. The formal statement is what makes the coefficient formula of Section 6 an identity, independent of any convergence question.

5.2 Consequences

Near-identity form. With $\Phi(q) = q$ and $\eta = 1$,

$$f(u) = u + h(u) \quad\Longrightarrow\quad g(y) = y + \sum_{N\geq 1} \frac{(-1)^N}{N!} \frac{\mathrm{d}^{N-1}}{\mathrm{d}y^{N-1}} [h(y)^N], \tag{5.3}$$

formally; Sections 7 and 9 say when setting $\eta = 1$ is legitimate as an asymptotic statement. The formula (5.3) is a complete answer for the log example: with $h(y) = y^2(1+\log y)$, the N-th term is $\frac{(-1)^N}{N!}\partial_y^{N-1}[y^{2N}(1+\log y)^N]$, computed by Lemma 2.6; for $N=2$, $\frac{1}{2}\partial_y[y^4(1+L)^2] = y^3(2(1+L)^2 + (1+L)) = y^3(2L^2+5L+3)$.

Several markers. If $h = \sum_i \eta_i h_i$ with independent markers, expanding h^N multinomially in (5.2) gives the coefficient of $\prod \eta_i^{k_i}$, $|\boldsymbol{k}| = N$, as

$$\frac{(-1)^N}{\boldsymbol{k}!}\frac{\mathrm{d}^{N-1}}{\mathrm{d}q^{N-1}}\Big[\Phi'(q)\prod_i h_i(q)^{k_i}\Big], \tag{5.4}$$

because the multinomial coefficient $N!/\boldsymbol{k}!$ cancels the $1/N!$.

Transport of observables. Formula (5.2) gives at once the expansion of any analytic function of the inverse. With $\Phi(u) = u^r$ one obtains powers of the inverse; with $\Phi = \log$ its logarithm; differentiating the identity for u^r with respect to r at $r = 0$ gives the logarithm again. These are used in Section 6.

General core. Suppose $F = F_0 + R$ where F_0 has a known inverse ϕ on the branch of interest and R is a perturbation. Put $t = F_0(x)$, so that $F(x) = y$ becomes $t + \eta\, R(\phi(t)) = y$ with $\eta = 1$; Theorem 5.1 with $\Phi = \phi$ yields

$$x = \phi(y) + \sum_{N\ge 1}\frac{(-1)^N}{N!}\frac{\mathrm{d}^{N-1}}{\mathrm{d}y^{N-1}}\Big[\phi'(y)\,R(\phi(y))^N\Big]. \tag{5.5}$$

This is the "phase-coordinate" inversion of [11]; it is the natural coefficient identity when the dominant part of F is not a monomial (Section 12). For the monomial core $F_0(x) = ax^p$, $\phi(y) = (y/a)^{1/p}$, it is equivalent to the substitution $t = x^p$ used in the next section.

6 The coefficient theorem

6.1 Statement

Let f be the germ (4.1), z the uniformizer (4.2), $L = \log z$, $\mathcal{D} = \mathrm{d}/\mathrm{d}L$. For a multi-index $\boldsymbol{k} \in \mathbb{N}^m$ write $n = |\boldsymbol{k}|$, $\sigma = \boldsymbol{k}\cdot\boldsymbol{\delta}$ and $B^{\boldsymbol{k}} = \prod_i B_i^{k_i}$.

Theorem 6.1 (Euler-operator coefficient formula)**.** *For every real r, the branch g of Theorem 4.1 satisfies, formally and (Theorem 7.1) as a convergent expansion for small z,*

$$\boxed{g(y)^r = z^r + \sum_{\boldsymbol{k}\ne 0} z^{r+\boldsymbol{k}\cdot\boldsymbol{\delta}}\, C_{\boldsymbol{k}}^{(r)}(L), \qquad C_{\boldsymbol{k}}^{(r)}(L) = \frac{(-1)^n\, r}{p^n\, \boldsymbol{k}!}\prod_{j=1}^{n-1}(\mathcal{D} + r + \sigma + pj)\, B^{\boldsymbol{k}}(L).} \tag{6.1}$$

The empty product ($n = 1$) is the identity. Multi-indices of equal weight are summed: the block of g^r at $z^{r+\sigma}$ is $C_\sigma^{(r)} = \sum_{\boldsymbol{k}\cdot\boldsymbol{\delta}=\sigma} C_{\boldsymbol{k}}^{(r)}$. In particular

$$\begin{aligned} g(y) &= z\Big(1 + \sum_{\boldsymbol{k}\ne 0} z^{\boldsymbol{k}\cdot\boldsymbol{\delta}} C_{\boldsymbol{k}}(L)\Big), \qquad C_{\boldsymbol{k}} = C_{\boldsymbol{k}}^{(1)}, \\ \log g(y) &= L + \sum_{\boldsymbol{k}\ne 0} z^{\boldsymbol{k}\cdot\boldsymbol{\delta}}\frac{(-1)^n}{p^n \boldsymbol{k}!}\prod_{j=1}^{n-1}(\mathcal{D} + \sigma + pj)B^{\boldsymbol{k}}(L). \end{aligned} \tag{6.2}$$

Proof. Put $t = u^p$ and $q = v/a$ (so $q = z^p$). The equation $f(u) = y$ becomes

$$q = t + \sum_i t^{1+\delta_i/p} B_i\Big(\frac{\log t}{p}\Big),$$

which is (5.1) with markers: $t + \sum_i \eta_i h_i(t) = q$, $h_i(t) = t^{1+\delta_i/p} B_i(\log t/p)$, at $\eta_i = 1$. The observable is $\Phi(t) = t^{r/p} = u^r$. All h_i and Φ are analytic near any $q > 0$. By (5.4) the coefficient of $\prod \eta_i^{k_i}$ is

$$\frac{(-1)^n}{\boldsymbol{k}!}\frac{\mathrm{d}^{n-1}}{\mathrm{d}q^{n-1}}\Big[\frac{r}{p}\, q^{r/p-1} \prod_i \big(q^{1+\delta_i/p} B_i(\tfrac{\log q}{p})\big)^{k_i}\Big] = \frac{(-1)^n r}{p\,\boldsymbol{k}!}\frac{\mathrm{d}^{n-1}}{\mathrm{d}q^{n-1}}\Big[q^{\,n-1+(r+\sigma)/p}\, B^{\boldsymbol{k}}(\tfrac{\log q}{p})\Big].$$

Apply the repeated Euler identity (2.2) in the variable q, with $\Lambda = \log q$ and the polynomial $\tilde{P}(\Lambda) = B^{\boldsymbol{k}}(\Lambda/p)$: the derivative of order $n-1$ of $q^b \tilde{P}(\Lambda)$, $b = n-1+(r+\sigma)/p$, is $q^{b-n+1}\prod_{j=0}^{n-2}(b-j+\mathrm{d}/\mathrm{d}\Lambda)\tilde{P}$. Since $L = \Lambda/p$, $\mathrm{d}/\mathrm{d}\Lambda = \mathcal{D}/p$, and $b - j = (r+\sigma+p(n-1-j))/p$; hence each factor is $\frac{1}{p}(\mathcal{D} + r + \sigma + p(n-1-j))$, and with $j' = n-1-j$ running from 1 to $n-1$ the product is $p^{-(n-1)}\prod_{j'=1}^{n-1}(\mathcal{D}+r+\sigma+pj')B^{\boldsymbol{k}}(L)$. Finally $q^{b-n+1} = q^{(r+\sigma)/p} = z^{r+\sigma}$, and the prefactor r/p combines with $p^{-(n-1)}$ into r/p^n. The formal validity is Theorem 5.1's formal clause, applied to the finitely many markers; setting all $\eta_i = 1$ is legitimate coefficientwise because by Lemma 2.4 only finitely many multi-indices contribute to any exponent. The formula for $\log g$ is the derivative of (6.1) with respect to r at $r = 0$, where the factor r vanishes and only the term with the factor differentiated survives; equivalently it is Theorem 5.1 with $\Phi = \log$. ☐

Remark 6.2. The normalization in (6.1) contains four essential features: sign $(-1)^n$, the multi-index factorial $\boldsymbol{k}!$ (not $n!$: the multinomial coefficient from expanding h^n has been absorbed), the factor p^{-n} (one $1/p$ from Φ' and one from each of the $n-1$ derivatives, through $\mathrm{d}/\mathrm{d}\Lambda = \mathcal{D}/p$), and the operator constants $r + \sigma + pj$. Forgetting the substitution $L = \log(v/a)/p$ in the coefficients, or the factor p^{-n}, are the two classical mistakes for nonunit p.

6.2 First orders and cross terms

For a single block ($m = 1$, gap δ, polynomial B):

$$g = z - \frac{z^{1+\delta}}{p}B + \frac{z^{1+2\delta}}{2p^2}\big((1+2\delta+p)B^2 + 2BB'\big) - \frac{z^{1+3\delta}}{6p^3}(\mathcal{D}+1+3\delta+p)(\mathcal{D}+1+3\delta+2p)B^3 + \cdots. \tag{6.3}$$

For two blocks the mixed multi-index $e_i + e_j$ contributes

$$\frac{z^{1+\delta_i+\delta_j}}{p^2}(\mathcal{D}+1+\delta_i+\delta_j+p)(B_iB_j) = \frac{z^{1+\delta_i+\delta_j}}{p^2}\big((1+\delta_i+\delta_j+p)B_iB_j + B_i'B_j + B_iB_j'\big), \tag{6.4}$$

which shows why the inverse of a sum of perturbations is not the sum of the separately computed inverse corrections.

6.3 Degrees and leading logarithms

Proposition 6.3 (Degrees)**.** *Let $d_i = \deg B_i$ and $D(\boldsymbol{k}) = \sum_i k_i d_i$. Then $\deg C_{\boldsymbol{k}}^{(r)} \le D(\boldsymbol{k})$, with coefficient at the largest possible degree*

$$[L^{D(\boldsymbol{k})}]\, C_{\boldsymbol{k}}^{(r)} = \frac{(-1)^n r}{p^n \boldsymbol{k}!}\prod_{j=1}^{n-1}(r+\sigma+pj)\prod_i\big([L^{d_i}]B_i\big)^{k_i}, \tag{6.5}$$

which is nonzero exactly when $r \neq 0$ and $r+\sigma+pj \neq 0$ for $1 \le j \le n-1$ (always true for $r>0$, $p>0$). Under those conditions the degree is exactly $D(\boldsymbol{k})$. Zero operator constants differentiate and may lower the degree or annihilate the polynomial, while $r=0$ annihilates every correction. An aggregated block $C^{(r)}_\sigma$ has degree at most $\max\{D(\boldsymbol{k}) : \boldsymbol{k}\cdot\boldsymbol{\delta}=\sigma\}$; equality requires a nonzero sum of its top-degree coefficients.

Proof. Each factor $\mathcal{D}+c$ with $c \neq 0$ preserves the degree and multiplies the top coefficient by c; a factor with $c=0$ differentiates. The product $B^{\boldsymbol{k}}$ has degree $D(\boldsymbol{k})$, and the prefactor contains r. □

For the log example ($a=p=\delta=1$, $B=1+L$, single index n) the formula reads

$$P_n(L) = \frac{(-1)^n}{n!}\prod_{j=n+2}^{2n}(\mathcal{D}+j)\,(1+L)^n,$$
$$[L^n]P_n = \frac{(-1)^n}{n!}\prod_{j=n+2}^{2n} j = (-1)^n\frac{(2n)!}{n!\,(n+1)!} = (-1)^n\,\mathrm{Cat}_n, \tag{6.6}$$

the signed Catalan numbers. The next coefficient is also explicit: selecting the derivative in exactly one factor of the product gives

$$[L^{n-1}]P_n = (-1)^n\,\mathrm{Cat}_n\; n\,(1+H_{2n}-H_{n+1}), \qquad H_k = \textstyle\sum_{j\le k} 1/j, \tag{6.7}$$

because $\prod_{j=n+2}^{2n}(\mathcal{D}+j)(1+L)^n$ has L^{n-1}-coefficient $\left(\prod j\right)\left(n+\sum_{j=n+2}^{2n} n/j\right)$ (the term n from expanding $(1+L)^n$ and the terms n/j from applying $\mathcal{D}$ in the factor j). Every P_n is divisible by $1+L$: in the variable $t=1+L$ the operator product $\prod_{j=n+2}^{2n}(\mathcal{D}+j)$ consists of $n-1$ factors, each of which either differentiates (lowering the power of t by one) or multiplies by a constant, so every monomial of $\prod(\mathcal{D}+j)\,t^n$ has degree at least $n-(n-1)=1$ in t. Thus

$$P_1 = -(1+L), \qquad P_2 = (1+L)(2L+3), \qquad P_3 = -\tfrac{1}{2}(1+L)(10L^2+31L+23),$$
$$P_4 = \tfrac{1}{6}(1+L)(84L^3+398L^2+607L+299),$$

which are the blocks of (1.3).

6.4 Constant coefficients: Fuss–Catalan numbers

When every B_i is a constant b_i, the operators act as scalars and

$$C^{(r)}_{\boldsymbol{k}} = \frac{(-1)^n r}{p^n\boldsymbol{k}!}\prod_{j=1}^{n-1}(r+\sigma+pj)\prod_i b_i^{k_i} = \frac{r}{r+\sigma}\binom{-(r+\sigma)/p}{n}\frac{n!}{\boldsymbol{k}!}\prod_i b_i^{k_i}. \tag{6.8}$$

The binomial expression in (6.8) is read by its removable limit when $r+\sigma=0$; the operator-product expression has no division by $r+\sigma$ and remains directly valid. For $f(x)=x+bx^\alpha$ ($p=1$, $\delta=\alpha-1$, $r=1$) this is the generalised Fuss–Catalan law

$$g(y) = \sum_{n\ge 0} c_n\, b^n\, y^{1+n(\alpha-1)},$$
$$c_n = \frac{(-1)^n}{n!}\prod_{j=0}^{n-2}(n\alpha-j) = \frac{(-1)^n}{1+n(\alpha-1)}\binom{n\alpha}{n} = (-1)^n\frac{\Gamma(n\alpha+1)}{\Gamma(n+1)\Gamma(n(\alpha-1)+2)}, \tag{6.9}$$

valid for every real $\alpha>1$ (no rationality is used), with $c_0=1$ understood separately in the product formula. For $\alpha=2$ the c_n are the signed Catalan numbers and, when $b\neq 0$, $g=(\sqrt{1+4by}-1)/(2b)$; the limit at $b=0$ is $g=y$. For $b=1$ and $\alpha=\sqrt{2}$,

$$c_1 = -1, \qquad c_2 = \sqrt{2}, \qquad c_3 = -\frac{6-\sqrt{2}}{2},$$

$$c_4 = \frac{17\sqrt{2}}{3} - 4, \qquad c_5 = \frac{51\sqrt{2}}{4} - \frac{305}{12}, \qquad c_6 = \frac{322\sqrt{2}}{5} - 77,$$

which gives (1.4). In this one-gap case $g(y) = yV(by^{\alpha-1})$ with V the solution of $V + tV^\alpha = 1$, $V(0) = 1$, analytic at $t = 0$ with radius of convergence

$$R_\alpha = \frac{(\alpha-1)^{\alpha-1}}{\alpha^\alpha}, \qquad |c_n| \sim \frac{\sqrt{\alpha}}{\sqrt{2\pi}\,(\alpha-1)^{3/2}}\, n^{-3/2} R_\alpha^{-n}, \tag{6.10}$$

by Stirling's formula, so the expansion (1.4) converges (and equals the positive inverse, by analytic continuation of the implicit equation) for $0 < y < R_{\sqrt{2}}^{1/(\sqrt{2}-1)} \approx 0.12686$. The singularity at $t = -R_\alpha$ is the fold $V = \alpha/(\alpha-1)$ of the implicit equation on the negative axis.

6.5 Formal existence, uniqueness, and iterative construction

Theorem 6.1 produces the coefficients; the following independent argument shows that they are the only possible ones and provides two alternative algorithms. Write $g = z(1+U)$ with U an unknown jet of positive valuation. Then $f(g) = y$ is equivalent to

$$E(U) := (1+U)^p \Big(1 + \sum_i z^{\delta_i}(1+U)^{\delta_i} B_i(L + \log(1+U))\Big) - 1 = 0, \tag{6.11}$$

an equation in the jet algebra (Lemma 2.7 evaluates every term).

Proposition 6.4 (Triangular structure)**.** *Equation (6.11) has exactly one formal solution U of positive valuation with support in the semigroup S generated by the gaps. Its coefficient at z^σ is determined by the coefficients at smaller weights through one division by p.*

Proof. $E(U) = pU + \sum_i z^{\delta_i} B_i(L) + (\text{terms of weight} \geq \min(2\,\mathrm{val}\,U, \mathrm{val}\,U + \delta_*))$, because $(1+U)^p = 1 + pU + \mathcal{O}(U^2)$ and every correction carries a factor z^{δ_i}. Order the elements of S increasingly (Lemma 2.4); at a new weight σ the coefficient of z^σ in $E(U)$ is pU_σ plus an expression in the $U_{\sigma'}$ with $\sigma' < \sigma$, which fixes U_σ. If U, V are two solutions with $U - V$ of least weight σ, then $E(U) - E(V) = (U - V)(p + \text{positive weight})$ has a nonzero block at z^σ, a contradiction. ☐

Picard iteration. Rewriting (6.11) as $U = \Psi(U) := \big(1 + \sum_i z^{\delta_i}(1+U)^{\delta_i} B_i(L + \log(1+U))\big)^{-1/p} - 1$, the map Ψ gains one gap per step: $\mathrm{val}(\Psi(U) - \Psi(V)) \geq \mathrm{val}(U - V) + \delta_*$. Hence $U_{k+1} = \Psi(U_k)$, $U_0 = 0$, agrees with the solution below weight $(k+1)\delta_*$, and $\lceil H/\delta_* \rceil$ iterations give all blocks below H.

Newton iteration. With $E'(U) = p(1+U)^{p-1} + \sum_i z^{\delta_i}(1+U)^{p+\delta_i-1}\big((p+\delta_i)B_i + B_i'\big)(L + \log(1+U))$, a unit of the jet algebra with constant leading coefficient p, the iteration $U_{k+1} = U_k - E(U_k)/E'(U_k)$ doubles the valuation of the error: if $\mathrm{val}(U_k - U) \geq s$ then $\mathrm{val}(U_{k+1} - U) \geq 2s$, because $E(U_k) = E'(U)(U_k - U) + \mathcal{O}((U_k - U)^2)$ and $E'(U_k)^{-1} = E'(U)^{-1}(1 + \mathcal{O}(U_k - U))$. Starting from $U_0 = 0$ with $s = \delta_*$, about $\log_2(H/\delta_*)$ iterations suffice. Formal uniqueness implies that Newton iteration and the multi-index formula give the same coefficients at every weight.

7 Convergence

A formal identity such as (6.1) says nothing, by itself, about the function g: the series might diverge, or converge to something else. This section proves that the expansion converges absolutely for small z and equals the branch of Theorem 4.1, first by an implicit-function argument that also yields the coefficient bounds needed for the remainder theory, then by a Rouché argument that yields explicit constants.

7.1 Convergence in finitely many small parameters

Theorem 7.1 (Convergence)**.** *Let f be the germ (4.1) and g its branch. For every real r the series (6.1) converges absolutely for all sufficiently small $z > 0$, its sum is $g(y)^r$, and there are constants M, K such that*

$$|C_{\boldsymbol{k}}^{(r)}(L)| \le M\, K^{|\boldsymbol{k}|}\, \mathrm{M}(z)^{D(\boldsymbol{k})} \qquad (L \le \log z_1), \quad D(\boldsymbol{k}) = \textstyle\sum_i k_i \deg B_i. \tag{7.1}$$

In particular $g^r \in \mathcal{E}$ (Definition 3.1) with support $r + S$.

Proof. Write $g = z\, w$ with $w \to 1$; by (6.11) the unknown w solves

$$w^p\Big(1 + \sum_i z^{\delta_i} w^{\delta_i} B_i(L + \log w)\Big) = 1.$$

Expand each $B_i(L + \log w) = \sum_{j=0}^{d_i} L^j A_{ij}(\log w)$ with $A_{ij}(v) = \sum_{s\ge j} b_{is}\binom{s}{j} v^{s-j}$, where $B_i = \sum_s b_{is} L^s$. Introduce independent complex parameters T_{ij} in place of $z^{\delta_i} L^j$ and consider

$$\mathcal{E}(w, T) = w^p\Big(1 + \sum_{i,j} T_{ij}\, w^{\delta_i} A_{ij}(\log w)\Big) - 1. \tag{7.2}$$

On the disc $|w - 1| < 1$ with the principal logarithm, $\mathcal{E}$ is analytic in (w, T); $\mathcal{E}(1, 0) = 0$ and $\partial_w \mathcal{E}(1, 0) = p \ne 0$. The analytic implicit function theorem gives an analytic $W(T)$ on a polydisc $|T_{ij}| < \rho$ with $W(0) = 1$ and $\mathcal{E}(W(T), T) = 0$, and W^r is analytic there too. Its Taylor series $W(T)^r = \sum_{\boldsymbol{\nu}} c_{\boldsymbol{\nu}} T^{\boldsymbol{\nu}}$ converges absolutely on the smaller polydisc $|T_{ij}| \le \rho/2$, with the Cauchy estimates $|c_{\boldsymbol{\nu}}| \le M(2/\rho)^{|\boldsymbol{\nu}|}$.

Now specialise $T_{ij} = z^{\delta_i} L^j$. Since $z^{\delta_i}\mathrm{M}(z)^{d_i} \to 0$, the point lies in the polydisc for small z, and $W(T(z))$ is real (all data are real and the solution is unique) and close to 1, so $zW(T(z))$ is a solution of $f(u) = y$ with $u \sim z$; by Theorem 4.1 it is $g(y)$. Group the absolutely convergent series $\sum c_{\boldsymbol{\nu}} \prod_{ij} (z^{\delta_i} L^j)^{\nu_{ij}}$ by $k_i = \sum_j \nu_{ij}$. There are at most $\prod_i (d_i + 1)^{k_i}$ terms for a given $\boldsymbol{k}$, with total logarithmic degree $\sum_{i,j} j\nu_{ij} \le D(\boldsymbol{k})$. Their sum is $z^{\boldsymbol{k}\cdot\boldsymbol{\delta}} \tilde{C}_{\boldsymbol{k}}(L)$, and the Cauchy bounds give both (7.1) and $\|\tilde{C}_{\boldsymbol{k}}\|_1 \le MK^{|\boldsymbol{k}|}$ for $K = (2/\rho)\max_i(d_i + 1)$. To identify the individual multi-index coefficients even when weights coincide, retain independent markers η_i by substituting $T_{ij} = \eta_i z^{\delta_i} L^j$. Both constructions solve the same formal implicit equation in the markers, whose linearization in w is $p \ne 0$, so its solution is unique coefficientwise. Thus $\tilde{C}_{\boldsymbol{k}} = C_{\boldsymbol{k}}^{(r)}$. Finally, aggregation by equal weights preserves the bounds of Definition 3.1: all representations of weight σ have length between $\sigma/\delta_{\max}$ and σ/δ_*, so the shortest representation controls their lengths up to a fixed factor; their count is at most exponential in that length. □

Remark 7.2. The theorem gives convergence of a multivariate power series after the substitution $T_{ij} = z^{\delta_i}(\log z)^j$. The normalized series g^r/z^r converges uniformly after shrinking z_1 so all these parameters remain in a smaller polydisc; for $r < 0$ this does not assert absolute uniform convergence of g^r itself on $(0, z_1)$. A single-variable radius exists in special cases such as (6.10). Nothing is uniform in a, p, δ_i, B_i: for $x + x^\alpha$ as $\alpha \downarrow 1$, the radius R_α in $t = y^{\alpha-1}$ tends to 1, whereas the corresponding endpoint $R_\alpha^{1/(\alpha-1)}$ in y tends to 0.

7.2 An explicit conditional majorant

The implicit-function argument has no explicit constants. Rouché's theorem in the marker equation gives them, at the price of a hypothesis that must be checked at each y.

Theorem 7.3 (Explicit tail bound). *Consider $f(t) = t + h(t)$, $h(t) = \sum_i t^{1+\alpha_i} P_i(\log t)$ with real α_i and $P_i(L) = \sum_s p_{is} L^s$. The general germ reduces to this form by $t = u^p$, $q = v/a$, with $\alpha_i = \delta_i/p$; these are positive at $t \to 0$ when $p > 0$ and negative at $t \to \infty$ when $p < 0$. Fix $0 < r < 1$, put $c_r = -\log(1-r)$,*

$$A_r(b) = \max\{(1-r)^b, (1+r)^b\},$$

and

$$\mathcal{Q}_r(t) = \frac{1}{r} \sum_i t^{\alpha_i} A_r(1+\alpha_i) \sum_s |p_{is}| \big(|\log t| + c_r\big)^s. \tag{7.3}$$

Let $t > 0$ satisfy $\mathcal{Q}_r(t) < 1$. Then:

1. *for every complex λ with $|\lambda| < 1/\mathcal{Q}_r(t)$ the equation $u + \lambda h(u) = t$ has exactly one solution $u(\lambda)$ in the disc $|u - t| < rt$; it is analytic in λ, $u(0) = t$, and $u(\lambda)$ is real for real λ;*
2. *the Lagrange terms $u_N = \frac{(-1)^N}{N!} \frac{\mathrm{d}^{N-1}}{\mathrm{d}t^{N-1}}[h(t)^N]$ satisfy $|u_N| \le \frac{rt}{N} \mathcal{Q}_r(t)^N$, and $u(\lambda) = t + \sum_{N \ge 1} u_N \lambda^N$ converges absolutely for $|\lambda| < 1/\mathcal{Q}_r(t)$, in particular at $\lambda = 1$;*
3. *$u(1)$ is the positive solution of $f(u) = t$ in $(t(1-r), t(1+r))$. It coincides with any local branch $g(t) \sim t$ near the endpoint under consideration, once t is sufficiently close to that endpoint, and*
$$\Big|u(1) - t - \sum_{N=1}^{N_0} u_N\Big| \le \frac{r\,t}{N_0+1} \frac{\mathcal{Q}_r(t)^{N_0+1}}{1 - \mathcal{Q}_r(t)}. \tag{7.4}$$

Proof. On the circle $|s - t| = rt$ write $s = t(1+\zeta)$ with $|\zeta| = r < 1$. For a real exponent b, $|s^b| = |s|^b \le t^b A_r(b)$; using only the upper bound on $|s|$ would be incorrect for $b < 0$. The principal logarithm satisfies $|\log s| \le |\log t| + |\log(1+\zeta)| \le |\log t| + c_r$. Hence $|h(s)| \le rt\mathcal{Q}_r(t)$. Rouché's theorem gives exactly one zero, counted with multiplicity, whenever $|\lambda|\mathcal{Q}_r(t) < 1$. That zero is simple, analytic in λ, and real for real λ by conjugation and uniqueness. Cauchy's estimate in the marker formula gives

$$|u_N| \le \frac{1}{N!} \frac{(N-1)!\,(rt\mathcal{Q}_r)^N}{(rt)^{N-1}} = \frac{rt}{N} \mathcal{Q}_r^N.$$

The root at $\lambda = 1$ is positive and lies in the indicated interval. Any branch with $g(t) \sim t$ eventually lies in that interval too, so the roots agree. Finally sum the bound over $N > N_0$, using $1/N \le 1/(N_0+1)$. □

Corollary 7.4 (Exponent cutoff). *Assume $\alpha_i > 0$ and $B \ge 0$. Let G_B retain all multi-indices of weight $\boldsymbol{k} \cdot \boldsymbol{\alpha} \le B$. Every multi-index with $|\boldsymbol{k}| \le N_0 := \lfloor B/\alpha_{\max} \rfloor$ is retained, so under $\mathcal{Q}_r(t) < 1$*

$$|g(t) - G_B(t)| \le \frac{rt}{N_0+1} \frac{\mathcal{Q}_r(t)^{N_0+1}}{1 - \mathcal{Q}_r(t)}.$$

Proof. The omitted part of the multi-marker expansion is a subseries of $\sum_{N > N_0} u_N$ (the expansion of u_N into multi-indices is absolutely convergent by the same Cauchy estimates, since $\sum_{|\boldsymbol{k}|=N} \frac{\prod M_i^{k_i}}{\boldsymbol{k}!} = \frac{(\sum M_i)^N}{N!}$). □

Example 7.5. For $f_1(x) = x + x^2(1 + \log x)$ and $r = \frac{1}{2}$: $\mathcal{Q}_{1/2}(y) = \frac{9}{2}\, y\,(1 + |\log y| + \log 2)$, so for $y \le \frac{1}{100}$ (where $\mathcal{Q}_{1/2} \le 0.284$) the expansion converges and the error after N_0 blocks is at most $\frac{y}{2(N_0+1)} \mathcal{Q}_{1/2}(y)^{N_0+1}/(1 - \mathcal{Q}_{1/2}(y))$; at $y = 10^{-2}$ and $N_0 = 3$ this is $1.1 \cdot 10^{-5}$ while the actual error is $1.5 \cdot 10^{-7}$. The bound is conditional (it says nothing when $\mathcal{Q}_r \ge 1$) and not tight, but it is rigorous, cheap, and turns the asymptotic statements into inequalities at specific points.

8 Remainders after a cutoff

8.1 Exponent cutoff: the first omitted block

Let the target cutoff be an exponent q of the local variable of the result, exclusive. In terms of weights, with r the observable power, a block $z^{r+\sigma} C(L)$ has exponent $(r + \sigma)/|p|$ in the local variable of y (Section 10), so the retained set is

$$I_H = \{\boldsymbol{k} \in \mathbb{N}^m : \boldsymbol{k} \cdot \boldsymbol{\delta} < H\}, \qquad H = |p|\, q - r, \tag{8.1}$$

a finite downward-closed set (Lemma 2.4). Set $C_0^{(r)} = 1$. When $H > 0$ its *one-step boundary* is

$$\partial I_H = \{\boldsymbol{k} + e_i : \boldsymbol{k} \in I_H,\ 1 \le i \le m\} \setminus I_H, \tag{8.2}$$

also finite. When $H \le 0$, I_H is empty and we define its boundary to be $\{0\}$, since the leading monomial is already omitted. With at least one correction gap, put

$$\sigma_* = \min\{\boldsymbol{\nu}\cdot\boldsymbol{\delta} : \boldsymbol{\nu} \in \partial I_H\}, \qquad C_{\sigma_*}^{(r)} = \sum_{\boldsymbol{k}\cdot\boldsymbol{\delta}=\sigma_*} C_{\boldsymbol{k}}^{(r)}, \qquad k_* = \deg C_{\sigma_*}^{(r)}\ (k_* = 0 \text{ if } C_{\sigma_*}^{(r)} = 0). \tag{8.3}$$

Lemma 8.1. *$\sigma_* = \min(S \setminus [0, H))$, and every multi-index of weight σ_* lies in ∂I_H. Thus the complete coefficient at the first omitted weight is obtained from the boundary alone.*

Proof. For $H \le 0$ the assertion is immediate, with $\sigma_* = 0$. For $H > 0$, let $\boldsymbol{\nu} \notin I_H$ have weight σ. Along a path from 0 to $\boldsymbol{\nu}$ that increases one coordinate at a time, the weights increase strictly, and the first point outside I_H is a boundary point of weight $\le \sigma$; hence $\sigma_* \le \sigma$, so σ_* is the least weight outside $[0, H)$. If $\sigma = \sigma_*$ the first point outside I_H has weight exactly σ_* by minimality, and no later point on the path can have the same weight, so $\boldsymbol{\nu}$ is that boundary point. ☐

Theorem 8.2 (Remainder after an exponent cutoff)**.** *Let $G_H^{(r)} = z^r \sum_{\boldsymbol{k}\in I_H} z^{\boldsymbol{k}\cdot\boldsymbol{\delta}} C_{\boldsymbol{k}}^{(r)}(L)$. Then, as $z \to 0^+$,*

$$g(y)^r - G_H^{(r)}(y) = z^{r+\sigma_*} C_{\sigma_*}^{(r)}(L) + o(z^{r+\sigma_*}) = \mathcal{O}(z^{r+\sigma_*} \mathrm{M}(z)^{k_*}). \tag{8.4}$$

If $C_{\sigma_*}^{(r)} \ne 0$ the bound is attained by its top logarithm; if $C_{\sigma_*}^{(r)} = 0$ (a resonant cancellation) the remainder is $o(z^{r+\sigma_*})$, and the true order is that of the next nonzero block, which the theorem does not identify.

Proof. By Theorem 7.1, $g^r \in \mathcal{E}$ with the bound (7.1), and the omitted multi-indices are those of weight $\ge \sigma_*$ by Lemma 8.1; Lemma 3.2 (refined form) gives the claim. ☐

Corollary 8.3 (In the variable of the result)**.** *With w the local variable of y ($w = |y - y_0|$ for finite y_0, $w = 1/|y|$ for infinite y_0),*

$$g(y)^r - G_H^{(r)}(y) = \mathcal{O}(w^{(r+\sigma_*)/|p|} \mathrm{M}(w)^{k_*}), \qquad \frac{r + \sigma_*}{|p|} \ge q,$$

since $z = (w/|a|)^{1/p}$ for $p > 0$, $z = (|a|w)^{1/|p|}$ for $p < 0$, and $\mathrm{M}(z) \asymp \mathrm{M}(w)$.

For a pure monomial ($m = 0$), the inverse observable is exactly z^r: the tail is zero if this term is retained and equals z^r otherwise. The pair $\big((r+\sigma_*)/|p|, k_*\big)$ specifies the remainder class in the target coordinate, while $z^{r+\sigma_*}C^{(r)}_{\sigma_*}$ is its complete first omitted block. If that coefficient vanishes, a sharper order requires computing further weights or proving that the remaining series terminates.

Example 8.4 (Resonances)**.** For $f = x + x^2 + x^3$ the gaps are 1 and 2; at weight 2 the indices $(2, 0)$ and $(0, 1)$ contribute 2 and -1, at weight 3 the indices $(3, 0)$ and $(1, 1)$ contribute -5 and $+5$, so

$$g(y) = y - y^2 + y^3 + 0 \cdot y^4 - 4y^5 + 14y^6 - 30y^7 + 33y^8 + \cdots ,$$

The coarse boundary estimate at cutoff 4 is $\mathcal{O}(y^4)$; summing the weight-3 coefficients improves it to $\mathcal{O}(y^5)$. For $f = x + x^2 + 2x^3$ the block at y^3 cancels instead. The cancellation concerns the sum of all contributions at an exactly equal weight, not each multi-index separately.

Example 8.5 (The log example)**.** Here $m = 1$, $\delta = 1$, $B = 1 + L$, so $I_H = \{0, \ldots , N\}$ for $H = N + 1$, the boundary is $\{N + 1\}$, and by (6.6) the frontier block $y^{N+2}P_{N+1}(\log y)$ has degree exactly $N + 1$ with top coefficient $(-1)^{N+1}\,\mathrm{Cat}_{N+1}$. Hence

$$\begin{gathered} g_1(y) = y + \sum_{n=1}^{N} y^{n+1}P_n(\log y) + \mathcal{O}\big(y^{N+2}\mathrm{M}(y)^{N+1}\big), \\ \frac{g_1(y) - y - \sum_{n\le N} y^{n+1}P_n(\log y)}{y^{N+2}(\log y)^{N+1}} \to (-1)^{N+1}\,\mathrm{Cat}_{N+1}, \end{gathered} \tag{8.5}$$

and in particular after $y - y^2(1 + \log y)$ the error is $y^3(2\log^2 y + 5\log y + 3) + \cdots$, not $\mathcal{O}(y^3)$. The ratio in (8.5) converges slowly (like $1 + c/\log y$): for $N = 1$ it is 1.89 at $y = 10^{-20}$ and 1.98 at $y = 10^{-100}$.

8.2 Depth truncation

When the exponents are symbolic (Section 12) their order may be undecidable, but the total degree $|\boldsymbol{k}|$ of a multi-index is always known.

Theorem 8.6 (Remainder after a depth cutoff)**.** *Let* $G^{(r)}_{\le N} = z^r \sum_{|\boldsymbol{k}|\le N} z^{\boldsymbol{k}\cdot\boldsymbol{\delta}}C^{(r)}_{\boldsymbol{k}}(L)$, $\delta_* = \min_i \delta_i$ *and* $d_* = \max_i \deg B_i$. *Then*

$$g(y)^r - G^{(r)}_{\le N}(y) = \mathcal{O}\big(z^{r+(N+1)\delta_*}\mathrm{M}(z)^{(N+1)d_*}\big). \tag{8.6}$$

Proof. The omitted terms have $|\boldsymbol{k}| \ge N + 1$; use (7.1) and the count m^n of multi-indices of degree n exactly as in the proof of Lemma 3.2. □

Depth truncation retains every block of exponent below $r + (N + 1)\delta_*$, but it may also retain blocks above that bound while omitting smaller ones when several gaps are present: for $f = x + x^2 + x^4$ depth 1 gives $y - y^2 - y^4$ although the omitted depth-two term $2y^3$ dominates y^4. The remainder (8.6) is nevertheless correct as stated.

9 Transport of remainders and finite smoothness

The theory so far concerns an exact finite germ. In practice the forward function is often known only through a truncated expansion, and one wants to know how the omitted part affects the inverse; conversely, one wants to turn a small residual $f(G(y)) - y$ of an approximation G into a bound for $G - g$. Both are mean-value arguments, but the scaling factor z^{1-p} of the derivative must not be forgotten.

9.1 From the forward function to the inverse

Theorem 9.1 (Transport of a forward remainder)**.** *Let f be the germ (4.1) with $p \neq 0$ and let $F = f + R$ on $(0, u_0)$, where*

$$R(u) = \mathcal{O}(u^{\rho}\mathrm{M}(u)^k), \qquad R'(u) = \mathcal{O}(u^{\rho-1}\mathrm{M}(u)^k), \qquad \rho > p,\ k \in \mathbb{N}. \tag{9.1}$$

Then F has a local inverse G on the same branch as g, and

$$G(y) - g(y) = \mathcal{O}(z^{\rho-p+1}\mathrm{M}(z)^k) = \mathcal{O}(w^{(\rho-p+1)/|p|}\mathrm{M}(w)^k), \tag{9.2}$$

where $w = |y - y_0|$ for $p > 0$ and $w = 1/|y|$ for $p < 0$. For any fixed real observable power r,

$$G(y)^r - g(y)^r = \mathcal{O}(z^{\rho-p+r}\mathrm{M}(z)^k) = \mathcal{O}(w^{(\rho-p+r)/|p|}\mathrm{M}(w)^k). \tag{9.3}$$

For $r = 0$ this difference is identically zero. The same conclusions hold if the derivative hypothesis is replaced by the assumption that F is continuous and strictly monotone near 0.

Proof. From (9.1) and Lemma 2.2, $F(u) - y_0 \sim au^p$ and $F'(u) \sim apu^{p-1}$, so F is strictly monotone near 0 and its inverse G satisfies $G \sim z$ (Theorem 4.1 applies verbatim). Put $u = G(y)$, $v = g(y)$; both are $\sim z$. Then $F(v) - y = R(v) = \mathcal{O}(z^{\rho}\mathrm{M}(z)^k)$, while $F(u) - y = 0$. On the interval between u and v, which lies in $[z/2, 2z]$ for small z, F' is at least $\frac{1}{2}|ap|z^{p-1}2^{-|p-1|}$ in absolute value. The mean value theorem gives (9.2). With continuous strict monotonicity instead of the derivative bound, the leading equivalence still supplies a continuous local inverse with $G \sim z$, and $f(u) - f(v) = -R(u)$ together with $|f'| \asymp z^{p-1}$ gives the same estimate. Finally the derivative of s^r on $[z/2, 2z]$ is $\mathcal{O}(z^{r-1})$, giving (9.3). $\square$

Remark 9.2. The derivative hypothesis is not a consequence of the value bound: $R(u) = u^2\sin(u^{-2})$ is $\mathcal{O}(u^2)$, but $R'(u) = 2u\sin(u^{-2}) - 2u^{-1}\cos(u^{-2})$ destroys the monotonicity of $u + R(u)$. Nor does non-strict monotonicity alone ensure an inverse, since it permits plateaus. Expansions obtained by the convergent structural operations above admit corresponding derivative estimates. For a general remainder, either the derivative estimate or continuous strict monotonicity must be established independently.

Corollary 9.3 (Cutoff cap)**.** *The data justify every block of G^r strictly below the exponent $(\rho-p+r)/|p|$ in w. Thus a requested cutoff q is certified from those data when $q \leq (\rho-p+r)/|p|$; the resulting remainder is the coarser of the truncation remainder and (9.3), compared by exponent first and by logarithmic degree at equal exponents. A larger requested cutoff requires a sharper forward remainder or additional information. For powers of the original variable at an infinite source endpoint, replace r here by the local-coordinate power $-r$.*

For example $\sin x + x^2\log x = x + x^2\log x - x^3/6 + \mathcal{O}(x^5)$ is a germ with $p = 1$, gaps $1, 2$ and $\rho = 5$, $k = 0$; the inverse is determined below cutoff $q = 5$ and has the expansion $y - y^2\log y + y^3(\frac{1}{6} + \log y + 2\log^2 y) + \mathcal{O}(y^4\mathrm{M}^3)$ for $q = 4$.

9.2 From a residual to the error of the inverse

Theorem 9.4 (Residual bracket)**.** *Let F be continuous on $[A, B]$ and differentiable inside with $F' \geq \mu > 0$. Let $x_0 \in (A, B)$, $y \in \mathbb{R}$, $\varrho = F(x_0) - y$. If $[x_0 - |\varrho|/\mu,\ x_0 + |\varrho|/\mu] \subseteq [A, B]$, then $F(x) = y$ has exactly one solution x_* in $[A, B]$, it lies in that interval, and*

$$|x_* - x_0| \ \leq\ \frac{|\varrho|}{\mu}. \tag{9.4}$$

If also $F' \leq M_1 < \infty$ on (A, B), then $|x_* - x_0| \geq |\varrho|/M_1$.

Proof. $F(x_0 + |\varrho|/\mu) \geq F(x_0) + |\varrho| \geq y$ and $F(x_0 - |\varrho|/\mu) \leq y$; the intermediate value theorem gives a root in the interval, monotonicity its uniqueness in $[A, B]$, and the mean value theorem gives the error bounds. A finite derivative maximum was not part of the hypotheses, so the lower error bound is stated separately. $\square$

This gives an a posteriori certificate that needs no expansion theory, only a lower bound on the derivative. For the two examples the global bounds (4.5) give, for every $y > 0$ and every $G > 0$,

$$|G - g_1(y)| \leq \frac{|G + G^2(1 + \log G) - y|}{1 - 2e^{-5/2}}, \qquad |G - g_2(y)| \leq |G + G^{\sqrt{2}} - y|. \tag{9.5}$$

For a general germ, $|f'| \asymp z^{p-1}$. For an approximation $G \sim z$, apply the bracket on $[z/2, 2z]$ with $\mu = \frac{1}{2}|ap|z^{p-1}2^{-|p-1|}$, after replacing (f, y) by $(-f, -y)$ if the derivative is negative. A residual small enough that the bracket lies in this interval yields $|G - g| \leq \frac{2^{1+|p-1|}}{|ap|} z^{1-p}|f(G) - y|$.

The residual of the truncated expansion. Combining Theorem 8.2 with the mean value theorem, the truncated inverse G_H satisfies

$$\frac{f(G_H(y)) - y_0}{a\, z^p} - 1 = -p\, z^{\sigma_*} C_{\sigma_*}(L) + o(z^{\sigma_*}), \tag{9.6}$$

Here $y_0 = 0$ for $p < 0$, and the truncation retains the leading term so $G_H \sim z$. Indeed $f'(s) = apz^{p-1}(1 + \mathcal{O}(z^{\delta_*}\mathrm{M}^D))$ between g and G_H, with a positive power margin that absorbs the logarithms in the error. Multiplying by $G_H - g$ proves the stated little-o term. The normalized residual has the first error block multiplied by $-p$, an identity in the jet algebra. If the inverse is retained to cutoff q in w, its unnormalized forward residual cutoff is $q + (p-1)/|p|$; the normalized cutoff subtracts $p/|p|$ from that value.

9.3 Finite smoothness suffices

The finite germ is analytic away from 0, and the convergence proof used that. For perturbations outside the finite power–log class (a coefficient $\sin(\log x)$, a term $e^{-1/x}$, a function known only through derivative bounds) the marker formula (5.3) can still be applied, and the following finite-smoothness result justifies the resulting expansion whenever the perturbation and finitely many of its derivatives are small in the power–log scale.

Theorem 9.5 (Tame near-identity inversion)**.** *Let $N \geq 0$, $\delta > 0$ and $M \geq 0$. Let φ be real-valued and C^{N+1} on $(0, x_0)$ with*

$$\varphi^{(j)}(x) = \mathcal{O}(x^{1+\delta-j}\mathrm{M}(x)^M) \qquad (x \to 0^+), \quad j = 0, 1, \ldots, N+1. \tag{9.7}$$

Then $f = x + \varphi$ has a unique local inverse g on $(0, y_0)$ with $g(y) \sim y$, and

$$g(y) = y + \sum_{n=1}^{N} \frac{(-1)^n}{n!} \frac{\mathrm{d}^{n-1}}{\mathrm{d}y^{n-1}}[\varphi(y)^n] + \mathcal{O}(y^{1+(N+1)\delta}\mathrm{M}(y)^{(N+1)M}). \tag{9.8}$$

Proof. Step 1: the homotopy. For $\lambda \in [0, 1]$ put $F_\lambda(x) = x + \lambda\varphi(x)$. By (9.7) with $j = 1$, $\varphi'(x) \to 0$, so there is x_1 with $F'_\lambda \geq \frac{1}{2}$ on $(0, x_1)$ for all $\lambda \in [0, 1]$; each F_λ is an increasing bijection of $(0, x_1)$ onto an interval containing $(0, y_1)$ for some y_1 independent of λ. Let $G(\lambda, y)$ be its inverse, so that

$$G + \lambda\varphi(G) = y, \qquad G(0, y) = y. \tag{9.9}$$

Since $|\varphi(x)| \le x/2$ for $x < x_1$ (after shrinking x_1), $F_\lambda(y/2) < y < F_\lambda(2y)$ and therefore $G(\lambda, y) \in [y/2, 2y]$ for all λ and all small y. The map $(\lambda, x) \mapsto x + \lambda\varphi(x) - y$ is C^{N+1} with x-derivative $\ge \frac{1}{2}$, so by the implicit function theorem G is C^{N+1} in (λ, y), with

$$\partial_y G = \frac{1}{1+\lambda\varphi'(G)} \in \left[\tfrac{2}{3}, 2\right], \qquad \partial_\lambda G = -\frac{\varphi(G)}{1+\lambda\varphi'(G)} = \psi(G)\,\partial_y G, \quad \psi := -\varphi. \tag{9.10}$$

Step 2: the transport identity. For every C^n function Ψ and $1 \le n \le N+1$,

$$\partial_\lambda^n[\Psi(G)] = \partial_y^{n-1}[\psi(G)^n\,\partial_y\Psi(G)]. \tag{9.11}$$

For $n = 1$: $\partial_\lambda\Psi(G) = \Psi'(G)\partial_\lambda G = \Psi'(G)\psi(G)\partial_y G = \psi(G)\partial_y[\Psi(G)]$. Assume (9.11) for n and all Ψ. Then, using it once more for the functions ψ^n and Ψ in the second step,

$$\begin{aligned}
\partial_\lambda^{n+1}[\Psi(G)] &= \partial_y^{n-1}\,\partial_\lambda[\psi(G)^n\partial_y\Psi(G)] \\
&= \partial_y^{n-1}\big[\psi(G)\,\partial_y(\psi(G)^n)\,\partial_y\Psi(G) \\
&\qquad\qquad + \psi(G)^n\,\partial_y\big(\psi(G)\partial_y\Psi(G)\big)\big] \\
&= \partial_y^{n-1}\,\partial_y\big[\psi(G)^{n+1}\partial_y\Psi(G)\big],
\end{aligned}$$

where the last equality is the product rule read backwards: $\partial_y[\psi^{n+1}\Psi_y] = (n+1)\psi^n\psi_y\Psi_y + \psi^{n+1}\Psi_{yy}$ and the bracket equals $n\psi^n\psi_y\Psi_y + \psi^n(\psi_y\Psi_y + \psi\Psi_{yy})$, the same thing. (Here ψ_y stands for $\partial_y[\psi(G)]$.)

Step 3: the coefficients. At $\lambda = 0$, $G = y$, so (9.11) with $\Psi = \mathrm{id}$ gives $\partial_\lambda^n G(0, y) = \partial_y^{n-1}[\psi(y)^n] = (-1)^n\partial_y^{n-1}[\varphi(y)^n]$, the Lagrange terms. Taylor's formula in λ with integral remainder gives

$$g(y) = G(1, y) = y + \sum_{n=1}^{N}\frac{(-1)^n}{n!}\partial_y^{n-1}[\varphi^n] + \frac{1}{N!}\int_0^1 (1-\lambda)^N\,\partial_\lambda^{N+1}G(\lambda, y)\,\mathrm{d}\lambda.$$

Step 4: the remainder. Put $\kappa(y) = y^\delta \mathrm{M}(y)^M$. Since $G \in [y/2, 2y]$ and $\mathrm{M}(G) \le 2\mathrm{M}(y)$, hypothesis (9.7) gives $\varphi^{(j)}(G) = \mathcal{O}(\kappa\, y^{1-j})$ uniformly in λ, for $j \le N+1$. We claim that for $1 \le j \le N+1$,

$$\partial_y^j G = \mathcal{O}(y^{1-j}) \quad (j \ge 1), \qquad \partial_y^j[\varphi(G)] = \mathcal{O}(\kappa\, y^{1-j}) \quad (j \ge 0), \tag{9.12}$$

uniformly in $\lambda \in [0, 1]$. Differentiating (9.9) j times in y and solving for $\partial_y^j G$: $(1 + \lambda\varphi'(G))\partial_y^j G$ equals δ_{j1} minus λ times a sum of terms $\varphi^{(m)}(G)\prod\partial_y^{i_s}G$ with $2 \le m$, $\sum i_s = j$, $i_s \ge 1$ (Faà di Bruno). By induction on j each such term is $\mathcal{O}(\kappa y^{1-m})\prod\mathcal{O}(y^{1-i_s}) = \mathcal{O}(\kappa y^{1-j})$, so $\partial_y^j G = \mathcal{O}(\kappa y^{1-j}) \subseteq \mathcal{O}(y^{1-j})$ for $j \ge 2$ (and $\partial_y G = \mathcal{O}(1)$). The same Faà di Bruno expansion with $m \ge 1$ gives the second claim. By Leibniz, $\partial_y^j[\psi(G)^{N+1}] = \mathcal{O}(\kappa^{N+1}y^{N+1-j})$, and $\partial_y^{N-j}[\partial_y G] = \partial_y^{N-j+1}G = \mathcal{O}(y^{-(N-j)})$. Applying (9.11) with $\Psi = \mathrm{id}$ and $n = N+1$,

$$\partial_\lambda^{N+1}G = \partial_y^N\left[\psi(G)^{N+1}\partial_y G\right] = \sum_{j=0}^{N}\binom{N}{j}\partial_y^j[\psi(G)^{N+1}]\,\partial_y^{N-j+1}G = \mathcal{O}(\kappa^{N+1}y)$$

uniformly in λ, and the integral remainder is $\mathcal{O}(y\kappa(y)^{N+1}) = \mathcal{O}(y^{1+(N+1)\delta}\mathrm{M}(y)^{(N+1)M})$. □

Remark 9.6. Applied to $\varphi(x) = x^2(1+\log x)$ ($\delta = 1$, $M = 1$) the theorem re-proves the remainder $\mathcal{O}(y^{N+2}\mathrm{M}^{N+1})$ of (8.5) with a proof that uses no analyticity at all; applied to $\varphi(x) = x^2\sin(\log x)$ ($\delta = 1$, $M = 0$) it justifies $g(y) = y - y^2\sin L + y^3\sin L(2\sin L + \cos L) + \mathcal{O}(y^4)$, $L = \log y$, a perturbation whose coefficients oscillate and which no power–log expansion describes. The marker identity determines the coefficients and the derivative hypotheses justify their complete remainder.

10 Finite endpoints, infinity, and powers of the inverse

The branch theorem, coefficient formula and parametric convergence proof only need $p \neq 0$ and $u \to 0^+$: the substitution $t = u^p$ and the marker theorem are applied at positive points regardless of the endpoint approached by t. Statements that use positivity explicitly, such as the exponent-cutoff corollary to the Rouché majorant, retain their stated restrictions. This section makes the endpoint bookkeeping explicit.

10.1 Local variables

Let f be given near x_0, and let $u \to 0^+$ be the local variable of x:

$$u = \begin{cases} x - x_0 & x \to x_0^+, \\ x_0 - x & x \to x_0^-, \\ 1/x & x \to +\infty, \\ -1/x & x \to -\infty. \end{cases} \tag{10.1}$$

Substituting into f and expanding (Section 3) gives a germ $y_0 + au^p(1 + \sum u^{\delta_i} B_i(\log u))$ in the sense of (4.1), where $y_0 = \lim f$ is finite when $p > 0$. For $p < 0$ the target endpoint is $\pm\infty$ with the sign of a, while the additive offset y_0 in (4.1) is set to 0; any constant term is a correction with gap $-p$. The local variable of y is

$$w = \begin{cases} y - y_0 & a > 0,\ p > 0, \\ -(y - y_0) & a < 0,\ p > 0, \\ 1/|y| & p < 0. \end{cases} \tag{10.2}$$

The uniformizer $z = (v/a)^{1/p}$ tends to 0^+ in all cases. Its relation to w is $w = |a| z^p$ for $p > 0$ and $w = z^{|p|}/|a|$ for $p < 0$.

10.2 The theorem at an arbitrary endpoint

Theorem 10.1 (Signed leading power)**.** *Theorems 4.1, 6.1, 7.1 and 8.2 hold verbatim for the germ (4.1) with $p < 0$: the unique branch with $u \to 0^+$ is $u = g(y)$ with $g \sim z$, g^r has the expansion (6.1) convergent for small z (i.e. for $|y|$ large), and the remainder after the exponent cutoff q in $w = 1/|y|$ is $\mathcal{O}(w^{(r+\sigma_*)/|p|}\mathrm{M}(w)^{k_*})$.*

Proof. The derivative $f'(u) = apu^{p-1}(1 + o(1))$ has a constant sign, so f is monotone on $(0, \varepsilon)$ with $f \to \pm\infty$; the inverse exists on a neighbourhood of $\pm\infty$ and $u \sim z$. The proof of Theorem 6.1 is a computation at a fixed positive $q = v/a$ and never uses the sign of p. The parametric equation (7.2) has $\partial_w \mathcal{E}(1, 0) = p \neq 0$ regardless of sign, and the specialisation $T_{ij} = z^{\delta_i} L^j \to 0$ as $z \to 0^+$ is the same. The remainder theorem is Lemma 3.2. □

In the variable y the blocks of g^r read $(v/a)^{(r+\sigma)/p} C_\sigma^{(r)}(\frac{1}{p}\log(v/a))$, with negative exponents of y when $p < 0$; the exponent of w is $(r + \sigma)/|p|$, which is the quantity the cutoff refers to.

10.3 Powers of the inverse and the original variable

The observable power r in (6.1) gives directly the expansion of u^r. In terms of x:

$$\begin{aligned} (x - x_0)^r &= g(y)^r \ (x \to x_0^+), & (x_0 - x)^r &= g(y)^r \ (x \to x_0^-), \\ x^r &= g(y)^{-r} \ (x \to +\infty), & x^r &= (-1)^r g(y)^{-r} \ (x \to -\infty), \end{aligned} \tag{10.3}$$

The last equality in (10.3), involving $(-1)^r$, is used here only for integer r. Thus x^r at infinity is obtained from (6.1) with local-coordinate power $-r$. Both signed distances $x - x_0$ from above and $x_0 - x$ from below are positive and admit arbitrary real powers. For $r = 1$ and a finite endpoint, the original inverse is $x = x_0 \pm g(y)$. A non-integer power of a negative original variable needs an additional branch convention and is outside the positive-real-power convention of this article.

10.4 Examples

Example 10.2 (The Wright omega function)**.** $f(x) = x + \log x$ as $x \to +\infty$: $u = 1/x$, $f = u^{-1} - \log u = u^{-1}(1 - u \log u)$, so $a = 1$, $p = -1$, one gap $\delta = 1$ with $B = -L$, and target endpoint $+\infty$. The uniformizer is $z = 1/y$, $L = -\log y$, and (6.1) with $r = -1$ gives the original variable $x = 1/g(y)$:

$$x = y - \log y + \frac{\log y}{y} + \frac{\frac{1}{2}\log^2 y - \log y}{y^2} + \frac{\frac{1}{3}\log^3 y - \frac{3}{2}\log^2 y + \log y}{y^3} + \mathcal{O}\Big(\frac{\log^4 y}{y^4}\Big), \tag{10.4}$$

the classical expansion of the Wright omega function $\omega(y) = \mathrm{W}(e^y)$ [10, 11]; the blocks are the Lambert polynomials $P_n(\log y)$ of [11].

Example 10.3 (Reciprocal corrections)**.** $f(x) = x + 1/x$ at $+\infty$: $u = 1/x$, $f = u^{-1}(1 + u^2)$, $p = -1$, gap 2, constant coefficient. The inverse is the larger root of $x^2 - yx + 1 = 0$, $x = \big(y + \sqrt{y^2 - 4}\big)/2 = y - y^{-1} - y^{-3} - 2y^{-5} - \cdots$ (the coefficients are Catalan numbers), which (6.8) reproduces.

Example 10.4 (Half-integer gaps)**.** $f(x) = x + \sqrt{x}$ at $+\infty$: $f = u^{-1}(1 + u^{1/2})$, gap $\frac{1}{2}$. The inverse is $x = \big(\sqrt{4y+1} - 1\big)^2/4 = y - \sqrt{y} + \frac{1}{2} - \frac{1}{8\sqrt{y}} + \frac{1}{128 y^{3/2}} + \mathcal{O}(y^{-5/2})$; the constant term $\frac{1}{2}$ is the block of weight $2 \cdot \frac{1}{2}$, i.e. exponent 0 in $w = 1/y$, despite the infinite target endpoint.

Example 10.5 (A finite endpoint from below)**.** $f(x) = x - x^2$ near $x_0 = 0$ with $x \to 0^-$: $u = -x$, $f = -u - u^2$, $a = -1$, $p = 1$, $y_0 = 0$, and $w = -y$ (the image is $y < 0$). The inverse is $u = -y - y^2 - 2y^3 - \cdots$, i.e. $x = y + y^2 + 2y^3 + \mathcal{O}(y^4)$, the branch of $x = (1 - \sqrt{1 - 4y})/2$ continued to negative y.

Example 10.6 (A pole)**.** $f(x) = 1 + 1/x$ near $x_0 = 0^+$: $f = u^{-1}(1 + u)$, $p = -1$, target endpoint $+\infty$, and $x = 1/(y - 1) = y^{-1} + y^{-2} + \cdots$ with remainder $\mathcal{O}(y^{-3})$ after two terms.

Example 10.7 (The second question, inverted)**.** For $F(x) = (1 + x + x^{\sqrt{2}})^{\sqrt{2}}$ at $+\infty$ the forward germ is $u^{-2}(1 + \sqrt{2}u^{\sqrt{2}-1} + \sqrt{2}u^{\sqrt{2}} + \cdots)$ from (1.5), with gaps generated by $\sqrt{2} - 1$ and $\sqrt{2}$, $p = -2$, target endpoint $+\infty$, and $z = y^{-1/2}$. The inverse is

$$\begin{aligned} x = \sqrt{y} &- \frac{1}{\sqrt{2}}\, y^{(2-\sqrt{2})/2} + \Big(\frac{3}{4} - \frac{1}{2\sqrt{2}}\Big) y^{(3-2\sqrt{2})/2} \\ &+ \Big(1 - \frac{5}{3\sqrt{2}}\Big) y^{(4-3\sqrt{2})/2} - \frac{1}{\sqrt{2}}\, y^{(1-\sqrt{2})/2} + \mathcal{O}\big(y^{(5-4\sqrt{2})/2}\big), \end{aligned} \tag{10.5}$$

the terms being ordered by decreasing exponent $-(r + \sigma)/2$ of y, with local-coordinate power $r = -1$. The forward remainder is transported by (9.3) with that value of r; the available forward order must be sufficient for the desired inverse cutoff. An independent derivation removes the outer power exactly: $1 + x + x^{\sqrt{2}} = y^{1/\sqrt{2}}$, then applies the finite-germ coefficient formula.

11 Worked examples

The following examples apply the coefficient formula and the remainder transport of Section 9. All logarithms have the indicated positive real argument.

11.1 The two examples of the first question, once more

For $f_1 = x + x^2(1+\log x)$, formula (6.6) yields (1.3) with remainder $\mathcal{O}(y^6 \mathrm{M}(y)^5)$. The coefficient of the next block $y^6 P_5(L)$ is

$$\begin{aligned} P_5(L) &= -42L^5 - \tfrac{3727}{12}L^4 - \tfrac{5365}{6}L^3 - 1256L^2 - \tfrac{5177}{6}L - \tfrac{2789}{12} \\ &= -\tfrac{1}{12}(1+L)(504L^4 + 3223L^3 + 7507L^2 + 7565L + 2789), \end{aligned}$$

and the seventh $P_6 = \frac{1}{60}(1+L)(7920L^5 + 63852L^4 + 200833L^3 + 308537L^2 + 231883L + 68307)$. The remainder retains its logarithmic degree even though the power exponents happen to be integers.

For $f_2 = x + x^{\sqrt{2}}$, retaining four blocks in (6.9) gives remainder $\mathcal{O}(y^{4\sqrt{2}-3})$. The corresponding exclusive exponent cutoff is $4\sqrt{2} - 3$.

11.2 Nonunit leading power

$f(x) = 3x^2(1 + x(1 + \log x))$: $a = 3$, $p = 2$, $\delta = 1$, $B = 1 + L$. With $z = \sqrt{y/3}$ and $L = \log z = \frac{1}{2}\log(y/3)$, (6.3) gives

$$g(y) = z - \frac{z^2}{2}(1+L) + \frac{z^3}{8}(5L^2 + 12L + 7) - \frac{z^4}{48}(\mathcal{D}+6)(\mathcal{D}+8)(1+L)^3 + \mathcal{O}(z^5\mathrm{M}^4), \quad (11.1)$$

where $\frac{1}{48}(\mathcal{D}+6)(\mathcal{D}+8)(1+L)^3 = (1+L)^3 + \frac{7}{8}(1+L)^2 + \frac{1}{8}(1+L)$. Two ingredients are easy to forget: the factor $p^{-n} = 2^{-n}$ and the logarithm $L = \frac{1}{2}\log(y/3)$, not $\log y$. In the variable y the expansion reads

$$\begin{aligned} g(y) = \frac{\sqrt{y}}{\sqrt{3}} &- \frac{y}{3}\Big(\frac{1}{2} + \frac{\log(y/3)}{4}\Big) + \frac{y^{3/2}}{3\sqrt{3}}\Big(\frac{7}{8} + \frac{3}{4}\log\frac{y}{3} + \frac{5}{32}\log^2\frac{y}{3}\Big) \\ &- \frac{y^2}{9}\Big(2 + \frac{39}{16}\log\frac{y}{3} + \frac{31}{32}\log^2\frac{y}{3} + \frac{1}{8}\log^3\frac{y}{3}\Big) + \mathcal{O}(y^{5/2}\mathrm{M}^4). \end{aligned}$$

11.3 Several gaps

Irrational and logarithmic gaps together. $f(x) = x + x^{\sqrt{2}} + x^2(1 + \log x)$ has gaps $\delta_1 = \sqrt{2} - 1$ and $\delta_2 = 1$. Below the exponent 3 the support is $\{1, \sqrt{2}, 2\sqrt{2}-1, 2, 3\sqrt{2}-2, 1+\sqrt{2}, 4\sqrt{2}-3, 2\sqrt{2}\}$ and

$$\begin{aligned} g(y) = y - y^{\sqrt{2}} + \sqrt{2}\,y^{2\sqrt{2}-1} - y^2(1+L) - \frac{6-\sqrt{2}}{2}y^{3\sqrt{2}-2} + y^{1+\sqrt{2}}((\sqrt{2}+2)L + \sqrt{2} + 3) \\ + \Big(\frac{17\sqrt{2}}{3} - 4\Big)y^{4\sqrt{2}-3} - y^{2\sqrt{2}}\Big((3\sqrt{2}+5)L + 5\sqrt{2} + \frac{13}{2}\Big) + \mathcal{O}(y^3\mathrm{M}^2), \qquad L = \log y, \end{aligned}$$

the mixed blocks coming from (6.4): the index $(1,1)$ gives $(\mathcal{D}+2+\sqrt{2})(1+L) = (\sqrt{2}+2)(1+L)+1$. The remainder is the block $y^3(2L^2 + 5L + 3)$ of the index $(0,2)$, of degree 2.

Two irrational gaps. For $f(x) = x + x^{\sqrt{2}}(1 + \log x) + 2x^{\sqrt{3}}$ the support weights are $k_1(\sqrt{2}-1) + k_2(\sqrt{3}-1)$. Local finiteness gives finitely many blocks below every cutoff; below exponent 3 there are eleven possible weights. Algebraic identities among coefficient expressions must be respected when forming the complete polynomial of a block. For example, $\sqrt{5 + 2\sqrt{6}} = \sqrt{2} + \sqrt{3}$ because both sides are positive and their squares agree. Coefficient cancellation is a question about exact real numbers, independent of their chosen radical expressions.

11.4 Forward functions given by convergent series

$\sin x + x^2 \log x$ is not a finite power–log sum. Its Taylor expansion gives $x + x^2 \log x - \frac{1}{6}x^3 + \frac{1}{120}x^5 + \mathcal{O}(x^7)$, with $(\rho, k) = (7, 0)$, and the inverse below the exponent 4 is

$$g(y) = y - y^2 \log y + y^3\Big(\frac{1}{6} + \log y + 2\log^2 y\Big) + \mathcal{O}(y^4 \mathrm{M}^3);$$

here the y^3 block combines $(\mathcal{D}+4)L^2/2 = 2L^2 + L$ from the index $(2,0)$ (gap 1, $B_1 = L$) with $+\frac{1}{6}$ from the index $(0,1)$ (gap 2, $B_2 = -\frac{1}{6}$). Similarly the inverse of $\tan x$ is $\arctan y = y - \frac{1}{3}y^3 + \frac{1}{5}y^5 - \frac{1}{7}y^7 + \mathcal{O}(y^9)$. Determining the term of degree seven requires enough terms of the forward Taylor series to transport its remainder beyond that degree.

11.5 Symbolic data

With symbolic coefficients the exponents remain numeric and everything goes through under assumptions that make the leading coefficient's sign and the reality of the coefficients provable: $f = ax + bx^2$ with $a > 0$, $b \in \mathbb{R}$ gives $g = y/a - by^2/a^3 + 2b^2y^3/a^5 + \mathcal{O}(y^4)$. With a symbolic exponent the ordering of the support is undecidable in general and depth truncation (Theorem 8.6) is used: $f = x + x^\alpha$ with $\alpha > 1$ gives, at depth 3,

$$g = y - y^\alpha + \alpha y^{2\alpha-1} - \frac{\alpha(3\alpha-1)}{2} y^{3\alpha-2} + \mathcal{O}(y^{4\alpha-3}),$$

which is (6.9) with a symbolic α.

11.6 Powers and logarithms of the inverse

For $f = x + x^2$, (6.1) with $r = 2$ gives $g^2 = y^2 - 2y^3 + 5y^4 - 14y^5 + \cdots$ (Catalan numbers again), with $r = -1$ it gives $1/g = y^{-1} + 1 - y + 2y^2 - 5y^3 + \cdots$, and (6.2) gives $\log g = \log y - y + \frac{3}{2}y^2 - \frac{10}{3}y^3 + \cdots$. These expansions follow from the same coefficient identity. Equivalently, one can apply the arithmetic of jets while transporting the remainder, including the valuation shifts required by negative powers.

12 Lambert coordinates and boundaries of the theory

The convergent power–log class is closed under the operations of Theorem 3.3 and inversion of germs with a monomial leading block. A leading logarithm or exponential changes the natural coordinate. For exact Lambert cores, that coordinate change reduces the problem to the same polynomial–logarithmic arithmetic. More general perturbations require additional bookkeeping, described below.

12.1 A leading logarithm: Lambert W and the $\log\log$ scale

Let $a \neq 0$, $p \neq 0$, $b \neq 0$, and $x \to 0^+$ with $x < 1$. The exact core $ax^p(-\log x)^b = y$ has $Y = y/a > 0$. With $\tau = -\log x \to +\infty$, it reads $\tau^b e^{-p\tau} = Y$, hence

$$x = \exp\Big[\frac{b}{p}\,\mathrm{W}_k\Big(-\frac{p}{b}\,Y^{1/b}\Big)\Big], \qquad k = \begin{cases} -1, & p/b > 0, \\ 0, & p/b < 0. \end{cases} \tag{12.1}$$

The branch is forced by $\tau \to \infty$: when $p/b > 0$ the argument approaches 0 from below and $\mathrm{W}_{-1} \to -\infty$; when $p/b < 0$ it tends to $+\infty$ and $\mathrm{W}_0 \to +\infty$. For $p, b > 0$, the other real root on W_0 approaches $x = 1$ as $Y \to 0$, while the two branches meet at the maximum $x = e^{-b/p}$. For instance the inverse of $x \log(1/x)$ near 0^+ is $-y/\mathrm{W}_{-1}(-y)$, and that of $x \log x$ on the same source branch is $y/\mathrm{W}_{-1}(y)$ for $y \to 0^-$. For $p > 0$, with $\Lambda = \log(a/y) \to \infty$ and $\Lambda_2 = \log \Lambda$, the expansion begins

$$x = \Big(\frac{y}{a}\Big)^{1/p} \Big(\frac{\Lambda}{p}\Big)^{-b/p} \Big[1 - \frac{b^2}{p\,\Lambda} \log \frac{\Lambda}{p} + \mathcal{O}\Big(\frac{\Lambda_2^2}{\Lambda^2}\Big)\Big], \tag{12.2}$$

a series in $1/\Lambda$ with polynomial coefficients in Λ_2, multiplying the leading factor. The following construction computes all its logarithmic orders, with convergence and a remainder bound.

12.2 A convergent universal Lambert expansion

Let Z approach $+\infty$ on the principal real branch or 0^- on the lower real branch. Set

$$s = \begin{cases} 1, & \mathrm{W}_0(Z),\ Z \to +\infty, \\ -1, & \mathrm{W}_{-1}(Z),\ Z \to 0^-, \end{cases} \qquad A = |\log |Z||, \quad t = A^{-1}, \quad \ell = \log t.$$

Then $\log |\mathrm{W}| + \mathrm{W} = \log |Z| = sA$. Writing $\mathrm{W} = s(1 + V)/t$ gives the normalized equation

$$V + st \log(1 + V) - st\ell = 0. \tag{12.3}$$

Theorem 12.1 (Lambert expansion in a logarithmic coordinate)**.** *Equation* (12.3) *has a unique solution $V \to 0$ of the form $\sum_{n\ge1} t^n P_n(\ell)$ with $P_1 = s\ell$ and $\deg P_n \le n-1$ for $n \ge 2$. This series converges for sufficiently small $t > 0$. For $N \ge 1$, truncating V after t^N leaves $\mathcal{O}(t^{N+1}\mathrm{M}(t)^N)$, and multiplying by s/t gives the corresponding remainder for* W*. In particular,*

$$\mathrm{W} = \frac{s}{t} + \ell - s\ell t + \Big(\ell + \frac{\ell^2}{2}\Big)t^2 - s\Big(\ell + \frac{3\ell^2}{2} + \frac{\ell^3}{3}\Big)t^3 + \mathcal{O}(t^4\mathrm{M}(t)^4). \tag{12.4}$$

Proof. Replace $t\ell$ by an independent parameter T. The function $V + st\log(1+V) - sT$ is analytic near $(V, t, T) = (0, 0, 0)$ and its V-derivative there is 1. The analytic implicit function theorem gives a convergent Taylor series $V(t, T)$. Moreover $V = sT + tH(t, T)$ for an analytic H with $H(0, 0) = 0$. After $T = t\log t$, every term of total degree $n \ge 2$ therefore has logarithmic degree at most $n - 1$, and $P_1 = s\ell$. Cauchy estimates bound the coefficient norms exponentially. The tail starting at degree $N + 1$ is bounded by a geometric series with first term $t^{N+1}\mathrm{M}(t)^N$, proving the remainder. Coefficients can be computed by the iteration $V \leftarrow st\ell - st\log(1 + V)$, which gains one power of t each time, or by Newton iteration. The real solution has $\mathrm{W} \sim sA$ and hence belongs to the specified real Lambert branch. Expanding the equation gives (12.4). $\square$

An explicit sufficient convergence condition is

$$q_L = 4t(1 + |\log t|) < 1. \tag{12.5}$$

Indeed, for complex $|t|, |T| \le 1/4$, the map $V \mapsto sT - st\log(1 + V)$ sends $|V| \le 1/2$ into itself, since $(1 + \log 2)/4 < 1/2$, and has derivative bounded by $1/2$. Its fixed point is jointly analytic and bounded by $1/2$, so its Taylor coefficients are bounded by $4^{a+b}/2$. Beyond the leading term sT, every coefficient has $a \ge 1$ because $V(0, T) = sT$. If V_N retains total degrees through $N \ge 1$, the geometric majorant therefore gives

$$|V - V_N| \le \frac{2t\,q_L^N}{1 - q_L}, \qquad \Big|\mathrm{W} - \frac{s}{t}(1 + V_N)\Big| \le \frac{2q_L^N}{1 - q_L}. \tag{12.6}$$

The construction follows the two-small-parameter method in the transseries notes [11]; it establishes a sufficient domain and does not claim convergence from a fixed-logarithm implicit-function argument alone.

For the leading-log core, put $Z = -(p/b)Y^{1/b}$ and choose s from its branch. Since $-bs/p > 0$, an arbitrary real observable power is best assembled as

$$x^r = Y^{r/p}\Big(-\frac{bs}{p}\Big)^{-br/p} t^{br/p}(1+V)^{-br/p}. \tag{12.7}$$

This uses a positive real power of the unit $1+V$. Directly exponentiating a truncated W would shift the required precision by one order; (12.7) makes that precision shift explicit.

Example 12.2 (The inverse of $x \log x$)**.** For $y \to 0^-$ let $A = \log(1/(-y))$ and $B = \log A$. The branch with $x \to 0^+$ is

$$x = \frac{-y}{A}\left[1 - \frac{B}{A} + \frac{B^2 - B}{A^2} + \frac{-B^3 + \frac{5}{2}B^2 - B}{A^3} + \mathcal{O}\left(\frac{(1+B)^4}{A^4}\right)\right].$$

The endpoint $x \to 1$ selects a different inverse germ and is not represented by this expansion.

12.3 Exponential cores at infinity

For a positive source coordinate x with $X = x^p \to +\infty$, consider $y = ax^b e^{cx^p}$, where $a \neq 0$, $p \neq 0$, $c \neq 0$, and $Y = y/a > 0$. If $b \neq 0$, taking the power p/b gives

$$X = \frac{b}{cp}\,\mathrm{W}_k\left(\frac{cp}{b}Y^{p/b}\right), \qquad k = \begin{cases} 0, & cp/b > 0, \\ -1, & cp/b < 0. \end{cases} \tag{12.8}$$

These branch choices ensure $X > 0$ and $X \to \infty$: the argument tends to $+\infty$ in the first case and 0^- in the second. If $b = 0$, inversion is elementary, $x = (\log Y/c)^{1/p}$. In the Lambert cases, with s, t, V as above,

$$x^r = \left(\frac{bs}{cp}\right)^{r/p} t^{-r/p}(1+V)^{r/p},$$

whose positive prefactor and unit series again have explicit precision tracking.

Example 12.3 (The inverse of xe^x at infinity)**.** Here $x = \mathrm{W}_0(y)$. With $A = \log y$ and $B = \log A$,

$$x = A - B + \frac{B}{A} + \frac{B(B-2)}{2A^2} + \frac{B(2B^2 - 9B + 6)}{6A^3} + \mathcal{O}\left(\frac{(1+B)^4}{A^4}\right).$$

Both this expansion and the $x \log x$ expansion use $t = 1/A$ as the local variable. An exponent cutoff in t specifies logarithmic accuracy; it is not a cutoff in y or $1/y$. A nonconstant external factor such as $-y$ must also multiply the remainder bound.

12.4 Perturbations of a Lambert core

Relative perturbations $F(x) = ax^p(-\log x)^b(1 + \sum_i x^{\delta_i} B_i(\log x))$ can be approached with the general-core formula (5.5). If ϕ is the exact core inverse and $\tau = -\log \phi(y)$, then

$$\phi'(y) = \Big[ax^{p-1}\tau^{b-1}(p\tau - b)\Big]^{-1}_{x=\phi(y)}.$$

The marker coefficients have powers of $\phi(y)$ and rational functions of τ, with denominators arising from $p\tau - b$. A convergent marker expansion requires analytic control on the chosen branch, and re-expanding its coefficients introduces a second order of truncation: perturbation power and logarithmic depth. The exact-core Lambert construction proves the logarithmic expansion. Sections 17 and 25 retain the exact core and compute separate perturbation corrections for the admitted finite families.

The leading logarithmic polynomial itself can also be generalized before introducing a second perturbation scale.

Corollary 12.4 (An arbitrary polynomial at the leading exponent)**.** *Let $F_0(u) = au^pQ(\log u)$, where $p \neq 0$ and Q is a nonconstant real polynomial of degree $q \geq 1$, on the real branch $u \to 0^+$. Its inverse has a convergent relative power–log expansion in a reciprocal-logarithm coordinate, even when Q is not a power of an affine polynomial.*

Proof. Absorb the leading coefficient of Q into a, so Q is monic, and set $b = [L^{q-1}]Q/q$ and $\tau = -\log u - b$. Then

$$Q(-\tau - b) = (-\tau)^q R(1/\tau), \qquad R(v) = 1 + \sum_{j=2}^{q} r_j v^j.$$

With $Y = y/(a(-1)^q) > 0$ (after subtracting any constant target offset), $k = -p/q$, $s = \operatorname{sign} k$ and $Z = kY^{1/q}e^{pb/q}$, take $A = s\log|Z|$, $t = 1/A$, and $\tau = (1+V)/(|k|t)$. The inverse equation becomes

$$V + st\left[-\ell + \log(1+V) + \frac{1}{q}\log R\left(\frac{|k|t}{1+V}\right)\right] = 0. \tag{12.9}$$

Replacing $t\ell$ by an independent parameter again gives an analytic equation with V-derivative 1 at the origin. Its solution is jointly analytic, and $R(0) = 1$ makes its real powers analytic too. For any real local-coordinate observable power r,

$$u^r = Y^{r/p}\left(\frac{A}{|k|}\right)^{-qr/p}(1+V)^{-qr/p}R\left(\frac{|k|t}{1+V}\right)^{-r/p}.$$

This yields convergence and the same type of finite-order logarithmic remainder as before. The sufficient numerical bound (12.6) was proved for $R = 1$ and is not asserted unchanged for a general R. □

The polynomial correction modifies the logarithmic coefficients and must be included in the equation; it is not beyond all logarithmic orders. In general this inverse need not be expressible using a single Lambert W function. The construction also applies to logarithmic cores at either infinite source endpoint after $u = 1/x$ or $u = -1/x$, using the observable and sign rules of Section 10.

Proposition 12.5 (Perturbations beyond every logarithmic order)**.** *Suppose a finite leading-log core $F_0(u) = au^pQ(\log u)$ from Corollary 12.4, or an affine-log power core, is supplemented by finitely many polynomial–logarithmic terms of strictly larger u-exponent. Alternatively, suppose a growing exponential core $F_0(x) = ax^be^{cx^p}$ with $c, p > 0$ at $x \to +\infty$ is supplemented by a finite power–log sum. Let g_0 and g be the corresponding inverse branches, with any common target offset subtracted, and let t be the Lambert coordinate of g_0. For every fixed real r and every $N > 0$,*

$$g(y)^r - g_0(y)^r = o(g_0(y)^r t^N).$$

Consequently every finite relative logarithmic expansion of the core inverse is also an expansion of the full inverse, with the same finite-order remainder class.

Proof. In the leading-log case $F_0'(u) \sim pF_0(u)/u$, and the perturbation and its derivative are smaller by a factor $\mathcal{O}(u^\delta \mathrm{M}(u)^D)$ for some $\delta > 0$ and finite D. A mean-value bracket near g_0 yields $(g - g_0)/g_0 = \mathcal{O}(g_0^\delta \mathrm{M}(g_0)^D)$. Here $1/t$ is a positive constant times $|\log g_0|$ plus lower logarithmic terms, so this bound is $o(t^N)$ for every N.

In the exponential case $F_0'/F_0 \sim cpx^{p-1}$, whereas the additive perturbation R and its derivative have only power–log growth. A bracket of width $\mathcal{O}(|R(g_0)|/|F_0'(g_0)|)$ has slope comparable to $F_0'(g_0)$: the logarithm of that derivative changes by $\mathcal{O}(R/F_0) = o(1)$ across the bracket. Thus $(g - g_0)/g_0$ is exponentially small in g_0^p, while t is comparable to g_0^{-p}. Finally apply the mean value theorem to the positive real power u^r. Continuous local inverse branches exist because the perturbation derivative is smaller than the nonzero core derivative. $\square$

These perturbations affect an exponentially smaller scale while leaving every finite coefficient in the logarithmic expansion unchanged. The exact core inverse therefore determines all logarithmic orders, but does not equal the full inverse in general.

12.5 Zero gaps: corrections of relative size $1/\log$

For $f(x) = x + x/\log x$ the correction has power gap 0 and relative size $1/\log x$. All corrections of the inverse then sit at the same power y with increasing negative powers of $\log y$, the index set I_H is infinite, and the exponent cutoff has no meaning. The right coordinate is $\tau = 1/\log y$: with $g = yW$ the equation becomes $W(1 + \tau/(1 + \tau \log W)) = 1$, analytic near $(W, \tau) = (1, 0)$, so W is an ordinary convergent power series in τ, $W = 1 - \tau + \tau^2 - 2\tau^3 + \frac{7}{2}\tau^4 - \cdots$, i.e. $g(y) = y(1 - 1/\log y + 1/\log^2 y - 2/\log^3 y + \cdots)$. The general-core formula produces these terms, but the scale is the reciprocal-logarithm scale, not the power–log scale, and Section 20 develops the required reciprocal-logarithm valuation.

12.6 Flat terms

$e^{-1/x}$ is smaller than every power of x; a germ $x + e^{-1/x}$ has the same asymptotic power–log expansion as x. Its inverse obeys $g = y - e^{-1/g}$, so $y - e^{-1/y} \le g < y$ and $(y - g)/y^2 \to 0$. Consequently $e^{-1/g}/e^{-1/y} \to 1$, proving $g - y = -e^{-1/y}(1 + o(1))$. The marker formula begins $y - e^{-1/y} + y^{-2}e^{-2/y} + (y^{-3} - \frac{3}{2}y^{-4})e^{-3/y} + \cdots$. These exponential sectors are invisible to a pure power–log expansion. Section 21 computes them with a separate sector cutoff and a complete-tail bound; this differs from the exact exponential cores above, whose inverses reduce to Lambert coordinates.

12.7 Oscillatory coefficients

$x^2 \sin(\log x)$ is $\mathcal{O}(x^2)$ with all derivatives of the right size, so Theorem 9.5 applies and the marker formula gives a valid expansion $y - y^2 \sin L + y^3 \sin L(2 \sin L + \cos L) + \mathcal{O}(y^4)$ whose "coefficients" are bounded oscillating functions of $L = \log y$. They are not polynomials in L, the coefficient algebra is different, and the polynomial-coefficient remainder theory of Section 8 requires an extension. The tame theorem already provides a valid remainder estimate. Section 22 supplies a finite Fourier coefficient algebra and a nonoscillatory remainder bound.

12.8 Symbolic exponents and nonuniformity

When a gap δ is a symbol, the order of the weights $k \cdot \boldsymbol{\delta}$ depends on the parameter: the order of $2\delta_1$ and δ_2 changes at $\delta_2 = 2\delta_1$. Exponent truncation needs provable comparisons; depth truncation (Theorem 8.6) only needs positive gaps after a valid leading term and real branch have been selected. Nothing is uniform as a gap tends to 0: the y-radius $R_\alpha^{1/(\alpha-1)}$ of the $x + x^\alpha$ inverse tends to 0 as $\alpha \downarrow 1$, although $R_\alpha \to 1$, and at $\alpha = 1$ the inverse is $y/2$.

12.9 Nested powers of the logarithm

Non-integer powers $(-\log x)^s$, $\log(-\log x)$, and negative powers of $\log x$ in a correction leave the polynomial coefficient algebra. For example $\varphi(x) = x^{1+\delta}(-\log x)^{1/2}$ is real for small positive x and satisfies Theorem 9.5 with $M = 1$, but repeated differentiation produces negative half-integer as well as positive half-integer powers of $-L$. A suitable coefficient algebra therefore contains $(-L)^{1/2}$ and its inverse. The Euler identity and composition recurrence still apply when the coefficient functions are differentiable or analytic as needed; closure and remainder bounds in the enlarged scale must also be established.

The finite hierarchy of Section 20 establishes these closure and remainder estimates for generalized Laurent monomials in finitely many positive logarithmic levels. More general coefficient functions still need an independently justified algebra and ordering.

13 Inversion through explicit coordinate transformations

The leading expression in the original variable need not belong to the power–log scale. A change of source or target coordinate can put the equation into a suitable scale while preserving the selected real branch. The transformation is part of the mathematical result: it determines the expansion variable, how the cutoff is interpreted, and how an inner remainder becomes an error in the original variable.

13.1 Target transformations

Theorem 13.1 (An exact change of target coordinate)**.** *Let F be continuous and strictly monotone on a source interval I, and let J be continuous and one-to-one on $F(I)$. Put $\Phi = J \circ F$. If h is the inverse of Φ on $J(F(I))$, then the inverse of F on $F(I)$ is*

$$g(y) = h(J(y)).$$

If an observable of that inverse has an expansion $h(v)^r = A(v) + \mathcal{O}(R(v))$ as v approaches the transformed target endpoint, then

$$g(y)^r = A(J(y)) + \mathcal{O}(R(J(y)))$$

on the corresponding source branch, provided the same real power is defined there.

Proof. $\Phi(h(J(y))) = J(y)$ gives $J(F(h(J(y)))) = J(y)$. Injectivity of J gives $F(h(J(y))) = y$, and the source interval fixes the inverse branch. The remainder statement is direct substitution along the transformed endpoint; no output reconstruction is involved. $\square$

The theorem distinguishes two operations that are sometimes conflated. Changing the target of the equation does not amplify the error of its inverse observable. Changing the source variable and then reconstructing it generally does, as discussed below.

13.2 Exponential phases

Suppose an expression can be written on the selected real source branch as

$$F(x) = y_0 + \varepsilon e^{\Phi(x)}, \qquad \varepsilon \in \{1, -1\}. \tag{13.1}$$

An eventually signed amplitude is included in the phase: if the original expression is $y_0 + A(x)e^{P(x)}$, prove the eventual sign ε of A, then set $\Phi = P + \log(\varepsilon A)$. The inverse equation becomes

$$\Phi(x) = v, \qquad v = \log(\varepsilon(y - y_0)), \qquad \varepsilon(y - y_0) > 0. \tag{13.2}$$

The phase may contain several powers and polynomial–logarithmic terms even when the original function has exponential growth or decay. It is inverted on the same source interval. If $\Phi \to +\infty$, then $F \to \varepsilon\infty$; if $\Phi \to -\infty$, the target approaches y_0 from the side with sign ε.

Proposition 13.2 (The original residual and the phase residual)**.** *For a real approximation X on that source interval, put $D = \Phi(X) - v$. Then*

$$\frac{F(X) - y}{y - y_0} = e^D - 1. \tag{13.3}$$

In particular, if $D \to 0$, the relative original residual is $D + \mathcal{O}(D^2)$. If $|\Phi'| \geq \mu > 0$ between X and the true inverse $g(y)$, then $|X - g(y)| \leq |D|/\mu$.

Proof. Since $y - y_0 = \varepsilon e^v$, divide $\varepsilon(e^{\Phi(X)} - e^v)$ by εe^v. The remaining statements follow from the Taylor expansion of e^D at 0 and the mean value theorem for Φ. $\square$

Thus a phase residual can be numerically useful without forming enormous exponentials. A numerical solve in the phase remains numerical evidence unless the derivative and residual have rigorous bounds; it nevertheless refers to the original equation through the exact identity (13.3).

Example 13.3 (Several powers in an exponential)**.** For $F(x) = e^{x^2+x}$ at $+\infty$, put $v = \log y$. The inverse is

$$x = \frac{-1 + \sqrt{1 + 4v}}{2} = \sqrt{v} - \frac{1}{2} + \frac{1}{8\sqrt{v}} - \frac{1}{128v^{3/2}} + \mathcal{O}(v^{-5/2}).$$

This is an ordinary inverse expansion in the phase variable v. Its local target variable is $1/v = 1/\log y$, and its remainder is measured in that coordinate. For e^{-x^2-x} at the same source endpoint, replace v by $-\log y$ and take $y \to 0^+$.

Example 13.4 (A nonconstant amplitude)**.** For $F(x) = xe^{x^2+x}$ at $+\infty$, the phase is $\Phi = x^2 + x + \log x$. In terms of $v = \log y$, inversion begins

$$x = \sqrt{v} - \frac{1}{2} + \frac{\frac{1}{8} - \frac{1}{4}\log v}{\sqrt{v}} + \mathcal{O}(v^{-1}(1 + |\log v|)^2).$$

The logarithm of the amplitude contributes at the displayed order. Omitting it would preserve the leading term but give the wrong next coefficient. Translating x to $x - c$ translates the reconstructed inverse by c and leaves the phase calculation in that shifted source coordinate.

Example 13.5 (A flat forward function with an elementary inverse)**.** $F(x) = e^{-1/x}$ on $x \to 0^+$ has no nonzero power–log forward expansion, but its inverse is exactly

$$x = -\frac{1}{\log y}, \qquad 0 < y < 1.$$

The logarithmic target transformation turns the equation into $-1/x = \log y$. This example does not require an expansion into flat perturbation sectors: the exponential is part of the exact coordinate transformation.

Variable powers can enter the same route when their real domain is established. For x^x on $x > 0$, the phase is $x \log x$, so its inverse at $+\infty$ reduces to the Lambert-coordinate inversion of that phase. The identity $A(x)^{B(x)} = \exp(B(x) \log A(x))$ is used only where $A(x) > 0$.

13.3 Source reconstruction and its error

Theorem 13.6 (A change of source coordinate)**.** *Let $x = H(t)$ be a continuously differentiable one-to-one parametrization of the selected source branch, and write $K(t) = F(H(t))$. Let h be its inverse and suppose*

$$h(y) = A(y) + E(y), \qquad E(y) = \mathcal{O}(R(y)).$$

If H is defined on the interval between A and $A + E$, and $|H'| \leq B(y)$ there, then

$$F^{-1}(y) = H(A(y)) + \mathcal{O}(B(y)R(y)).$$

The same conclusion applies to an observable after replacing H by that observable composed with H.

Proof. The branch identity gives $F^{-1} = H \circ h$. Apply the mean value theorem between A and $h = A + E$. □

For the coordinate transformations used here, more informative exact identities specify when a small error survives reconstruction.

Corollary 13.7 (Exponential reconstruction)**.** *For a positive source coordinate $u = e^{st}$, $s \neq 0$, if $h = A + E$ and $E \to 0$, then*

$$u = e^{sA}(1 + \mathcal{O}(E)), \qquad u^r = e^{rsA}(1 + \mathcal{O}(E)) \tag{13.4}$$

for every fixed real r. If E does not tend to zero, an additive expansion for h alone does not justify these relative error assertions.

Proof. $u = e^{sA}e^{sE}$, and $e^{sE} = 1 + \mathcal{O}(E)$ when $E \to 0$. Replace s by rs for the observable. □

Corollary 13.8 (Logarithmic reconstruction)**.** *For $x = -\log t/\lambda$, $\lambda \neq 0$, suppose the positive inner inverse satisfies $h = A + E$ with $A > 0$ and $E/A \to 0$. Then*

$$x = -\frac{\log A}{\lambda} - \frac{1}{\lambda}\log\left(1 + \frac{E}{A}\right) = -\frac{\log A}{\lambda} + \mathcal{O}(E/A). \tag{13.5}$$

Thus the inner remainder must be divided by the inner leading scale before it becomes an error in x.

Proof. Factor $h = A(1 + E/A)$ and expand the logarithm of the positive unit factor. The hypothesis $E/A \to 0$ also guarantees $h > 0$ eventually. □

13.4 Pure logarithmic functions

For $x \to +\infty$, set $t = \log x \to +\infty$. For a positive source distance $u \to 0^+$, set $t = -\log u \to +\infty$, then reconstruct $u = e^{-t}$ and the original point $x = x_0 \pm u$. A pure logarithmic forward expression becomes a power–log expression in t. Its inverse is computed there before exponential reconstruction.

Example 13.9 (The constant term in an exponent matters)**.** For $F(x) = (\log x)^2 + \log x$ at $+\infty$, the transformed equation is $t^2 + t = y$. Hence

$$t = \sqrt{y} - \frac{1}{2} + \frac{1}{8\sqrt{y}} + \mathcal{O}(y^{-3/2}),$$

and one valid reconstruction is

$$x = \exp\left(\sqrt{y} - \frac{1}{2} + \frac{1}{8\sqrt{y}}\right)(1 + \mathcal{O}(y^{-3/2})).$$

In contrast, $t = \sqrt{y} + \mathcal{O}(1)$ does not justify $x = e^{\sqrt{y}}(1 + o(1))$: the omitted constant would change the leading multiplicative factor by $e^{-1/2}$. The inner expansion must retain enough terms to make its omitted additive error tend to zero before it yields a vanishing relative error after reconstruction.

This precision requirement belongs to reconstruction, not to the mere existence of the inner asymptotic expansion. A specified relative order in the reconstructed expression therefore determines the required inner additive precision; these two orders refer to different representations of the same approximation.

13.5 Finite sums of exponentials

Let

$$F(x) = \sum_{j=1}^{m} a_j e^{\mu_j x}, \qquad a_j, \mu_j \in \mathbb{R},$$

with equal rates merged. At $x \to +\infty$ choose $\lambda > 0$, and at $x \to -\infty$ choose $\lambda < 0$, so $t = e^{-\lambda x} \to 0^+$ in either case. Then

$$F\left(-\frac{\log t}{\lambda}\right) = \sum_j a_j t^{-\mu_j/\lambda}. \tag{13.6}$$

This is a finite real-power germ. Rational relations among the rates are unnecessary: local finiteness of the positive correction gaps supplies finite exponent truncations. Its leading coefficient and exponent determine the target side, while the chosen sign of λ preserves the source endpoint. After inversion, Corollary 13.8 transports the error back to x.

Example 13.10 (An elementary exact inverse)**.** For $F(x) = e^x + e^{2x}$ at $+\infty$, take $t = e^{-x}$. Then $F = t^{-1} + t^{-2}$, and an exact inverse is

$$x = \log\left(\frac{\sqrt{1+4y} - 1}{2}\right) = \frac{1}{2}\log y - \frac{1}{2\sqrt{y}} + \frac{1}{48y^{3/2}} + \mathcal{O}(y^{-5/2}).$$

The inverse of $e^x + e^{\sqrt{2}x}$ uses the same coordinate with irrational exponents. Constant terms, duplicate rates and cancellations are handled before extracting the leading transformed monomial.

13.6 Transport of an additional forward error

Suppose the equation is $F(x)+R(x) = y$ and J is an exact target coordinate. If J is continuously differentiable between $F(x)$ and $F(x) + R(x)$, with $|J'| \le B(x)$ there, then the transformed forward error is

$$J(F(x) + R(x)) - J(F(x)) = \mathcal{O}(B(x)|R(x)|).$$

For the logarithmic coordinate on a positive target branch, a relatively small error $R/F \to 0$ gives $\log(F + R) - \log F = \mathcal{O}(R/F)$. Thus an additive error in the original function and an additive error in its phase have different scales. The derivative estimates needed for inverse remainder transport must be transformed as well. Exact coordinate identities identify the equation being inverted, while these estimates supply the additional analytic control for an approximate equation.

14 Weighted enumeration and grouped coefficient formulas

The coefficient theorem admits several finite constructions at a prescribed cutoff. This section organizes them by exact weight, by increasing Newton precision, and by marker degree. All constructions use finitely many positive gaps δ_i and preserve the meaning of a complete power–log block.

14.1 Enumeration by complete weight layers

Let I be the set of retained multi-indices, initially empty, and let its candidate frontier initially consist of the zero multi-index. At each step, select the smallest candidate weight σ, promote *every* candidate of that weight into I, and add all their one-step successors $\boldsymbol{k} + e_i$ to the candidate set. The frontier is a set of integer vectors, so a candidate reached through several predecessors occurs only once. Distinct candidates of equal weight remain distinct until their coefficients are summed.

Proposition 14.1 (Complete layers and coefficient reuse)**.** *After the layer at weight σ is processed, I is exactly $\{\boldsymbol{k} \in \mathbb{N}^m : \boldsymbol{k} \cdot \boldsymbol{\delta} \leq \sigma\}$, and the candidate set is its one-step outer boundary. This construction determines every retained multi-index coefficient once. Summing the coefficients of an entire layer before counting a block preserves all resonances and exact cancellations.*

Proof. Every nonzero multi-index has a predecessor of strictly smaller weight because all gaps are positive. Inductively, every index of the next candidate weight has already been generated by a processed predecessor. A successor of an index in the current layer has strictly larger weight, so processing one member of the layer cannot expose a previously unseen member of the same layer. The set union identifies repeated occurrences of a candidate through different predecessors. The coefficient of each weight is the complete finite sum in Theorem 6.1, whether or not that sum is zero. □

Extending a cutoff requires only subsequent layers: coefficients below the previous cutoff do not depend on the new cutoff. A prescribed exponent cutoff is reached in finitely many layers by local finiteness. A prescribed number of nonzero blocks is a different requirement, because complete layers can cancel. Without a separate nontermination or exact-termination argument, enumeration alone does not prove that another nonzero block exists.

14.2 Newton iteration with growing precision

Write $E(U) = 0$ for (6.11), and let $\delta_* = \min_i \delta_i$. The zero jet is correct below δ_*. If U_s is correct strictly below weight s, one Newton step performed strictly below

$$h = \min(2s, H), \qquad U_h = \left[U_s - \frac{E(U_s)}{E'(U_s)}\right]_{<h},$$

is correct strictly below h. Indeed, $E'(U_s)$ has nonzero constant coefficient p, and the Newton error is quadratic in the old error. Products, unit powers, and logarithmic composition do not decrease the weight of the error. The argument uses exponent valuation, with polynomial logarithms as coefficients; logarithmic degree need not stay bounded as h increases.

The working cutoffs are $\min(2\delta_*, H), \min(4\delta_*, H), \ldots, H$, rather than H at every step. Their construction uses exact comparisons, with no floating-point estimate of the required iteration count. The identity $[E(U_h)]_{<h} = 0$ characterizes the formal accuracy of each step. Together with

triangular uniqueness it proves that the retained coefficients agree with those of the inverse. A numerical bound at a particular nonzero z additionally requires an analytic remainder estimate.

If a unit jet is already correct below s, refinement to $H > s$ starts with that same jet and the same doubling argument. Truncation to a smaller cutoff preserves correctness. For a pure monomial forward function the unit correction is exactly zero, so no refinement is needed.

14.3 Combine resonances before applying Euler operators

Put

$$H_1(z, L) = \sum_i z^{\delta_i} B_i(L), \qquad H_1(z, L)^n = \sum_\sigma z^\sigma A_{n,\sigma}(L).$$

The multinomial theorem gives

$$A_{n,\sigma} = \sum_{\substack{|\boldsymbol{k}|=n \\ \boldsymbol{k}\cdot\boldsymbol{\delta}=\sigma}} \frac{n!}{\boldsymbol{k}!} B^{\boldsymbol{k}}.$$

At fixed depth n and weight σ, all summands of (6.1) have the same differential operator. By linearity, the unit bracket of g^r can therefore be computed as

$$\frac{g(y)^r}{z^r} = 1 + \sum_{n\geq 1} \sum_\sigma z^\sigma \frac{(-1)^n r}{p^n n!} \prod_{j=1}^{n-1} (\mathcal{D} + r + \sigma + pj) A_{n,\sigma}(L). \tag{14.1}$$

For cutoff H, successive sparse products compute $[H_1^n]_{<H}$. Only n with $n\delta_* < H$ can contribute, so the computation is finite. Coefficients of equal weight are combined during each sparse product, before differentiation. Contributions at the same weight but different depths are then combined in the result; they too can cancel.

Equation (14.1) is especially useful when the generators have many additive relations. It does not require commensurable gaps, and it does not replace the individual-coefficient formula: that formula retains the distinct perturbation markers and their provenance. Grouping replaces several applications of the same linear differential operator by one application to the sum. The boundary argument of Section 8 is still needed to identify a complete omitted weight. Neither grouping nor marker-depth truncation changes the ordering of the final nonzero blocks. The equivalence of the grouped formula, the individual multi-index formula and Newton iteration follows from linearity and formal uniqueness.

15 Refinement by complete weight layers

A finite generalized-power expansion is a projection of a locally finite formal series. Refinement means determining additional complete coefficients of the same inverse germ while respecting the precision of the forward data. The positive source gaps provide a natural filtration for this process.

15.1 The parameter domain belongs to the germ

Let θ denote the fixed parameters and let A be the domain on which the chosen real branch and an estimate $F_\theta = T_\theta + \mathcal{O}(w^P \mathrm{M}^D)$ have been established. This means that for each $\theta \in A$ there are constants $C_\theta, w_\theta > 0$ such that

$$|F_\theta(w) - T_\theta(w)| \leq C_\theta w^P \mathrm{M}(w)^D, \qquad 0 < w < w_\theta.$$

The domain A remains a hypothesis when coefficients are reused, recombined, or refined. Combining conclusions proved on A and B is justified on $A \cap B$, further restricted by the domain conditions of the operation. A simplification proved only on a subset $A' \subset A$ likewise gives a conclusion on A'. Neither an algebraic cancellation nor the determination of additional coefficients extends a theorem to parameters outside its original hypotheses; such an extension needs a separate proof. These assertions follow by applying the corresponding identities and error estimates at each fixed admissible θ.

For example, the positive inverse of $ax + x^2$ for $a > 0$ starts with $y/a - y^2/a^3$. Those coefficients cannot be reused at $a = 0$, where the positive inverse is $\sqrt{y}$. Nor does validity for every fixed $a > 0$ give a bound uniform as $a \downarrow 0$. Uniformity on a parameter set requires common constants and a common deleted neighborhood, as well as the branch and nondegeneracy hypotheses used by the relevant theorem. Source and target restrictions retain their separate roles described in Section 28.

15.2 Complete Lagrange prefixes

Write the normalized inverse observable as

$$z^r \sum_{\omega<H} z^\omega Q_\omega(\log z), \qquad \omega = \boldsymbol{k}\cdot\boldsymbol{d}, \quad \boldsymbol{k} \in \mathbb{N}^m, \quad d_j > 0.$$

The indices of weight less than H form a finite downward-closed set. Its one-step outside boundary contains every index which can next enter as the cutoff increases. All boundary indices of the smallest weight must be added together. Every predecessor has smaller weight, so no undiscovered index of that same weight can lie beyond the boundary. Thus all resonant contributions are combined before a new coefficient block is determined.

A zero coefficient does not erase its index from this argument. Several successive complete layers may cancel, and the first nonzero omitted block can therefore occur beyond the smallest possible excluded weight. A complete-tail envelope remains valid at that smaller weight even without locating the first nonzero block. Conversely, a finite sequence of cancelled layers does not prove exact termination of the series.

15.3 Recovering and extending a Newton approximation

Suppose a normalized observable is $(1+U)^r$ with $r \neq 0$ and U of positive valuation. In the filtered unit algebra its formal power with exponent $1/r$ uniquely recovers $1+U$. The constant term 1 selects this formal branch, including when r is negative or noninteger.

If the recovered unit solves the normalized implicit equation modulo the current precision, Newton correction increases the precision to at least twice its former value under the usual unit-derivative hypothesis. Each step therefore extends the same inverse germ. The residual, rather than the number of retained displayed terms, determines the precision available for the next step. A coarser projection of a more accurate approximation remains consistent with every finer projection of the same formal series.

15.4 The forward-data precision ceiling

Let the leading source exponent be $p \neq 0$ and the internal observable exponent be $r \neq 0$. Under the value and derivative hypotheses of the inverse error-transport theorem, a forward error $\mathcal{O}(u^\rho \mathrm{M}(u)^k)$ generally guarantees target power accuracy through

$$\frac{\rho - p + r}{|p|}.$$

At equal powers its logarithmic degree must be combined with that of the model-inverse truncation error. Computing more coefficients of a fixed finite model does not remove the uncertainty inherited from the forward data. A stronger inverse statement requires a more accurate forward model or an independent exact identity establishing cancellation of the unknown error. This distinction separates refinement of a finite approximation from refinement of the mathematical information on which it is based.

16 Calculus of expansions with transported precision

Arithmetic, composition and differentiation must transport the uncertainty of an approximation together with its finite expression. A useful general form is

$$F(y) = b(y) + c(y)\left[J(w(y)) + \mathcal{O}(w(y)^P \mathrm{M}(w(y))^D)\right], \qquad w(y) \longrightarrow 0^+, \tag{16.1}$$

where b, c, w are exact expressions and J is a finite power–log jet. The offset b and prefactor c are treated as exact. Thus a cutoff in the jet describes relative precision with respect to c, and the absolute remainder contains $|c|$. This form includes ordinary forward and inverse expansions, Lambert expansions and exact coordinate substitutions. A transformed observable generally satisfies a different equation from the original inverse; a residual estimate must refer to the equation that the approximation actually represents.

For addition and multiplication, the precision rules of Section 2 are used after expressing both operands in the same positive coordinate. An exact prefactor may be absorbed into the jet when it is a finite power–log expression. Distinct prefactors can be aligned when their ratio lies in that algebra. Independently ordered exponential factors or incompatible coordinates require a larger common asymptotic scale before a single cutoff can be assigned.

Powers and logarithms. For a positive monomial leading term aw^α, write $J = aw^\alpha(1+U)$. A constant power and a logarithm reduce to the unit series of $(1+U)^r$ and $\log(1+U)$. The input uncertainty first becomes $\mathcal{O}(w^{P-\alpha}\mathrm{M}^D)$ in U. Multiplication by the output prefactor then transports it to the correct absolute scale. Real noninteger powers require positivity on the selected branch. These operations retain any earlier input precision ceiling; requesting a higher output cutoff does not create unknown coefficients.

Exponentials and exact carriers. Suppose

$$H = H_{\le 0} + H_{>0} + \mathcal{O}(w^P \mathrm{M}^D), \qquad P > 0,$$

where $H_{\le 0}$ contains the nonpositive-weight blocks. Then

$$e^H = e^{H_{\le 0}}\left[e^{H_{>0}} + \mathcal{O}(w^P \mathrm{M}^D)\right].$$

Indeed the omitted argument tends to zero and $e^r = 1 + \mathcal{O}(r)$ there. The operation keeps $e^{H_{\le 0}}$ as an exact carrier and expands only the vanishing part. For example $H = w^{-1} + \log w + w$ yields $we^{1/w}(1 + w + w^2/2 + \cdots)$. This also supplies the precision rule needed when a source logarithm is inverted and the source variable must be reconstructed by exponentiation. A nonvanishing argument uncertainty cannot in general justify a relative error tending to zero after exponentiation.

Composition and regular observables. The outer local coordinate must become $aw^\alpha(1 + o(1))$ with $a, \alpha > 0$ after substitution of the inner expansion. This condition checks both the limiting point and its selected side. An outer remainder $\mathcal{O}(v^P\mathrm{M}(v)^D)$ becomes $\mathcal{O}(w^{\alpha P}\mathrm{M}(w)^D)$, since $\log v = \alpha \log w + \mathcal{O}(1)$. The uncertainty of the inner argument is propagated independently through every jet operation, and the worse precision is retained. Regular analytic observables at a finite limiting argument use their Taylor series and the same precision arithmetic. Poles, branch points and logarithmic leading coefficient functions require their respective scale normalizations; the regular Taylor-composition rule does not apply at a singular point.

Differentiation requires derivative information. An estimate $R(w) = \mathcal{O}(w^P \mathrm{M}(w)^D)$ alone does not imply any estimate on R'. For example $w^P \sin(e^{1/w})$ has that magnitude bound and a much larger derivative. Therefore a nonzero remainder can be differentiated n times only with the additional hypotheses

$$R^{(j)}(w) = \mathcal{O}(w^{P-j}\mathrm{M}(w)^D), \qquad 0 \le j \le n.$$

Each differentiation uses one order of these derivative bounds. The jet derivative uses $D_w[w^\beta Q(\log w)] = w^{\beta-1}(\beta Q + Q')$; the derivatives of the exact coordinate, offset and prefactor in (16.1) are included by the chain and product rules. Exact finite expressions require no remainder hypothesis. A discarded tail of a known finite power–log expression also has derivative bounds of every order. In contrast, a derivative bound established for one unknown remainder is not automatically promoted to a stronger exponent when the source is refined; that stronger bound must be established separately.

Truncation and refinement. Truncation moves the first discarded block, including its logarithmic degree, into the remainder. It cannot improve uncertainty inherited from the input. Refinement instead determines more coefficients from the original function, implicit equation or composition. It remains subject to any error bound inherited from unspecified input data. Changing only the exponent written in a remainder symbol supplies no new mathematical information.

A remainder of a known analytic function and an error in an unspecified function play different roles. The Taylor expansion of $\sin x$ can be continued to arbitrary finite order because the original function is known. If only a finite approximation T and a bound $F - T = \mathcal{O}(w^P \mathrm{M}^D)$ are given, then the unknown coefficients of F at or beyond that precision are not determined. Higher-order operations on T cannot remove this uncertainty.

16.1 Separate error envelopes

A common ordered coefficient algebra is unnecessary for transporting magnitude bounds. Along one fixed real approach, suppose $F = A + \mathcal{O}(R)$ and $G = B + \mathcal{O}(S)$, where F, G, A, B are real-valued and $R, S \ge 0$ are error envelopes. Their coordinates and rates of decay may differ. The arithmetic of Section 3.2 then has the following form, without selecting a single exponent cutoff.

Proposition 16.1 (Error envelopes for compound functions)**.** *Under these hypotheses,*

$$\begin{aligned} F + G &= A + B + \mathcal{O}(R + S), \\ FG &= AB + \mathcal{O}(|A|S + |B|R + RS). \end{aligned} \tag{16.2}$$

For every fixed positive integer n,

$$F^n = A^n + \mathcal{O}\left(\sum_{k=1}^{n} \binom{n}{k} |A|^{n-k} R^k\right), \tag{16.3}$$

where the factor $|A|^0$ in the last summand is 1, including at $A = 0$. No nonvanishing or relative-smallness hypothesis is needed here.

If A is eventually nonzero and $R/|A| \to 0$, then for fixed real r

$$F^r = A^r + \mathcal{O}(|A|^{r-1}R),$$

provided $A > 0$ eventually when r is not an integer. In particular, $F^{-1} = A^{-1} + \mathcal{O}(R/|A|^2)$. The logarithm and exponential obey

$$\begin{aligned} \log F &= \log A + \mathcal{O}(R/A) \quad \text{if } A > 0 \text{ eventually and } R/A \to 0, \\ e^F &= e^A + \mathcal{O}(e^A R) \qquad \text{if } R \to 0. \end{aligned} \tag{16.4}$$

Finally $\phi(F) = \phi(A) + \mathcal{O}(R)$ for $\phi(t) = |t|, \sin t, \cos t$, without a smallness hypothesis on R or a finite-limit hypothesis on A.

Proof. Write $F = A + u$ and $G = B + v$, with $|u| \le CR$ and $|v| \le DS$ eventually. Addition and the identity $FG - AB = Av + Bu + uv$ give (16.2). The binomial formula gives (16.3), absorbing the finitely many factors C^k into one constant. Under relative smallness, $u/A \to 0$, so F/A stays near 1 and F has the same eventual sign as A. The functions $(1+t)^r$ and $\log(1+t)$ have bounded derivatives near $t = 0$. Apply their mean value bounds to $t = u/A$ and multiply by the exact prefactor. If $R \to 0$, then $u \to 0$ and $e^u - 1 = \mathcal{O}(u)$, giving the exponential bound. Absolute value, sine and cosine are globally 1-Lipschitz on the real line, giving the final assertion. As in Proposition 3.6, induction permits finite compositions whenever each operation satisfies its stated hypotheses. ☐

These are asymptotic bounds with unspecified constants and thresholds; the constants may depend on fixed data and powers. Cancellation of $A + B$ supplies no cancellation of unknown errors without additional relations between them. The retained expression need not be ordered as a Poincaré expansion, and a trigonometric bound may identify no leading term at all. Thus these assertions alone provide neither a single cutoff nor a pointwise numerical certificate, and they imply no derivative bounds. For a pure error, the integer-power rule gives $\mathcal{O}(R)^n = \mathcal{O}(R^n)$; a fractional power additionally needs a real-branch condition that a magnitude bound alone does not supply.

17 Perturbations of a retained exact inverse

An inverse such as a real branch of the Lambert function can be useful as an exact building block even when it also has a convergent inverse–logarithmic expansion. Retaining that building block separates two different refinements: expansion of the core inverse itself, and correction for terms outside the core. The second refinement is an expansion in powers of an auxiliary perturbation marker. Its coefficients are complete functions of the exact core inverse; they are not individual terms of an exponent-sorted power–logarithmic series.

17.1 The coefficient identity

Choose the positive source coordinate $u \to 0^+$ and write the original source point as

$$x = H(u), \qquad H(u) = x_0 + \sigma u \quad \text{or} \quad H(u) = \frac{\sigma}{u}, \qquad \sigma \in \{1, -1\}.$$

Let $F_0(u)$ be the chosen core and $R(u)$ its perturbation in that coordinate. Let $u_0(y)$ be the selected exact inverse of F_0, and set $\phi(y) = H(u_0(y))$. For a fixed real r, consider the following family of observables: $r = 1$ denotes the source point itself; at a finite endpoint, $r \neq 1$ denotes its displacement raised to r; at an infinite endpoint it denotes its rth power. Thus the local observable is

$$G_r(u) = \begin{cases} x_0 + \sigma u, & r = 1 \text{ at a finite endpoint}, \\ \sigma^r u^r, & r \neq 1 \text{ at a finite endpoint}, \\ \sigma^r u^{-r}, & \text{at an infinite endpoint}. \end{cases} \qquad r_* = \begin{cases} r, & \text{finite endpoint}, \\ -r, & \text{infinite endpoint}. \end{cases} \tag{17.1}$$

Let r be a fixed nonzero real number. For $\sigma = -1$, restrict to integer r; for $\sigma = 1$, all displayed bases are positive on the real source branch. Put $\Phi_r(y) = G_r(u_0(y))$; in particular $\Phi_1 = \phi$. Introduce a marker λ by

$$F_0(u_\lambda) + \lambda R(u_\lambda) = y, \qquad u_{\lambda=0} = u_0(y). \tag{17.2}$$

At any fixed point with $F_0'(u_0) \neq 0$, the inverse function theorem gives an analytic solution in λ near zero whenever the data are analytic locally. The Lagrange–Bürmann identity gives

$$G_r(u_\lambda) = \Phi_r(y) + \sum_{n\geq 1} \lambda^n C_n(u_0(y)), \tag{17.3}$$

where

$$C_n(u) = \frac{(-1)^n}{n!} \left(\frac{1}{F_0'(u)} \frac{d}{du} \right)^{n-1} \left(\frac{G_r'(u) R(u)^n}{F_0'(u)} \right). \tag{17.4}$$

Indeed, with $t = F_0(u)$, the equation becomes $t = y - \lambda R(u_0(t))$, and $d/dt = (F_0'(u))^{-1} d/du$. Applying the ordinary Lagrange identity to the observable $\Phi_r(t)$ proves (17.4). This form of the identity differentiates the explicit forward core. It expresses every coefficient using the forward core and its derivatives, even when the inverse is represented by a Lambert function.

For arbitrary data, this identity alone does not justify setting $\lambda = 1$, does not order the coefficients by their size as $u_0 \to 0$, and does not bound the tail by its first omitted coefficient. The following restricted class supplies those missing conclusions with a complete tail majorant.

17.2 A class with an asymptotic tail bound

Let $\mathrm{M}(u) = 1 + |\log u|$. Consider finite real power–logarithmic expressions

$$F_0(u) = y_0 + u^p Q_0(\log u) + \sum_{j=1}^{J} u^{p+\eta_j} Q_j(\log u), \qquad p \neq 0, \quad \eta_j > 0, \tag{17.5}$$

$$R(u) = \sum_{i=1}^{I} u^{p+\delta_i} P_i(\log u), \qquad \delta_i > 0. \tag{17.6}$$

All polynomials have real coefficients, Q_0 is nonzero, and all powers are fixed real numbers. Equal powers are merged before the leading term is selected. Put

$$\begin{aligned} q &= \deg Q_0, & \delta &= \min_i \delta_i, \\ D &= \max\left(0, \max_i \deg P_i - q\right), & \chi(u) &= u^\delta \mathrm{M}(u)^D. \end{aligned} \tag{17.7}$$

For a zero perturbation the observable remains exactly Φ_r, so these parameters and the following tail estimate are unnecessary.

Theorem 17.1 (An exact-core expansion with a complete marker tail)**.** *For fixed data satisfying* (17.5)–(17.6)*, the selected real inverse exists sufficiently near the source endpoint. Fix an observable satisfying* (17.1)*, and write* $g_r(y) = G_r(u_1(y))$ *for that observable of the inverse.*

There are positive constants C, K and a source threshold such that, for every integer $N \geq 0$ and all sufficiently small $u_0 > 0$ with $K\chi(u_0) < 1$, the marker series converges at $\lambda = 1$ and

$$\left| g_r(y) - \Phi_r(y) - \sum_{n=1}^{N} C_n(u_0(y)) \right| \leq C u_0^{r_*} \frac{(K\chi(u_0))^{N+1}}{1 - K\chi(u_0)}. \tag{17.8}$$

In particular, for every fixed N,

$$g_r(y) = \Phi_r(y) + \sum_{n=1}^{N} C_n(u_0(y)) + \mathcal{O}\Big(u_0^{r_*+(N+1)\delta} \mathrm{M}(u_0)^{(N+1)D}\Big). \tag{17.9}$$

The estimate concerns the complete omitted marker tail, including possible cancellations in its first coefficient.

Proof. Write $u = u_0 W$ and divide the equation $F_0(u_0W) + \lambda R(u_0W) = F_0(u_0)$ by the leading scale $u_0^p(\log u_0)^q$ and its nonzero leading coefficient. Use $\tau = 1/\log u_0$ and finitely many additional parameters for the higher-power terms. For example,

$$\frac{(\log(u_0W))^k}{(\log u_0)^q} = (\log u_0)^{k-q}(1 + \tau \log W)^k.$$

If $k > q$, include the factor $u_0^\eta(\log u_0)^{k-q}$ in its small parameter; if $k \leq q$, retain the nonnegative power τ^{q-k} as an analytic coefficient. All such parameters tend to zero. Powers of W and $\log W$ are analytic in a fixed complex neighborhood of $W = 1$ after fixing their branches.

At the limiting parameter origin, the normalized equation is $W^p - 1 = 0$, whose derivative at $W = 1$ is $p \neq 0$. The analytic implicit function theorem therefore gives a holomorphic solution in a common polydisc. The core parameters remain fixed when λ varies, whereas every perturbation parameter is multiplied by λ. Each perturbation parameter is bounded in magnitude by a constant times $\chi(u_0)$ for small u_0. Cauchy estimates in a smaller fixed polydisc consequently bound the total coefficient of marker degree n by $CK^n\chi(u_0)^n$. Equivalently, the solution is holomorphic and uniformly bounded on a marker disc whose radius is a fixed positive multiple of $1/\chi(u_0)$.

The observable in (17.1), after subtracting the constant x_0 when it occurs, is $\sigma^r u_0^{r_*} W^{r_*}$. The normalized factor W^{r_*} is analytic near $W = 1$ for every fixed real r_*, including negative and irrational values, giving the additional factor $u_0^{r_*}$. The sign convention ensures that this analytic continuation agrees with the stated real observable. Summing the coefficient bound gives (17.8). The local real branch is unique, and $F_0'(u) \sim pu^{p-1}Q_0(\log u)$ while $R'/F_0' \to 0$, so this analytic solution agrees with the selected real inverse near the endpoint. Finally $\chi(u_0) \to 0$, which proves (17.9). □

The constants in this theorem are existence bounds for fixed input data. The theorem does not specify numerical values for C, K or the source threshold. Obtaining a numerical enclosure at a fixed target requires explicit estimates of these quantities or a separate residual bracket. The radius on which an exact inverse identity holds need not be a radius on which the marker majorant has already been quantified.

The nonnegative value of D in (17.7) is deliberately conservative. A quotient by an extra power of $|\log u_0|$ can make an individual coefficient smaller than the stated polynomial–log bound. Keeping that improvement would require an additional logarithmic ordering. Also, if the perturbation contains several different powers, one complete marker coefficient may contain contributions lying beyond the leading contribution of a later marker coefficient. Marker depth does not promise a globally sorted list of all resulting monomials.

17.3 The Lambert core and its first correction

For the exact core

$$F_0(u) = y_0 + au^p(\log u + b)^q, \qquad p \neq 0, \qquad q \in \mathbb{Z}_{\geq 0},$$

the leading logarithmic polynomial is an exact power of an affine function. For $q = 0$, the positive local inverse is $u_0 = ((y - y_0)/a)^{1/p}$. For $q > 0$, put

$$A = a(-1)^q, \qquad Y = \frac{y - y_0}{A} > 0, \qquad k = -\frac{p}{q}, \qquad z = kY^{1/q}e^{pb/q}.$$

Then

$$u_0 = Y^{1/p}\left(\frac{W_\nu(z)}{k}\right)^{-q/p}, \qquad \nu = \begin{cases} -1, & p > 0, \\ 0, & p < 0. \end{cases} \tag{17.10}$$

These are the real branches for which $u_0 \to 0^+$. On the lower branch, the real argument condition is $-1/e \leq z < 0$. The quotient $W_\nu(z)/k$ is positive on the selected eventual branch, so all displayed real powers have positive bases.

Example 17.2 (Correcting $u \log u$ without expanding its exact inverse)**.** For $F(u) = u\log u + u^2$ as $u \to 0^+$, let $u_0 = y/W_{-1}(y)$. Formula (17.4) gives

$$C_1(u_0) = -\frac{u_0^2}{1 + \log u_0}, \tag{17.11}$$

$$C_2(u_0) = \frac{u_0^3(4\log u_0 + 3)}{2(1 + \log u_0)^3}. \tag{17.12}$$

Here $p = q = \delta = 1$ and $D = 0$. Thus truncation after C_1 has the proved remainder $\mathcal{O}(u_0^3)$, and truncation after C_2 has remainder $\mathcal{O}(u_0^4)$. The explicit omitted coefficient can exhibit a sharper logarithmic scale, while the theorem bounds the complete tail. Each displayed correction retains the exact W_{-1} expression in u_0.

17.4 A divergent power plus a logarithm

The exact core may itself contain smaller source powers and logarithms. Another explicitly invertible class is

$$F_0(u) = y_0 + au^p + b\log u, \qquad p < 0, \quad ab \neq 0. \tag{17.13}$$

It belongs to (17.5): the logarithm has the strictly higher source exponent zero. The leading logarithmic degree is $q = 0$. Set $X = u^p$, $d = b/p$, and $k = a/d$. The equation becomes $aX + d\log X = y - y_0$, whence

$$X = \frac{W_\nu\left(ke^{(y-y_0)/d}\right)}{k}, \qquad u_0 = X^{1/p}, \qquad \nu = \begin{cases} 0, & k > 0, \\ -1, & k < 0. \end{cases} \tag{17.14}$$

Indeed $kXe^{kX} = ke^{(y-y_0)/d}$; conversely, taking real logarithms of the positive quantities in this identity recovers the core equation. These branches give $X \to +\infty$, hence $u_0 \to 0^+$. The signs of the coefficients, positivity of the local inverse, and the lower branch's real argument interval determine the admissible branch. A constant in the zero-power logarithmic polynomial is extracted into y_0 before applying this identity, so affine target shifts are preserved.

Example 17.3 (An exact core for $x + \log x + 1/x$)**.** As $x \to +\infty$, retain the exact inverse $w = W_0(e^y)$ of $x + \log x$. The source coordinate is $u = 1/x$; the core is $u^{-1} - \log u$ and the perturbation is u. Thus $p=-1$, $q=0$, $\delta=2$, $D=0$, and $u_0=1/w$. For the source-point observable $r=1$, formula (17.4) gives

$$x = w - \frac{1}{w+1} - \frac{2w+1}{2w(w+1)^3} + \mathcal{O}(w^{-5}). \tag{17.15}$$

The two corrections are complete marker coefficients; the exact $W_0(e^y)$ has not been expanded into logarithms. For the reciprocal observable, $r=-1$ and $r_*=1$, and the corresponding result is

$$\frac{1}{x} = \frac{1}{w} + \frac{1}{w^2(w+1)} + \frac{4w+3}{2w^3(w+1)^3} + \mathcal{O}(w^{-7}). \tag{17.16}$$

The differing exponents 5 and 7 follow from the same complete-tail theorem with the appropriate observable derivative. A direct expansion of $X - w + \log(X/w) + \lambda/X$ verifies the first result through marker degree two independently of the coefficient generator.

17.5 An additional forward remainder

Suppose the actual forward function also contains an unknown error $E(u)$, satisfying the bounds

$$E(u) = \mathcal{O}(u^{\rho}\mathrm{M}(u)^k), \qquad E'(u) = \mathcal{O}(u^{\rho-1}\mathrm{M}(u)^k), \qquad \rho > p. \tag{17.17}$$

The derivative bound is an independent hypothesis, not a consequence of the first bound alone. Since the retained forward derivative has scale $u^{p-1}\mathrm{M}(u)^q$, the mean value argument of Section 9 gives an additional error in the selected observable of

$$\mathcal{O}\Big(u_0^{r_*+\rho-p}\mathrm{M}(u_0)^{\max(0,k-q)}\Big). \tag{17.18}$$

The gap $\rho > p$ ensures that this error is relatively small in the positive source coordinate. Both the perturbed solution and the core solution are asymptotic to the same u_0, so they can be used interchangeably in these scale estimates. The total error combines (17.18) with the marker-tail bound by taking the smaller source power, and the larger logarithmic degree if those powers coincide. Increasing marker depth cannot improve this inherited accuracy ceiling. The displayed marker coefficients still describe the specified finite model; coefficients smaller than the unknown input error are not asserted to be coefficients of every admissible actual function. The transport formula remains valid when $r_* < 0$ or when the resulting absolute error does not tend to zero: the forward gap $\rho > p$ still makes the error small relative to the observable's leading scale $u_0^{r_*}$. For example, at $x \to 0^+$, an unknown $\mathcal{O}(x^3\mathrm{M}(x)^2)$ error in the model $x + x^2$ gives an $\mathcal{O}(\mathrm{M}(u_0)^2)$ ceiling for the observable x^{-2}. At $x \to +\infty$, a declared $\mathcal{O}(x^{-2}\mathrm{M}(1/x))$ error in the model $x + 1/x$ gives an $\mathcal{O}(u_0^4\mathrm{M}(u_0))$ ceiling for x^{-1}.

17.6 Scope of the marker theorem

The identity (17.4) applies to any locally analytic core with a specified inverse branch and nonzero derivative. The complete tail theorem proved here additionally requires strictly positive source-power gaps in the perturbation. Perturbations of equal source power but smaller logarithmic size, purely logarithmic leading cores and exponential exact cores call for their own small parameters and error estimates. The formal marker identity alone supplies neither the ordering nor the complete-tail estimate in those larger scales.

18 Rigorous residual enclosures

An asymptotic remainder describes a limiting scale; it need not supply a constant or a numerical neighborhood. A root obtained by numerical iteration is useful independent evidence, but is not a verified enclosure. A residual bracket converts an approximation into a proved enclosure once the derivative and residual have explicit bounds. The resulting statement concerns a unique root on a specified interval and a specified center; it does not depend on how the center was chosen.

18.1 The certificate theorem with interval data

Let $I = [A, B]$, let $c \in (A, B)$, and suppose it is established that F is continuous on I and differentiable inside, with

$$F'(I) \subseteq [d_-, d_+], \qquad F(c) - y \in [\rho_-, \rho_+].$$

Assume either $d_- > 0$ or $d_+ < 0$, and put

$$\mu = \min(|d_-|, |d_+|), \quad M = \max(|d_-|, |d_+|), \quad \epsilon = \max(|\rho_-|, |\rho_+|), \quad h = \epsilon/\mu.$$

If $[c - h, c + h] \subseteq I$, Theorem 9.4, after changing the sign of F if necessary, proves a unique root x_* in I and

$$|x_* - c| \leq h.$$

The mean-value identity further gives

$$x_* \in I \cap [c - h, c + h] \cap \left(c - \frac{[\rho_-, \rho_+]}{[d_-, d_+]} \right).$$

Division is interval division; its denominator is separated from zero. This intersection is a sharpened root enclosure. If the residual interval does not contain zero, there is also the lower bound

$$|x_* - c| \geq \frac{\min(|\rho_-|, |\rho_+|)}{M}.$$

It can prove that a requested accuracy is impossible for a fixed center.

If the symmetric residual bracket does not fit inside I, the residual may instead be enclosed at A and B. Proved opposite signs in the orientation of F' establish existence by continuity, and the separated derivative still establishes uniqueness. Once existence in I is known, the mean-value theorem gives the same bound $|x_* - c| \leq h$ and the same intersection enclosure, without requiring the symmetric bracket to be contained in I. This is a second sufficient condition for existence, based on endpoint signs. This case matters for a coarse asymptotic seed close to one end of a valid interval: increasing arithmetic precision cannot repair an unsuitable symmetric-bracket geometry.

The theorem asserts uniqueness on the specified interval. To enclose the selected local inverse, that interval must lie on its source branch. It does not certify global injectivity or identify disconnected monotone intervals as the same continuation of an inverse germ.

18.2 Rational arithmetic and elementary enclosures

Interval bounds can be constructed with rational endpoints. Addition, multiplication, reciprocals and integer powers use the usual enclosing interval operations. If a finite dyadic grid of mesh $\eta > 0$ is desired, replace a lower endpoint q by $\lfloor q/\eta \rfloor \eta$ and an upper endpoint by $\lceil q/\eta \rceil \eta$. These directed roundings preserve containment by exact inequalities. A rational endpoint can first be scaled by an integer power of two to control the relative grid size.

For $0 \leq t \leq \frac{1}{2}$, the exponential has the bounds

$$S_N(t) = \sum_{j=0}^{N} \frac{t^j}{j!}, \qquad 0 \leq e^t - S_N(t) \leq \frac{t^{N+1}}{(N+1)!} \frac{1}{1 - t/(N+2)}.$$

The bound follows by majorizing the ratio of successive omitted terms by $t/(N+2)$. A positive argument is reduced by powers of two, and the enclosure is squared back using directed rounding. Negative arguments use the reciprocal of the positive exponential enclosure.

For $1 \le q \le 2$, put $v = (q-1)/(q+1) \in [0, \frac{1}{3}]$. Then

$$T_N(q) = 2\sum_{j=0}^{N-1} \frac{v^{2j+1}}{2j+1}, \qquad 0 \le \log q - T_N(q) \le \frac{2v^{2N+1}}{(2N+1)(1-v^2)}.$$

Every positive rational argument is written exactly as $2^k q$ with $1 \le q < 2$, by comparison with integer powers of two. The identity $\log(2^k q) = k\log 2 + \log q$ therefore reduces all logarithms to the displayed bound. Both tails are positive rational expressions. Increasing N improves their analytic precision independently of the directed arithmetic precision.

Exact real identities can simplify an enclosure before these bounds are applied: $\log(e^a) = a$ for real a, and $\log(ab) = \log a + \log b$ for $a, b > 0$. Positivity is therefore checked before distributing logarithms. If, for example, a positive product has negative factors, the product must be enclosed without distributing its logarithm. This avoids materializing a huge exponential only to take its logarithm while preserving the real-branch conditions.

Noninteger powers on a strictly positive base reduce to $a^b = \exp(b \log a)$, provided b also has a rigorous enclosure. For a fixed positive rational power, continuity additionally permits a zero lower base endpoint. Structural composition of these interval operations provides an enclosure whenever all domains hold on the entire interval. A reciprocal requires its denominator interval to avoid zero, and a logarithm requires a strictly positive argument. The same rules applied to an explicit derivative give a derivative enclosure. An approximate value at finitely many sample points supplies neither a function enclosure nor a sign proof on the intervening interval.

18.3 Centers, phase coordinates and improved accuracy

Example 18.1 (An enclosure independent of asymptotic accuracy)**.** For $F(x) = x + x^2$, $y = 6$, $I = [1, 3]$ and $c = 21/10$, the derivative enclosure is $[3, 7]$ and the exact residual is $51/100$. Hence $\mu = 3$ and $h = 17/100$, so the symmetric residual bracket $[193/100, 227/100]$ lies in I. There is one root in I, and

$$x_* \in [193/100, 227/100] \cap \left(\frac{21}{10} - \frac{51/100}{[3,7]}\right) = \left[\frac{193}{100}, \frac{1419}{700}\right].$$

The independently known root $x_* = 2$ lies in this enclosure. The residual theorem applies regardless of whether an asymptotic truncation is accurate at the target $y = 6$.

A center may be chosen from an asymptotic approximation, a numerical iteration or the midpoint of an interval. Its selection supplies a candidate; the derivative and residual inequalities supply the proof. The error estimate applies to the specified center. If the center is rounded before verification, the estimate must use that rounded center.

Suppose a forward model has an unspecified additional error E known only through $E = \mathcal{O}(R)$ as the endpoint is approached. This limiting statement has no numerical constant or interval threshold. An enclosure for the explicit model therefore does not automatically enclose roots of every function in that error class. Such a conclusion requires interval bounds for E and its derivative, or another explicit monotonicity argument.

For a phase representation $F(x) = y_0 + \varepsilon e^{\Phi(x)}$, the target $v = \log(\varepsilon(y - y_0))$ gives the equivalent equation $\Phi(x) = v$ on its real domain. The domain condition must hold on the whole verification interval. Since the target transformation is one-to-one, a unique root enclosure for the phase equation is also an enclosure for the original equation. Proposition 13.2 connects the two residuals without subtracting large exponentials.

Once existence and uniqueness have been proved in an interval I, a sharper interval-Newton enclosure may be formed by choosing $c \in I$ and intersecting I with $c - (F(c) - y)/F'(I)$. New derivative and residual bounds can then be formed on the smaller interval. Every enclosure still contains the root already proved to exist; there is no need to repeat the initial symmetric-bracket existence argument. Higher-order asymptotic approximations may supply better centers, but enclosure refinement is mathematically independent of expansion refinement.

For a fixed center, a positive lower error bound proves an accuracy floor. Increasing the precision of an enclosure cannot alter the actual error of that center. Failure to prove a desired bound, however, does not prove that the bound is false: a broad derivative enclosure, an unsuitable interval or an overestimate of the residual may be responsible.

18.4 Absolute and relative error

Suppose the root enclosure I is separated from zero, and set

$$m_I = \min_{z \in I} |z| > 0, \qquad M_I = \max_{z \in I} |z|.$$

An absolute error bound r gives the relative bound r/m_I. For prescribed tolerances $\epsilon, \tau \geq 0$, the sufficient condition

$$r \leq \max(\epsilon, \tau m_I)$$

proves $|c - x_*| \leq \max(\epsilon, \tau |x_*|)$. If $x_* = 0$, a relative quotient is undefined, so a positive absolute tolerance must be used for a nonzero error bound. If an enclosure merely crosses zero, a sharper enclosure may establish a positive lower root magnitude.

Conversely, to prove that a fixed center violates this combined tolerance, a certified lower error bound must exceed $\max(\epsilon, \tau M_I)$. Using m_I in this impossibility criterion would be incorrect: a sufficient success bound and a sufficient failure bound use opposite sides of the root-magnitude enclosure.

19 Source charts and reconstruction of inverse asymptotics

An exact source chart can convert a problem outside the power–logarithmic scale into one covered by the preceding inversion formulas. The inverse must then be reconstructed through that chart, with its error transported through the reconstruction function. Exponential and logarithmic reconstructions have substantially different sensitivity to inner errors.

19.1 Pure logarithmic source dependence

Choose the positive local source coordinate u associated with the endpoint and side. Substitute $u = e^{-h}$, so $h \to +\infty$. A purely logarithmic expression becomes an equation $\Phi(h) = y$ with no residual exponential dependence on h. At a finite endpoint x_0 and side $s \in \{-1, 1\}$,

$$x = x_0 + s e^{-h};$$

at a source infinity of sign s, the reconstruction is $x = s e^{h}$. Noninteger real powers $(-\log u)^q$ and polynomial or rational combinations of $\log u$ can therefore use the same chart whenever the transformed inverse has an appropriate asymptotic expansion.

For example, $F(u) = (\log u)^2 + \log u$ becomes $h^2 - h$. Its small positive inverse begins

$$u = e^{-\sqrt{y}-1/2}\left(1 - \frac{1}{8\sqrt{y}} + \mathcal{O}(y^{-1})\right).$$

The exponential factor multiplies the absolute remainder. Indeed, if $h = H + R$ with $R = o(1)$, then

$$e^{-h} = e^{-H}(1 + \mathcal{O}(R)).$$

The hypothesis $R = o(1)$ is essential. A finite relative expansion for a growing h need not have an absolute error tending to zero, and therefore need not justify this reconstruction. In particular, a fixed-depth inverse-logarithmic approximation to a growing Lambert function may fail this condition.

An exact transformed inverse has no such obstruction. Repeated exact charts give the exact inverse $u = \exp(-e^y)$ of $\log(-\log u)$. An exact candidate is identified by its defining equation and the selected local branch; equality to some other real or complex solution is insufficient.

19.2 Finite exponential families

At $x \to s\infty$, set $u = e^{-sx}$, so $x = -s \log u$. Finite sums of real exponentials become finite generalized-power germs in u. Their rates need not be commensurable. Equal powers and cancellations are combined before selecting the leading term. When the resulting expression has a nonzero leading power after subtraction of the target offset, its inverse can be constructed in the new coordinate. Logarithmic factors can be included whenever the transformed coefficient algebra has the required closure properties.

For example,

$$e^{-x} + e^{-2x} = y \quad \Longrightarrow \quad x = -\log y + y - \frac{3}{2}y^2 + \frac{10}{3}y^3 + \mathcal{O}(y^4), \qquad y \to 0^+.$$

The inverse of $e^x + e^{-x}$ at $+\infty$ begins $\log y - y^{-2} + \mathcal{O}(y^{-4})$. At $-\infty$ the selected inverse is its negative. The source endpoint is part of the branch specification.

19.3 Observables and exact carriers

At a finite source endpoint, a natural observable is $(x-x_0)^r$; the inverse itself additionally includes the offset when $r = 1$. At infinity the observable is x^r. On a negative source side, integer r gives a real observable without an additional real-power convention. For exponential reconstruction the exponent is multiplied by the corresponding signed power before exponentiation.

A noninteger power of a logarithmic reconstruction need not be a finite polynomial–logarithmic expression. It can instead be retained as an exact function of the finite approximation. Suppose $X > 0$, $X \sim c|\log w|$ with $c > 0$, and $x - X = \mathcal{O}(w^P \mathrm{M}(w)^D)$ with $P > 0$. Then $x/X \to 1$, and the mean value theorem yields

$$\frac{x^r - X^r}{X^r} = \mathcal{O}\left(\frac{x - X}{X}\right) = \mathcal{O}(w^P \mathrm{M}(w)^D).$$

The conservative final bound allows the carrier X^r to remain intact. Its internal retained corrections must still be distinguished from a single constant coefficient multiplying that carrier. Precision is determined by the approximation to X, rather than by merely counting outer summands.

19.4 Residuals after reconstruction

To evaluate a defining-equation residual, first recover the chart variable from the actual finite reconstructed observable. If that reconstructed variable is $\widehat{h}$, the normalized defect is

$$\frac{\Phi(\widehat{h}) - y}{y - y_0},$$

with denominator y at an infinite target. Using the unreconstructed inner approximation in place of $\widehat{h}$ would miss errors introduced by a subsequent truncation of the exponential or logarithm.

The residual is an exact expression even when its asymptotic expansion requires a larger coefficient algebra. An order assertion additionally requires a justified expansion in that algebra. Similarly, an unknown forward error must be transported through the source chart before applying an inverse-error theorem. Exact coordinate identities alone do not justify differentiating or exponentiating an unspecified magnitude remainder.

20 Finite logarithmic hierarchies and reciprocal-logarithmic order

A power gap and a logarithmic gap describe different orders of smallness. For example, the correction $u/\log u$ has the same source power as u, whereas $u^2\sqrt{-\log u}$ has a positive source-power gap and a nonpolynomial logarithmic coefficient. These two situations, and a leading product of logarithmic factors, require different normalizations. Exact source-coordinate reductions, including repeated logarithmic charts, are handled by the coordinate machinery of Section 13.

20.1 The finite coefficient hierarchy

In the positive source coordinate $u \to 0^+$, put

$$L_1(u) = -\log u, \qquad L_{j+1}(u) = \log L_j(u), \qquad 1 \le j < d. \tag{20.1}$$

The depth d is finite. All retained levels are positive and tend to infinity sufficiently near the endpoint. A generalized logarithmic monomial is

$$c\prod_{j=1}^{d} L_j^{\alpha_j}, \qquad c \in \mathbb{R}, \quad \alpha_j \in \mathbb{R}. \tag{20.2}$$

Finite sums of these monomials form an algebra under addition and multiplication. Powers in (20.2) have positive bases, including when an exponent is noninteger or negative. An arbitrary real power of a sum is not silently included in this finite algebra.

Proposition 20.1 (Closure under the source Euler derivative)**.** *For a differentiable coefficient $C(L_1, \ldots, L_d)$,*

$$u\frac{d}{du}C(L_1(u), \ldots, L_d(u)) = \mathcal{E}C, \qquad \mathcal{E} = -\sum_{j=1}^{d} \frac{1}{L_1 \cdots L_{j-1}}\frac{\partial}{\partial L_j}, \tag{20.3}$$

where the empty denominator is 1*. In particular, the finite generalized logarithmic algebra is closed under every derivative used in the inverse coefficient formula.*

Proof. The chain rule gives $uL_1'(u) = -1$ and $uL_j'(u) = -(L_1 \cdots L_{j-1})^{-1}$. Each summand in (20.3) decreases finitely many monomial exponents and creates no new logarithmic level. $\square$

This is the needed closure for coefficients such as $L_1^{1/2}$: its derivatives contain $L_1^{-1/2}, L_1^{-3/2}, \ldots$. Restricting the coefficient algebra to positive logarithmic powers would lose closure even in this simplest example. An expression containing $\sqrt{\log u}$ instead has an eventually negative base on $u \to 0^+$; the real hierarchy does not admit it.

20.2 Positive source-power gaps with logarithmic coefficients

Consider

$$F(u) = y_0 + au^p \left(1 + \sum_{i=1}^{m} u^{\delta_i} C_i(L_1(u), \ldots, L_d(u))\right), \qquad a \neq 0, \quad p \neq 0, \quad \delta_i > 0, \tag{20.4}$$

where every C_i belongs to the finite algebra above. All exponents are fixed real constants, and the coefficients and the sign of a are fixed on the parameter domain under consideration. Set $z = ((y - y_0)/a)^{1/p} > 0$, so $z \to 0^+$. For the observable u^r, the multi-index coefficient of weight $w = \boldsymbol{k} \cdot \boldsymbol{\delta}$ is

$$\frac{(-1)^{|\boldsymbol{k}|} r}{p^{|\boldsymbol{k}|} \prod_i k_i!} \prod_{j=1}^{|\boldsymbol{k}|-1} (\mathcal{E} + r + w + pj) \prod_i C_i^{k_i}, \tag{20.5}$$

evaluated at the logarithmic levels of z and multiplied by z^{r+w}. For $\boldsymbol{k} = 0$, the coefficient is 1. Formula (20.5) is the same Lagrange identity as in the polynomial case; Proposition 20.1 supplies its enlarged coefficient algebra.

Theorem 20.2 (Convergence and a conservative complete-tail bound)**.** *The inverse of* (20.4) *on its selected eventual real branch has the convergent multi-index expansion* (20.5)*. Choose integers $D_i \geq 0$ such that every monomial in C_i satisfies*

$$\prod_j L_j(u)^{\alpha_j} = \mathcal{O}(\mathrm{M}(u)^{D_i}).$$

For example, the ceiling of the largest sum of the positive monomial exponents is sufficient. Let $\mathcal{B}$ be the finite minimal excluded multi-index boundary of a positive source-weight cutoff, and put

$$\beta = \min_{\boldsymbol{k} \in \mathcal{B}} \boldsymbol{k} \cdot \boldsymbol{\delta}, \qquad D = \max_{\boldsymbol{k} \in \mathcal{B}} \sum_i k_i D_i. \tag{20.6}$$

Then the complete omitted expansion of u^r is $\mathcal{O}(z^{r+\beta} \mathrm{M}(z)^D)$. At a finite source endpoint the original inverse has $r = 1$; at an infinite source endpoint it has $r = -1$, with the selected source sign applied separately.

Proof. Write $u = zW$ near $W = 1$. For each level, the quotient $L_j(zW)/L_j(z)$ is analytic in W and the finite reciprocal logarithmic parameters near their limiting values. For instance, the first quotient is $1 - \log W / L_1(z)$; subsequent quotients are obtained by taking the logarithm of the preceding quotient and dividing by the next level. Each factor raised to a fixed real power is holomorphic near 1. Introduce a separate small parameter for every finite coefficient monomial $z^{\delta_i} \prod_j L_j(z)^{\alpha_j}$. All tend to zero. The normalized implicit equation has derivative $p \neq 0$ in W at its parameter origin, so the analytic implicit function theorem proves convergence in a fixed polydisc of these parameters.

Cauchy estimates, with the finite coefficient sums grouped by i, give a geometric multi-index majorant in $K_i z^{\delta_i} \mathrm{M}(z)^{D_i}$ for suitable constants K_i. The omitted upward-closed index set is covered by the translates of its finite minimal boundary. Sum the majorant over each translate. Its denominators remain bounded for sufficiently small z, and the boundary monomial is bounded by $z^{\beta}\mathrm{M}(z)^{D}$ using (20.6). Multiplication by z^r completes the estimate. The estimate does not assume that the first boundary coefficient is nonzero. $\square$

The resulting logarithmic degree can be larger than the optimal one. The finite boundary supplies a valid envelope for the entire omitted sum, while the retained coefficients remain exact generalized expressions. The cutoff uses the ordinary positive target coordinate: a source weight w contributes target power $(r+w)/|p|$. This convention preserves the nonunit leading-power and infinite-endpoint scaling.

Example 20.3 (Nonpolynomial logarithmic coefficients)**.** For $T = -\log y$, inversion at 0^+ gives

$$(u + u^2\sqrt{-\log u})^{-1}(y) = y - y^2\sqrt{T} + y^3(2T - \tfrac{1}{2}) + \mathcal{O}(y^4\mathrm{M}(y)^3), \tag{20.7}$$

$$(u + u^2\log(-\log u))^{-1}(y) = y - y^2\log T + y^3\left(2\log^2 T - \frac{\log T}{T}\right) + \mathcal{O}(y^4\mathrm{M}(y)^3). \tag{20.8}$$

Here the superscript -1 denotes function inversion. The displayed remainders are conservative complete-tail bounds. The derivative term in each cubic coefficient is essential and follows directly from (20.3).

20.3 Reciprocal-logarithmic units

Now consider a core of the form

$$F_0(u) = y_0 + au^pH(-1/\log u), \qquad p \neq 0, \quad H(0) = 1, \tag{20.9}$$

where H is a rational function analytic at 0 with real coefficients. Let $z = ((y-y_0)/a)^{1/p} > 0$, $t = -1/\log z > 0$, and write $u = zW$. The normalized inverse equation is exactly

$$W^pH\left(\frac{t}{1 - t\log W}\right) = 1. \tag{20.10}$$

Its derivative in W is p at $(t, W) = (0, 1)$, so its real solution $W(t)$ is analytic at 0. Iterating

$$W \longmapsto H\left(\frac{t}{1 - t\log W}\right)^{-1/p} \tag{20.11}$$

from $W = 1$ determines successively correct Taylor coefficients: dependence on the previous W carries a positive power of t. Exact finite Taylor arithmetic determines the finite bracket and its first omitted orders. An exclusive cutoff h retains exactly the integer degrees $n < h$. Its remainder has the form $\mathcal{O}(t^{\beta})$, where β is the first unretained nonzero degree, or any smaller degree beyond the retained terms when no sharper order has been established.

For the original source inverse, the bracket is multiplied by the exact prefactor σz at a finite endpoint or σ/z at infinity. An observable uses the corresponding power of W and transports the prefactor and remainder together. This is a logarithmic cutoff inside that prefactor, not a cutoff in y or $1/y$.

Example 20.4 (A zero source-power gap)**.** At $u \to 0^+$, the inverse of $u + u/\log u$ begins

$$y\left(1 + t + t^2 + 2t^3 + \frac{7}{2}t^4 + \mathcal{O}(t^5)\right), \qquad t = -\frac{1}{\log y}. \tag{20.12}$$

Equivalently, its bracket is $1 - 1/\log y + 1/\log^2 y - 2/\log^3 y + 7/(2\log^4 y)$. For the same forward expression at $+\infty$, use $t = 1/\log y$ and the bracket $1 - t + t^2 - 2t^3 + 7t^4/2$.

The expression $u + u/\log u - u/(1+\log u)$ has a cancelled first reciprocal-logarithmic correction. Its forward unit is $1+t^2/(1-t)$ in the source logarithmic coordinate, and its inverse bracket begins $1 - t^2 + \cdots$. Coefficients are merged before detecting the leading unit, so a cancelled coefficient does not create a fictitious retained term.

Higher source-power terms can accompany (20.9). If each has a strictly positive power gap and an admitted logarithmic coefficient envelope, their induced inverse correction is smaller than the prefactor times every fixed power of t. For $u(1+1/\log u)+u^2$, the logarithmic bracket therefore agrees with (20.12) at every finite logarithmic order. The omitted source-power expression nevertheless belongs to a distinct, smaller asymptotic sector. A residual computed solely from the logarithmic bracket refers to the leading core. Equality of those brackets does not assert that the full inverses are equal.

20.4 A leading logarithmic monomial

The same normalization extends to

$$F_0(u) = y_0 + au^p \prod_{j=1}^{d} L_j(u)^{q_j}, \qquad p \neq 0, \quad q_j \in \mathbb{R}. \tag{20.13}$$

Let $Y = (y-y_0)/a > 0$ and define

$$\begin{gathered} T = -\frac{\log Y}{p}, \quad t = \frac{1}{T}, \quad B_1 = T, \quad B_{j+1} = \log B_j, \\ d_0 = \frac{1}{p}\sum_j q_j \log B_j, \quad z = Y^{1/p}\prod_j B_j^{-q_j/p}. \end{gathered} \tag{20.14}$$

All B_j are positive sufficiently near the selected endpoint. Put

$$R_1(W) = 1 + t(d_0 - \log W), \tag{20.15}$$

$$R_{j+1}(W) = 1 + B_{j+1}^{-1}\log R_j(W). \tag{20.16}$$

Then $L_j(zW)/B_j = R_j(W)$ exactly, and the normalized inverse equation is

$$W^p \prod_j R_j(W)^{q_j} = 1. \tag{20.17}$$

Theorem 20.5 (A convergent leading-hierarchy bracket)**.** *Equation* (20.17) *has a unique real solution near $W=1$ for the selected asymptotic branch. Its expansion in powers of t, retaining the lower-level coefficients exactly, converges sufficiently near the endpoint. A truncation beginning its omitted degrees at β has bracket remainder*

$$\mathcal{O}\Big(t^\beta(1+|\log t|)^\beta\Big). \tag{20.18}$$

Proof. Treat t, td_0, and $B_2^{-1}, \ldots, B_d^{-1}$ as independent small parameters. The equation is analytic in these parameters and W near their origin and $W=1$, with derivative $p \neq 0$ in W. The analytic implicit function theorem gives a common polydisc and a convergent series there. Since $d_0 = \mathcal{O}(\log T)$, all the parameters indeed tend to zero on the actual asymptotic curve.

The total degree in the first two parameters equals the formal degree in t after substituting td_0. Cauchy estimates, uniform in the other parameters on a smaller polydisc, bound the tail of those degrees by a constant times $[t(1+|d_0|)]^\beta$ for sufficiently small t. This proves (20.18). The fixed-point expression $W = \prod_j R_j(W)^{-q_j/p}$ has positive formal valuation in its dependence on the previous W, so finite exact iteration computes the same coefficients. Real powers use the positive local factors near 1. $\square$

For $F(x) = x \log\log x$ at $+\infty$, let $T = \log y$, $B = \log T$, and $t = 1/T$. The reconstructed inverse starts

$$F^{-1}(y) = \frac{y}{B}\left(1 + \frac{\log B}{B}t + \mathcal{O}\Big(t^2(1 + |\log t|)^2\Big)\right). \tag{20.19}$$

Thus an iterated logarithm in the leading coefficient need not be replaced by a constant. Products of levels and real powers of those levels are covered by the same positive local normalization. Positive source-power perturbations again belong to separate smaller sectors.

20.5 Precision, exactness, and the boundary of the hierarchy

The three normalizations above have different precision meanings. Positive source-power gaps lead to a cutoff in the positive target power coordinate. Reciprocal-logarithmic units and leading logarithmic monomials instead use a cutoff in the small logarithmic coordinate inside an exact prefactor. All these cutoffs are exclusive. Terms at a common source weight must be combined before counting nonzero blocks; cancellation can remove a complete block without removing its contribution to the index boundary used for the tail estimate.

A finite run of zero coefficients does not establish termination. A sufficient exactness argument reconstructs a positive candidate source from the finite observable and verifies the original forward equation identically. If no smaller sector has been omitted, local uniqueness then identifies this candidate with the inverse germ. In particular, a polynomial source unit does not by itself make a truncated noninteger power of that unit exact.

The normalized equations also give direct residual identities. Substituting a finite unit into (20.10) or (20.17) determines its defect in the leading-core equation. Such a defect is not a residual for a different forward function containing unretained source-power sectors.

The reciprocal-logarithmic unit has a single ordinary Taylor bracket. The other normalizations retain an enlarged coefficient algebra, so their composition requires additional closure arguments. The convergence and tail theorems give existence of constants and neighborhoods; they do not supply explicit numerical values for them. An unknown forward remainder likewise requires a separate transport estimate through the selected logarithmic hierarchy.

These constructions form a finite hierarchy rather than a general transseries field. Arbitrary real powers of coefficient sums, sums of incomparable leading logarithmic monomials, and independent simultaneous cutoffs at every logarithmic level require further hypotheses and additional notions of precision. The formulas above establish closure under the Euler derivatives used in inversion, convergence for the stated families, and complete-tail estimates with the distinct precision meanings made explicit.

21 Finite commensurate flat sectors

An exponential that is smaller than every power of the source coordinate needs its own truncation index. A finite part of this larger scale arises from the following exact forward expression in a positive source coordinate $u \to 0^+$:

$$F(u) = y_0 + au^q + \sum_{j=1}^{J} A_j(u)e^{-k_j c/u^p}, \quad p, c > 0, \quad q \neq 0, \quad a \neq 0, \quad k_j \in \mathbb{N}_{>0}, \tag{21.1}$$

where each A_j is a finite real power–log expression. The phase power and rates are fixed real numbers, with all rates integer multiples of a common positive rate. Coefficients may depend on fixed real parameters subject to the stated sign assumptions. Different phase powers and incommensurate rates require more than one exponential coordinate and fall outside this single-phase construction.

The zero sector must be exact. An ordinary finite asymptotic approximation to a nonmonomial core leaves a power-sized error, which dominates every sector in (21.1). The exact shifted monomial core $y_0 + au^q$ avoids this obstruction: its zero-sector inverse is exact. Put

$$u_0 = \left(\frac{y - y_0}{a}\right)^{1/q} > 0, \qquad E_0 = e^{-c/u_0^p}.$$

At a finite source endpoint the observable is $H(u) = x_0 + \sigma u$ for power one and $H(u) = \sigma^r u^r$ otherwise. At an infinite endpoint it is $H(u) = \sigma^r u^{-r}$. For negative source branches, take an integer observable power.

21.1 Computing complete sectors

Introduce a formal symbol Z and the polynomial $R(u, Z) = \sum_j A_j(u)Z^{k_j}$. Differentiation of the actual exponential coordinate is represented by the exact operator

$$\mathcal{D} = \partial_u + cp\, u^{-p-1} Z\partial_Z, \qquad \frac{d}{du} e^{-c/u^p} = cp\, u^{-p-1} e^{-c/u^p}.$$

The marker identity of Section 17 now gives

$$\frac{(-1)^n}{n!} \left(\frac{\mathcal{D}}{aqu^{q-1}}\right)^{n-1} \left(\frac{H'(u)R(u, Z)^n}{aqu^{q-1}}\right). \tag{21.2}$$

Sum this expression over $1 \le n \le N$, group equal powers of Z, and evaluate $u = u_0$, $Z = E_0$. Since every $k_j \ge 1$, no marker degree larger than N contributes to an exponential degree at most N. Also $\mathcal{D}$ preserves exponential degree, so intermediate polynomial truncation discards no contribution to a retained sector. Each resulting coefficient is a complete finite power–log function of u_0.

For example, the inverse of $F(x) = x + e^{-1/x}$ at 0^+ has the first three exponential corrections

$$y - E + y^{-2}E^2 + \left(y^{-3} - \frac{3}{2}y^{-4}\right) E^3, \qquad E = e^{-1/y}. \tag{21.3}$$

For $F(x) = x + x^2 e^{-1/x}$, the first two sectors are $y - y^2E + (y^2 + 2y^3)E^2$. Adding $2e^{-2/x}$ to the first example changes the second coefficient to $y^{-2} - 2$. Thus the sector index distinguishes input sectors from powers of the perturbation marker.

21.2 A bound for the complete omitted tail

Write every coefficient as a finite sum $u^\alpha P(\log u)$, and choose

$$b = \max(0, \max_\alpha (q + p - \alpha)), \qquad d = \max(0, \max_P \deg P),$$
$$B(u) = u^{-b}\mathrm{M}(u)^d, \qquad \chi(u) = B(u)e^{-c/u^p}.$$

Let $r_* = r$ at a finite source endpoint and $r_* = -r$ at infinity. For fixed admitted data, there are $C, K > 0$ and a source threshold such that the complete omitted sector tail after degree N satisfies

$$|\operatorname{Tail}_N| \le C u_0^{r_*+p} \frac{(K\chi(u_0))^{N+1}}{1 - K\chi(u_0)} \quad \text{when } K\chi(u_0) < 1. \tag{21.4}$$

Consequently its asymptotic class is

$$\mathcal{O}\Big(E_0^{N+1} u_0^{r_*+p-(N+1)b} \mathrm{M}(u_0)^{(N+1)d}\Big).$$

This bound is allowed to be conservative; it is independent of whether the first omitted coefficient cancels.

To justify the estimate, set $u = u_0(1 + u_0^p V)$ and divide the forward equation by u_0^{q+p}. The normalized monomial difference has the analytic extension

$$a\frac{(1+\epsilon V)^q - 1}{\epsilon}, \qquad \epsilon = u_0^p,$$

whose derivative at $V = 0$, $\epsilon = 0$ is $aq \neq 0$. Each exponential factor, after extracting $E_0^{k_j}$, becomes

$$\exp\left(k_j c\,\frac{1-(1+\epsilon V)^{-p}}{\epsilon}\right),$$

which is analytic and uniformly bounded for small ϵ, V. The power–log factors have the same local analytic property after their powers of u_0 and $\log u_0$ are extracted. Each normalized coefficient is bounded by a constant times $B(u_0)$. Since $B \geq 1$ for small u_0, replacing E_0 by a complex variable Z gives a uniform implicit solution on $|Z| < 1/(KB(u_0))$. One may regard the bounded normalized coefficients as additional parameters in a compact polydisc; the derivative condition at $Z = 0$ is uniform. Cauchy estimates therefore bound the coefficient of Z^k by $C(KB)^k$. The nonconstant observable contributes the factor $u_0^{r_*+p}$. Summing the geometric tail proves (21.4), and $\chi(u_0) \to 0$ establishes applicability at $Z = E_0$.

The constants and threshold in (21.4) are existential. They are not explicit numerical error bounds until suitable values have been established. Sector depth is inclusive: a depth N retains $k \leq N$, whereas an ordinary power cutoff retains powers strictly below its boundary. The first omitted coefficient and a bound for the complete omitted tail are distinct mathematical objects.

The primary example also admits a direct formal check. Substitute $g = y + \sum_{k=1}^{N} C_k(y)Z^k$ into

$$g + Z\exp(1/y - 1/g) - y.$$

Vanishing through degree N in Z identifies the same sector coefficients by the uniqueness of the formal implicit solution. This identity verifies the finite coefficients; the uniform analytic argument above supplies the bound for the complete infinite tail.

22 Finite Fourier coefficients in a logarithmic variable

The polynomial coefficient algebra extends naturally to finite sums

$$B(L) = \sum_{\omega\in\Omega} P_\omega(L)e^{i\omega L}, \qquad L = \log u, \qquad \Omega \subset \mathbb{R} \text{ finite},$$

where every P_ω is a polynomial and the frequencies are fixed real numbers. This includes $\sin(\log u)$, several incommensurable frequencies, and polynomial amplitudes multiplying those oscillations. The source powers and the Fourier frequencies have different roles: source weights determine asymptotic truncation, while frequencies label coefficient modes. No Fourier mode is discarded merely because its frequency is large.

22.1 Coefficient arithmetic and reality

Multiplication convolves frequencies, and the Euler derivative is

$$\frac{\mathrm{d}}{\mathrm{d}L}\left(P_\omega(L)e^{i\omega L}\right) = \left(P'_\omega(L) + i\omega P_\omega(L)\right)e^{i\omega L}.$$

Thus this algebra is closed under precisely the operations used by Theorem 6.1. Equal frequencies are identified before their polynomial amplitudes are added. Frequency equality is an exact mathematical relation; an approximation of two close frequencies does not justify treating them as equal.

The real-valued condition is the conjugacy relation $P_{-\omega}(L) = \overline{P_\omega(L)}$ on real L. This relation is the condition imposed on any fixed parameters in the amplitudes. Complex exponential notation is compatible with a real-valued coefficient because conjugate modes combine into real trigonometric expressions. Affine trigonometric phases, such as $\sin(2L+1)$, contribute their constant phase factor to the polynomial amplitude.

22.2 Inverse coefficients and independent composition

The admitted forward family is

$$f(u) = y_0 + au^p \left(1 + \sum_j u^{\delta_j} B_j(\log u)\right), \qquad ap \neq 0, \quad \delta_j > 0,$$

with finitely many real Fourier-polynomial coefficients. The leading core is a monomial with a proved nonzero real coefficient. An oscillatory leading block, even one that happens to remain positive, requires a different inverse normalization and is not covered by this construction.

For a multi-index $\boldsymbol{k}$ of depth n and weight $\sigma = \boldsymbol{k} \cdot \boldsymbol{\delta}$, the unchanged coefficient formula is

$$C_{\boldsymbol{k}}^{(r)}(L) = \frac{(-1)^n r}{p^n \boldsymbol{k}!} \prod_{j=1}^{n-1} (\mathcal{D} + r + \sigma + pj) \prod_i B_i(L)^{k_i}.$$

Convolution forms the product, and the Fourier Euler operation applies each factor. Contributions at a common source weight must then be combined to obtain a complete asymptotic block.

Residual checking uses a separate composition recurrence. If $u = z(1+U)$, then

$$(1+U)^\alpha B(L + \log(1+U)) = \sum_{k\geq 0} Q_k(L) U^k, \qquad Q_0 = B, \qquad Q_{k+1} = \frac{(\alpha - k)Q_k + \mathcal{D}Q_k}{k+1}.$$

This follows by Taylor expansion in the multiplicative increment; it holds for each exponential-polynomial mode as well as for their sum. Its use in the normalized forward equation independently checks the inverse coefficients. For a nonunit observable power, the residual first reconstructs the source unit by raising the observable unit to its reciprocal power.

22.3 Envelopes, convergence, and omitted terms

For real L, Fourier modes have modulus one. If $P_\omega(L) = \sum_j c_{\omega,j} L^j$, then

$$|B(L)| \leq \sum_{\omega,j} |c_{\omega,j}| (1 + |L|)^j.$$

This gives a nonoscillatory envelope for every retained block. Euler differentiation preserves the polynomial degree; frequency factors affect coefficient constants, rather than the power of the logarithmic envelope. Consequently a multi-index coefficient has polynomial degree at most $\sum_i k_i \deg B_i$, just as in the polynomial case.

The convergence argument also extends. After replacing each finite source mode by an independent marker, logarithmic composition uses

$$e^{i\omega(L+\log W)} = e^{i\omega L} W^{i\omega},$$

which is analytic for W near 1 with its local logarithm. The implicit derivative at the origin is still $p \neq 0$. The finite-dimensional analytic implicit function theorem gives an absolutely convergent marker expansion for sufficiently small markers. Along the real source slice these markers are bounded by constants times $z^{\delta_i}(1 + |\log z|)^{d_i}$ and tend to zero. The positive source gaps also make the forward derivative asymptotic to apu^{p-1}, so the selected real inverse branch exists near the endpoint.

The complete one-step outer boundary of the retained multi-index region therefore bounds the omitted tail. Its least weight supplies the remainder power, and the maximum summed polynomial degree over that finite boundary supplies a conservative logarithmic envelope. This bound need not be sharp after cancellation, but remains valid even at zeros of every displayed oscillatory coefficient. A declared forward remainder is transported with the same observable-dependent exponent as Theorem 9.1; its matching derivative hypothesis remains required. Constants and source thresholds are not computed numerical certificates.

22.4 Examples and scope

For $f(x) = x + x^2 \sin(\log x)$ at 0^+, the inverse begins

$$y - y^2 \sin L + y^3 \sin L(2\sin L + \cos L) + \mathcal{O}(y^4), \qquad L = \log y.$$

With an additional $x^3 \cos(2\log x)$, the coefficient of y^3 acquires $-\cos(2L)$, combining a directly supplied frequency with frequencies generated by convolution. The identity

$$\sin(\sqrt{2}L)\sin(\sqrt{3}L) + \tfrac{1}{2}\cos\!\left(\sqrt{5+2\sqrt{6}}\,L\right) = \tfrac{1}{2}\cos\!\left((\sqrt{2}-\sqrt{3})L\right)$$

exhibits an algebraic resonance and the resulting exact cancellation.

The cutoff is exclusive in the positive target power coordinate, with the same finite-endpoint, infinite-endpoint, and observable conventions as in the ordinary power–logarithmic case. Frequencies are fixed real constants; parameter-dependent frequency collisions require their equality relations to be resolved before forming complete blocks. Nonlinear logarithmic phases require a different coefficient algebra.

The nonoscillatory envelope is essential even when a retained coefficient vanishes along a sequence of source points. For example, the zeros of $\sin(\log u)$ do not justify replacing a tail bound by an oscillatory factor which also vanishes there. Bounds must control the entire omitted sum uniformly near the endpoint, independently of such zeros.

23 Special-function expansions and absolute error envelopes

The preceding calculus does not depend on an elementary description of the function being expanded. Its analytic input is a finite approximation on a specified branch, together with a bound for the omitted part. Special functions supply such input through differential equations, integral representations, and convergent defining sums. These sources must be distinguished from exact identities: a finite asymptotic expansion need not determine contributions smaller than every power in its stated scale.

All parameters in this section are fixed while the argument approaches its endpoint. Unless an explicit bound is given, constants and thresholds in $\mathcal{O}$ may depend on those parameters and on the fixed truncation order. No uniformity in a growing order, a coalescing singularity, or a parameter approaching the boundary of its domain is inferred. We use the real positive axis for fractional powers and logarithms. Reflection or continuation to another ray requires an exact identity or a separate asymptotic theorem on that ray.

23.1 Finite composition, oscillation, and real projection

Let $w \to 0^+$, and suppose finitely many functions have representations

$$F_j(w) = A_j(w) + \mathcal{O}(R_j(w)), \qquad R_j(w) \geq 0. \tag{23.1}$$

The approximations may contain exact exponential factors and bounded oscillations. For example, a finite expression of the form

$$A(w) = A_0(w) + \sum_{j=1}^{m} P_j(w)T_j(w) \tag{23.2}$$

can keep the factors P_j exact while each T_j is a finite power–logarithmic sum with trigonometric coefficients. An error $\mathcal{O}(R_j)$ inside the jth bracket gives the absolute error $\mathcal{O}(|P_j|R_j)$ outside it. Several independent errors give their sum as a valid envelope, even when no common leading term is available.

Proposition 16.1 propagates these errors through finite arithmetic expressions. At a regular finite limiting argument, Taylor's theorem supplies composition with an analytic function. More generally, a derivative bound on a neighborhood containing the true and approximate arguments supplies the corresponding mean-value bound. Division requires separation from zero; a real noninteger power requires the appropriate positive branch; exponentiation of an uncertain phase requires its absolute error to tend to zero. Finitely many successive applications are justified by induction, with every earlier error included at the next step.

An oscillatory coefficient changes the interpretation of the result. For instance,

$$F(x) = x^{-1/2} \cos x + \mathcal{O}(x^{-3/2})$$

is a useful absolute approximation on the positive axis. It does not assert $F(x)/(x^{-1/2} \cos x) \to 1$, since the denominator vanishes arbitrarily far along that axis. Nor does it justify taking a reciprocal on a whole positive half-line. The boundedness of sine and cosine supports absolute error transport through their zeros.

Proposition 23.1 (Projection of a real-valued function)**.** *Suppose F is real-valued on the specified approach, A may be complex, and $F - A = \mathcal{O}(R)$ for a nonnegative real envelope R. Then*

$$F - \operatorname{Re} A = \mathcal{O}(R), \qquad \operatorname{Im} A = \mathcal{O}(R).$$

In particular, projecting the finite approximation to its real part does not increase an available pointwise absolute error bound.

Proof. Since $\operatorname{Re} F = F$ and $\operatorname{Im} F = 0$, both assertions follow from $|\operatorname{Re} z| \leq |z|$ and $|\operatorname{Im} z| \leq |z|$. The same inequalities hold before an explicit bound is replaced by asymptotic notation. □

The real-valuedness assumption refers to the exact function on its chosen branch. It cannot be deduced merely by discarding an imaginary part of an approximation. Likewise, algebraic cancellation of finite terms need not cancel their unknown errors. If $F = A + B + \mathcal{O}(R)$, $B = \mathcal{O}(S)$ and $R = o(S)$, then one may conclude $F = A + \mathcal{O}(S)$; the smaller independent error has been absorbed into a proved bound for the omitted expression.

23.2 An exponent allowance for logarithmic tails

A remainder exponent without a logarithmic degree is incomplete information in the scale of Section 2. If it is known independently that the degree is finite, a strict loss in exponent gives a bound that does not need that degree explicitly.

Lemma 23.2 (Absorbing an unspecified finite logarithmic degree)**.** *Suppose, for some fixed real p and finite $D \geq 0$,*

$$R(w) = \mathcal{O}(w^p \mathrm{M}(w)^D).$$

For every fixed $\varepsilon > 0$,

$$R(w) = o(w^{p-\varepsilon}).$$

In particular, $R(w) = \mathcal{O}(w^{p-\varepsilon})$.

Proof. Divide the assumed bound by $w^{p-\varepsilon}$ and apply Lemma 2.2 to $w^\varepsilon \mathrm{M}(w)^D$. □

For a lattice of exponent spacing $1/q$, choosing $\varepsilon = 1/(2q)$ leaves an intermediate exponent between adjacent lattice positions. This choice is only a conservative bound; it does not specify the actual first omitted coefficient. A subsequently established omitted block may give a sharper boundary by the absorption argument above. The finite-degree hypothesis is essential: a purely formal remainder symbol or the logarithmic degrees of retained coefficients do not prove it. These assertions concern values, and provide no derivative bounds without the additional hypotheses in Section 16; see also [3, §2.1(iii)].

More precisely, suppose a complete omitted block has been determined in an approximation

$$F(w) = T(w) + w^\beta C(\log w) + E(w), \qquad E(w) = \mathcal{O}(w^\rho \mathrm{M}(w)^D), \qquad \rho > \beta,$$

where C is a nonzero polynomial and D is fixed and finite. Choosing $0 < \varepsilon < \rho - \beta$ gives $E = o(w^\beta)$, and hence $F - T = w^\beta C(\log w) + o(w^\beta) = \mathcal{O}(w^\beta \mathrm{M}^{\deg C})$. The strict gap is essential to this argument: at $\rho = \beta$, the unknown degree D cannot be replaced by $\deg C$. Vanishing computed coefficients do not establish a zero tail. The exponent allowance and any sharpening of it concern the proved analytic bound, independently of the formal cutoff used to select retained blocks. The constants and thresholds may depend on fixed parameters; uniform estimates require corresponding uniform tail hypotheses. No assertion here permits a logarithmic degree that varies with w.

23.3 Exact identities preserve smaller exponential terms

On $x > 0$, define the finite polynomials

$$P_n(t) = \sum_{k=0}^{n} \frac{(n+k)!}{k!(n-k)!2^k}\, t^k, \qquad n = 0, 1, 2, \ldots.$$

The half-integer modified Bessel functions satisfy

$$\begin{aligned} K_{n+1/2}(x) &= \sqrt{\frac{\pi}{2x}} e^{-x} P_n(1/x), \\ I_{n+1/2}(x) &= \frac{e^x P_n(-1/x) + (-1)^{n+1} e^{-x} P_n(1/x)}{\sqrt{2\pi x}}. \end{aligned} \tag{23.3}$$

They follow from the half-order identities in [3, §10.39] and the order recurrences. Here both equations are exact. In particular,

$$e^{-x} I_{1/2}(x) = \frac{1 - e^{-2x}}{\sqrt{2\pi x}}.$$

Its inverse-power expansion contains only the first displayed algebraic term, although the difference from that term is nonzero. The vanishing of every subsequent algebraic coefficient therefore cannot establish an exact finite identity.

The same distinction occurs for confluent hypergeometric functions:

$$ {}_1F_1(1;2;x) = \int_0^1 e^{xt}\,\mathrm{d}t = \frac{e^x - 1}{x} \qquad (x \neq 0). \tag{23.4} $$

The value at zero is obtained by continuity. The integral is the specialization of the Euler representation in [3, (13.4.1)]. Both endpoints contribute to the exact answer. Keeping the term $-1/x$ matters after cancellation or rescaling, even though it is exponentially small relative to e^x/x.

Terminating defining sums give a different route to exactness. If $m \geq 0$ is an integer and $b \notin \{0, -1, -2, \ldots\}$, then

$$ {}_1F_1(-m;b;x) = \sum_{k=0}^{m} \frac{(-m)_k}{(b)_k\,k!} x^k. $$

This conclusion comes from the zero numerator Pochhammer symbols and the nonvanishing denominators in the defining series, rather than from observing several zero coefficients in an asymptotic expansion.

23.4 Exponential and oscillatory differential-equation families

For fixed $\nu \in \mathbb{R}$, set

$$ a_0(\nu) = 1, \qquad a_k(\nu) = \frac{\prod_{j=1}^{k}(4\nu^2 - (2j-1)^2)}{8^k k!}, \qquad \omega = x - \frac{\nu\pi}{2} - \frac{\pi}{4}. $$

The positive-axis modified Bessel expansions are, for fixed $N \geq 1$,

$$ \begin{aligned} I_\nu(x) &= \frac{e^x}{\sqrt{2\pi x}} \left(\sum_{k=0}^{N-1} \frac{(-1)^k a_k(\nu)}{x^k} + \mathcal{O}(x^{-N}) \right), \\ K_\nu(x) &= \sqrt{\frac{\pi}{2x}} e^{-x} \left(\sum_{k=0}^{N-1} \frac{a_k(\nu)}{x^k} + \mathcal{O}(x^{-N}) \right). \end{aligned} \tag{23.5} $$

The corresponding oscillatory examples begin

$$ \begin{aligned} J_\nu(x) &= \sqrt{\frac{2}{\pi x}} \left(\cos\omega - \frac{a_1(\nu)}{x} \sin\omega - \frac{a_2(\nu)}{x^2} \cos\omega + \mathcal{O}(x^{-3}) \right), \\ Y_\nu(x) &= \sqrt{\frac{2}{\pi x}} \left(\sin\omega + \frac{a_1(\nu)}{x} \cos\omega - \frac{a_2(\nu)}{x^2} \sin\omega + \mathcal{O}(x^{-3}) \right). \end{aligned} \tag{23.6} $$

These formulas and their finite-order bounds are given in [3, §§10.17, 10.40]. Each outer factor multiplies the absolute error as well as the finite bracket. The oscillatory formulas have this meaning throughout the positive axis, including zeros of their leading trigonometric terms.

The Airy functions illustrate a fractional power in the exponential phase. With $\xi = 2x^{3/2}/3$ and $x \to +\infty$,

$$ \begin{aligned} \mathrm{Ai}(x) &= \frac{e^{-\xi}}{2\sqrt{\pi}\,x^{1/4}} \left(1 - \frac{5}{48x^{3/2}} + \frac{385}{4608x^3} + \mathcal{O}(x^{-9/2}) \right), \\ \mathrm{Bi}(x) &= \frac{e^{\xi}}{\sqrt{\pi}\,x^{1/4}} \left(1 + \frac{5}{48x^{3/2}} + \frac{385}{4608x^3} + \mathcal{O}(x^{-9/2}) \right). \end{aligned} \tag{23.7} $$

On the reflected real ray they instead satisfy

$$ \begin{aligned} \mathrm{Ai}(-x) &= \frac{1}{\sqrt{\pi}\,x^{1/4}} \left(\sin(\xi + \pi/4) + \mathcal{O}(x^{-3/2}) \right), \\ \mathrm{Bi}(-x) &= \frac{1}{\sqrt{\pi}\,x^{1/4}} \left(\cos(\xi + \pi/4) + \mathcal{O}(x^{-3/2}) \right). \end{aligned} \tag{23.8} $$

See [3, §9.7]. Thus a change of ray can replace real exponential growth by oscillation. The derivative expansions in that reference are separate analytic assertions; the displayed value remainders alone would not justify differentiating them.

Exact substitutions such as $x = c/w^\alpha + d$, with $c, \alpha > 0$ and fixed real d, preserve a positive-axis asymptotic theorem once the argument is eventually positive. The remainder is first substituted in its full absolute form. Further expansion of the exact phase or prefactor then uses the finite-composition rules, with enough accuracy in a phase to control its exponential.

23.5 A finite-moment theorem for integral and sum representations

A single Taylor estimate treats several integral tails and discrete defining sums. Write $|\mu|$ for the total variation of a real signed measure μ on $[0,\infty)$, and $M_k(\mu) = \int t^k \, \mathrm{d}\mu(t)$ for its moments.

Theorem 23.3 (Expansion against a fixed measure)**.** *Fix $s \in \mathbb{R}$ and an integer $N \geq 0$, and set $D = \max(0, \lceil -s - N \rceil)$. Suppose*

$$\int_0^\infty (1+t)^{N+D} \, \mathrm{d}|\mu|(t) < \infty.$$

For $a \geq 1$, define

$$F_\mu(s,a) = \int_0^\infty (a+t)^{-s} \, \mathrm{d}\mu(t).$$

Then

$$F_\mu(s,a) = \sum_{k=0}^{N-1} \frac{(-1)^k (s)_k M_k(\mu)}{k!} \, a^{-s-k} + R_N(a), \tag{23.9}$$

with the empty sum understood as zero, and

$$\begin{aligned} |R_N(a)| &\leq C_N(\mu,s)\, a^{-s-N}, \\ C_N(\mu,s) &= \frac{|(s)_N|}{N!} \int_0^\infty t^N (1+t)^D \, \mathrm{d}|\mu|(t). \end{aligned} \tag{23.10}$$

Proof. For $h(v) = (1+v)^{-s}$ on $v \geq 0$, $h^{(N)}(v) = (-1)^N (s)_N (1+v)^{-s-N}$. If $N \geq 1$, Taylor's remainder on $[0,v]$ is therefore bounded by

$$\left| h(v) - \sum_{k=0}^{N-1} \frac{(-1)^k (s)_k}{k!} v^k \right| \leq \frac{|(s)_N|}{N!} \, v^N (1+v)^D.$$

For $N = 0$ the same inequality is simply $(1+v)^{-s} \leq (1+v)^D$. Substitute $v = t/a$, multiply by a^{-s}, and use $(1+t/a)^D \leq (1+t)^D$ for $a \geq 1$. Integrating the absolute bound against $|\mu|$ proves both formulas. The moment hypothesis ensures that the original integral, every retained term, and the remainder bound are finite. □

If all moments of $|\mu|$ are finite, the theorem gives every fixed order of a Poincaré expansion. It does not assert convergence as $N \to \infty$. If $s = -m$ for an integer $m \geq 0$, the binomial expression is a polynomial and the expansion is exact after degree m.

Exponential integrals and incomplete Gamma functions. For fixed real p and $a > 0$, the positive-axis integral

$$E_p(a) = \int_1^\infty e^{-at} t^{-p} \, \mathrm{d}t = \frac{e^{-a}}{a} \int_0^\infty e^{-v} (1 + v/a)^{-p} \, \mathrm{d}v$$

has moments $\int_0^\infty e^{-v} v^k \, \mathrm{d}v = k!$. Theorem 23.3 gives

$$E_p(a) = \frac{e^{-a}}{a} \left(\sum_{k=0}^{N-1} \frac{(-1)^k (p)_k}{a^k} + \mathcal{O}(a^{-N}) \right). \tag{23.11}$$

When $p + N \geq 0$, the absolute remainder inside the bracket is at most $|(p)_N| a^{-N}$ for $a \geq 1$. The identities

$$\Gamma(\alpha, a) = a^\alpha E_{1-\alpha}(a), \qquad \operatorname{erfc} x = \frac{\Gamma(1/2, x^2)}{\sqrt{\pi}} \quad (x > 0)$$

give incomplete-Gamma and complementary-error-function tails by the same argument. Their standard forms and sharper domains for bounds are recorded in [3, §§8.11(i), 7.12(i)]. Section 24 establishes the additional derivative bounds needed for complementary-error-function inversion.

The growing exponential integral has a different positive-axis form,

$$\operatorname{Ei}(x) = \frac{e^x}{x} \left(\sum_{k=0}^{N-1} \frac{k!}{x^k} + \mathcal{O}(x^{-N}) \right). \tag{23.12}$$

The real reflection identity $\operatorname{Ei}(-x) = -E_1(x)$ for $x > 0$ selects its negative-axis tail. The sine and cosine integrals instead begin

$$\begin{aligned} \operatorname{Si}(x) &= \frac{\pi}{2} - \frac{\cos x}{x} - \frac{\sin x}{x^2} + \mathcal{O}(x^{-3}), \\ \operatorname{Ci}(x) &= \frac{\sin x}{x} - \frac{\cos x}{x^2} + \mathcal{O}(x^{-3}). \end{aligned} \tag{23.13}$$

These positive-axis formulas are in [3, §6.12]. They again give absolute approximations through oscillatory zeros.

A confluent hypergeometric example. For fixed $b \in \mathbb{R}$, $\alpha > 0$ and $x > 0$, the integral representation [3, (13.4.4)], after rescaling its integration variable, gives

$$U(\alpha, b, x) = \frac{x^{-\alpha}}{\Gamma(\alpha)} \int_0^\infty e^{-t} t^{\alpha-1} (1 + t/x)^{-(1+\alpha-b)} \, \mathrm{d}t.$$

The positive measure here has moments $M_k = (\alpha)_k$. Taking $s = 1 + \alpha - b$ in the moment theorem yields

$$U(\alpha, b, x) = x^{-\alpha} \left(\sum_{k=0}^{N-1} \frac{(-1)^k (\alpha)_k (1 + \alpha - b)_k}{k! \, x^k} + \mathcal{O}(x^{-N}) \right). \tag{23.14}$$

The moment bound supplies an explicit constant at each fixed order. Extending this proof to other values of α requires a continuation or recurrence argument with its own domain conditions.

23.6 Lerch's transcendent with a large third argument

For fixed real z, s with $|z| < 1$ and $a > 0$, the defining sum is

$$\Phi(z, s, a) = \sum_{n=0}^\infty z^n (a + n)^{-s}. \tag{23.15}$$

It converges absolutely for every real s; all powers use positive bases. See [3, (25.14.1)]. The signed discrete measure $\mu_z = \sum_{n \geq 0} z^n \delta_n$ has finite absolute moments of every order. Define

$$M_0(z) = \frac{1}{1 - z}, \qquad M_k(z) = \sum_{n=1}^\infty n^k z^n = \operatorname{Li}_{-k}(z) \quad (k \geq 1). \tag{23.16}$$

The last equality is the polylogarithm defining sum [3, (25.12.10)]; the separate zeroth moment includes the mass at $n = 0$.

Corollary 23.4 (Lerch expansion and a computable tail constant)**.** *For every fixed integer $N \geq 0$,*

$$\Phi(z, s, a) = \sum_{k=0}^{N-1} \frac{(-1)^k (s)_k M_k(z)}{k!} a^{-s-k} + R_N(z, s, a), \tag{23.17}$$

and, for $a \geq 1$,

$$\begin{aligned} |R_N(z, s, a)| &\leq C_N(z, s)\, a^{-s-N}, \\ C_N(z, s) &= \frac{|(s)_N|}{N!} \sum_{j=0}^{D} \binom{D}{j} M_{N+j}(|z|), \qquad D = \max(0, \lceil -s - N \rceil). \end{aligned} \tag{23.18}$$

Proof. Apply Theorem 23.3 to μ_z. Its total variation is $\mu_{|z|}$. Expanding the finite polynomial $(1 + n)^D$ gives the stated constant. □

Negative values of z change the coefficients through cancellation; the bound uses $|z|$ and remains valid. Negative noninteger values of s give growing initial powers and are covered by the factor $(1 + t)^D$ in the proof. For example,

$$\begin{aligned} \Phi(1/2, 2, a) &= \frac{2}{a^2} - \frac{4}{a^3} + \frac{18}{a^4} + \mathcal{O}(a^{-5}), \\ \Phi(1/2, -1/2, a) &= 2\sqrt{a} + \frac{1}{\sqrt{a}} + \mathcal{O}(a^{-3/2}). \end{aligned} \tag{23.19}$$

The respective absolute remainder constants from (23.18) are 104 and 3/4 for $a \geq 1$. If $z = 0$, the exact answer is a^{-s}. If $s = -m$ with $m \geq 0$ an integer, the expansion terminates:

$$\Phi(z, -m, a) = \sum_{k=0}^{m} \binom{m}{k} a^{m-k} M_k(z).$$

For example $\Phi(1/2, -2, a) = 2a^2 + 4a + 6$.

The positive coordinate is $w = 1/a$, with absolute block exponents $s + k$. Zero moments or zero Pochhammer factors remove entire blocks; they do not authorize changing the scale of the remaining uncertainty. A strict cutoff H retains only nonzero blocks with $s + k < H$. The constants above need not stay bounded as $|z| \to 1$, and the finite-measure proof does not apply at $z = \pm 1$. At $z = 1$, for example, the function becomes the Hurwitz zeta function when $s > 1$, and integral comparison gives leading behavior $a^{1-s}/(s-1)$, a different power from the fixed-$|z| < 1$ leading term $a^{-s}/(1 - z)$.

23.7 The zeta function in an exponential coordinate

Large positive values of the argument of the Riemann zeta function lead to a different scale. Its convergent defining sum is [3, (25.2.1)]

$$\zeta(S) = \sum_{n=1}^{\infty} n^{-S}, \qquad S > 1.$$

With $w = e^{-S} \to 0^+$ this becomes

$$\zeta(-\log w) = \sum_{n=1}^{\infty} w^{\log n}, \qquad 0 < w < e^{-1}. \tag{23.20}$$

The exponents $\{\log n : n \geq 1\}$ are left-finite, since $\log n < H$ implies $n < e^H$. They form an additive semigroup under $\log m + \log n = \log(mn)$, and are not finitely generated: a finite set of integers involves only finitely many prime factors. Every finite cutoff still has finitely many terms, and multiplication has finite coefficient sums indexed by factorizations of an integer. Thus the finite arithmetic extends beyond a globally finitely generated exponent semigroup.

Proposition 23.5 (A two-sided zeta tail bound)**.** *For every integer $m \geq 1$ and every real $S > 1$,*

$$\begin{aligned} \zeta(S) &= \sum_{n=1}^{m-1} n^{-S} + R_m(S), \\ m^{-S} &\leq R_m(S) \leq m^{-S}\left(1 + \frac{m}{S-1}\right). \end{aligned} \tag{23.21}$$

Consequently, for fixed m, $R_m(S) \sim m^{-S}$ as $S \to +\infty$.

Proof. Every summand of R_m is positive, giving the lower bound. Since t^{-S} decreases on $t > 0$,

$$\sum_{n=m+1}^{\infty} n^{-S} \leq \int_m^{\infty} t^{-S}\,\mathrm{d}t = \frac{m^{1-S}}{S-1}.$$

Add m^{-S} and divide both bounds by it. The quotient of the remainder and its first term is squeezed to 1. ☐

For example, retaining five terms gives

$$\zeta(S) = 1 + 2^{-S} + 3^{-S} + 4^{-S} + 5^{-S} + \mathcal{O}(6^{-S}). \tag{23.22}$$

Here the first term is the constant corresponding to exponent $\log 1 = 0$. A strict cutoff in the coordinate $w = e^{-S}$ retains $\log n < H$ and has first omitted exponent $\log m$ as in (23.21). Substituting $S = ax + b \to +\infty$ preserves both bounds on $ax + b > 1$. In contrast, all corrections to 1 are flat in $1/S$; an inverse-power expansion in that coordinate cannot distinguish them.

The zeta representation is a convergent generalized power series on its stated interval. The Lerch expansion follows from a convergent defining sum but is asserted only at each fixed asymptotic order. This distinction persists under finite composition, together with the absolute envelopes, branch conditions, and fixed-parameter hypotheses used at each step.

24 Special functions, asymptotic forward models, and real thresholds

Special-function inversion illustrates two different mathematical situations. Tail expansions use a finite Poincaré forward model and transport its omitted error through an inverse. Expansions at a real threshold instead use an exact local equation in a ramified coordinate. Convergence of the inverse expansion of a finite model does not imply convergence of the original Poincaré forward series.

A real affine target transformation $y = b + aF(x)$, with $a \neq 0$, is handled by replacing the target of F with $(y - b)/a$. Its branch domain and its error bounds must be transformed at the same time. The formulas below use $a = 1, b = 0$ unless stated otherwise.

24.1 The complementary-error-function tail

For $x > 0$, define

$$S(x) = \sqrt{\pi}\, x e^{x^2} \operatorname{erfc} x, \qquad S_m(x) = \sum_{k=0}^{m-1} \frac{(-1)^k (1/2)_k}{x^{2k}}, \qquad A_m = (1/2)_m. \tag{24.1}$$

The real positive-axis expansion has a remainder bounded by its first neglected term; see DLMF §7.12(i). For inverse-error transport we also require a derivative bound.

Lemma 24.1 (Value and derivative bounds for the normalized tail)**.** *For every integer $m \geq 1$ and every $x > 0$,*

$$|S(x) - S_m(x)| \leq A_m x^{-2m}, \qquad |S'(x) - S'_m(x)| \leq 2m A_m x^{-2m-1}. \tag{24.2}$$

Proof. Starting with the integral definition of erfc in DLMF (7.2.2), substitute $s^2 = x^2 + t$ to obtain

$$S(x) = \int_0^\infty e^{-t} (1 + t/x^2)^{-1/2}\, dt. \tag{24.3}$$

Put $w = x^{-2}$. The derivatives of $(1 + tw)^{-1/2}$ alternate in sign, and their magnitudes for $w \geq 0$ are at most their values at $w = 0$. Taylor's integral remainder, followed by integration in t, bounds the remainder of the first m terms by $(1/2)_m w^m$. Applying the same argument to the derivative with respect to w bounds its remainder by $m(1/2)_m w^{m-1}$. For $m = 1$, the latter statement is the direct bound on the first derivative. Finally $|dw/dx| = 2x^{-3}$ gives the second inequality in (24.2). The exponential factor in the integral justifies these finite differentiations and integrations. □

Let $Q_m(w)$ be the Taylor polynomial of degree $m-1$ of $-\log S_m(w^{-1/2})$ at $w = 0$. Define the finite phase

$$\Phi_m(x) = x^2 + \log x + Q_m(x^{-2}), \qquad v = -\log(\sqrt{\pi}\, y). \tag{24.4}$$

The exact phase is $\Phi(x) = -\log(\sqrt{\pi} \operatorname{erfc} x)$. For fixed m, Lemma 24.1 implies

$$\Phi(x) - \Phi_m(x) = \mathcal{O}(x^{-2m}), \qquad \Phi'(x) - \Phi'_m(x) = \mathcal{O}(x^{-2m-1}). \tag{24.5}$$

These are value and derivative assertions; the second is not inferred merely by differentiating an unspecified big-O term.

Explicit conditional bounds follow as well. Write $\delta = A_m x^{-2m}$, $\delta_1 = 2m A_m x^{-2m-1}$, and $\widetilde{\Phi}_m = x^2 + \log x - \log S_m$. Wherever $S_m > \delta$,

$$|\Phi - \Phi_m| \leq \frac{\delta}{S_m - \delta} + |\widetilde{\Phi}_m - \Phi_m|, \tag{24.6}$$

$$|\Phi' - \Phi'_m| \leq \frac{\delta_1}{S_m - \delta} + \frac{|S'_m|\delta}{S_m(S_m - \delta)} + |\widetilde{\Phi}'_m - \Phi'_m|. \tag{24.7}$$

The terms involving $\widetilde{\Phi}_m$ are exact elementary expressions. The positivity condition is satisfied eventually because $S_m \to 1$ and $\delta \to 0$.

Now invert $\Phi_m(x) = v$ on $x \to +\infty$. Its leading source scale is x^2, so the inverse error transported from (24.5) is $\mathcal{O}(x^{-2m-1}) = \mathcal{O}(v^{-m-1/2})$. Thus the forward model supports inverse accuracy through power $m + 1/2$ in the coordinate $1/v$. A stronger assertion requires a more accurate forward model. This bound must be combined with the truncation error of the model inverse.

For example, with $v = -\log(\sqrt{\pi}y)$,

$$\operatorname{erfc}^{-1} y = \sqrt{v} - \frac{\log v}{4\sqrt{v}} + \frac{-\log^2 v + 4\log v - 8}{32v^{3/2}} + \mathcal{O}\Big(v^{-5/2}(1 + |\log v|)^3\Big). \tag{24.8}$$

The cubic numerator follows by inserting $x = \sqrt{v} + A/\sqrt{v} + B/v^{3/2}$ into $x^2 + \log x + (2x^2)^{-1} + \mathcal{O}(x^{-4}) = v$. The expansion has a power cutoff in $1/v$, although $y \to 0^+$.

The negative-source tail uses the exact identity $\operatorname{erfc}(-x) = 2 - \operatorname{erfc} x$ and changes the reconstructed source sign. For $b + a \operatorname{erfc} x$ at $-\infty$, the positive small tail is $2 - (y - b)/a$ and the target endpoint is $b + 2a$. These identities handle reflected, shifted and sign-scaled tails without changing the forward-error proof.

24.2 A Stirling model with a retained Lambert inverse

For $x > 0$, let

$$G_m(x) = x(\log x - 1) - \frac{1}{2}\log x + \frac{1}{2}\log(2\pi) + \sum_{k=1}^{m-1} \frac{B_{2k}}{2k(2k-1)x^{2k-1}}. \tag{24.9}$$

The positive-axis Stirling and digamma error bounds in DLMF §5.11(ii) give

$$|\log \Gamma(x) - G_m(x)| \le A_m x^{-\rho}, \tag{24.10}$$

$$|\psi(x) - G_m'(x)| \le \rho A_m x^{-\rho-1}, \tag{24.11}$$

$$\rho = 2m - 1, \qquad A_m = \frac{|B_{2m}|}{2m(2m-1)}.$$

The finite value and derivative bounds apply for all positive x. Select the increasing inverse branch $x > 2$; on this interval $\psi(x) \ge \log x - (2x)^{-1} - (12x^2)^{-1} > 0$. This branch choice excludes the separate behavior near the positive minimum of the gamma function.

Use the exact core

$$F_0(x) = x(\log x - 1), \qquad \phi(v) = \frac{v}{W_0(v/e)}. \tag{24.12}$$

The difference $G_m - F_0$ is a finite higher-power perturbation in $u = 1/x$. The formula of Section 17 therefore retains ϕ exactly and computes its complete marker corrections. With $u_0 = 1/\phi(v)$, the omitted Stirling error transports to

$$\mathcal{O}(u_0^{2m-1}) \tag{24.13}$$

in a conservative polynomial–log bound. The actual derivative of the core provides an additional logarithmic denominator, but discarding that improvement preserves a valid bound in the stated remainder class. The final error combines (24.13) with the complete marker tail.

For $m = 1$, the first marker correction gives

$$(\log \Gamma)^{-1}(v) = \phi(v) + \frac{1}{2} - \frac{\log(2\pi)}{2\log \phi(v)} + \mathcal{O}(\phi(v)^{-1}). \tag{24.14}$$

Increasing marker depth while holding m fixed cannot improve the inherited Stirling accuracy limit. Marker coefficients beyond that limit still describe the chosen finite model. Neither the finite marker expansion nor its analytic complete-tail theorem establishes convergence of the infinite Stirling series.

For inversion of $\Gamma(x)$, use the exact target transformation $v = \log y$. The defining equation is then $\log \Gamma(x) = \log y$, which is equivalent on the positive real branch. For a real affine target $b + a\Gamma(x)$, replace y by $(y - b)/a > 1$. For $b + a \log \Gamma(x)$, use $v = (y - b)/a > 0$ directly.

24.3 The two real Lambert branches at their common threshold

The two real inverses of $F(x) = xe^x$ meet at $x = -1$, $y = -1/e$. The branch description is given in DLMF §4.13. Writing $x = -1 + h$ gives the exact local equation

$$eF(-1+h) + 1 = \frac{h^2}{2} + \frac{h^3}{3} + \frac{h^4}{8} + \frac{h^5}{30} + \cdots . \tag{24.15}$$

The leading power is two. Ordinary analytic reversion in the positive ramified coordinate

$$t = \sqrt{2(1+ey)} \tag{24.16}$$

therefore yields a convergent local expansion. Its first coefficients are

$$W_\nu(y) = -1 + \sigma t - \frac{t^2}{3} + \sigma\frac{11t^3}{72} - \frac{43t^4}{540} + \mathcal{O}(t^5), \tag{24.17}$$

where $(\nu, \sigma) = (0, 1)$ or $(-1, -1)$. These coefficients follow by reversion of the exact local forward equation. The principal branch approaches -1 from above and the lower branch from below, with real target neighborhood $-1/e < y < 0$. A real affine target map carries this neighborhood to its corresponding image. There is no real inverse below the threshold. An exclusive power cutoff $5/2$ in the positive distance $y + 1/e$ retains the terms through t^4.

24.4 An exact coalescing quadratic branch

For an exact quadratic model

$$F(x) = b + ac(x - x_0)^2, \qquad ac \neq 0, \tag{24.18}$$

the two branches are exactly

$$x = x_0 + \sigma\sqrt{\frac{y-b}{ac}}, \qquad \sigma \in \{1, -1\}, \qquad \frac{y-b}{ac} > 0. \tag{24.19}$$

The source direction selects σ. The remainder is zero because this is an exact inverse identity. For $F(x) = x^2 - \mu$, the branches at target $y = 0$ are $\pm\sqrt{\mu}$ and coalesce as $\mu \to 0^+$. The same coordinate retains the selected sign throughout this limiting comparison. It does not choose a different branch because the two roots become close.

24.5 Distinct scales and regions of validity

The error-function inverse uses a power expansion in a logarithmic target coordinate. Stirling inversion retains a Lambert core and uses marker degree for its corrections. The two threshold examples use powers of a local target distance. Their precision parameters therefore describe different asymptotic orders and cannot be interchanged without the corresponding coordinate transformation.

The number of forward terms limits the inverse accuracy. In particular, a coarse Stirling model keeps the accuracy floor in (24.13) even when additional marker coefficients are computed. Local convergence at a threshold and tail asymptotics at infinity likewise have different domains. Agreement of finite approximations at selected points is useful numerical evidence, but does not establish a universal crossover point or a uniform error bound.

The arguments here concern the stated real branches. Complex continuation, the minimum of the gamma function and its two nearby inverse branches, and higher-order coalescing critical points require additional branch and error analysis. No automatic transition between those problems is implied by the common use of an implicit inverse equation.

25 Retaining an exact growing exponential core

An inverse–logarithmic expansion of the Lambert function does not resolve the much smaller correction caused by adding x^2 to xe^x. Retaining the Lambert inverse exactly makes it possible to expand those corrections in a separate small exponential scale. The zero sector is an exact inverse, and each retained coefficient is a complete function of that inverse.

25.1 The admitted core and source charts

Use a positive coordinate $v \to +\infty$ and consider the exact model

$$F(v) = F_0(v) + R(v), \qquad F_0(v) = y_0 + av^b e^{cv^p}, \qquad c, p > 0, \quad a \neq 0, \tag{25.1}$$

where b is a fixed real number and R is a finite real power–log expression $\sum_\alpha v^\alpha P_\alpha(\log v)$. Every such perturbation, including its finitely many derivatives, is smaller than the growing core sufficiently near the endpoint.

At an infinite original source endpoint, the reconstruction is $x = H(v) = s + \sigma v$ with $\sigma \in \{1, -1\}$. The source translation s is fixed. At a finite endpoint, use $x = H(v) = x_0 + \sigma/v$. Thus the same construction covers both $xe^x + x^2$ at infinity and $e^{1/x} + x$ at 0^+. Define $r_* = 1$ for the first reconstruction and $r_* = -1$ for the second.

Let $Y = (y - y_0)/a > 0$. The selected exact core inverse is

$$v_0 = \begin{cases} (\log Y/c)^{1/p}, & b = 0, \\ \left(\dfrac{b}{cp} W_\nu\left(\dfrac{cp}{b} Y^{p/b}\right)\right)^{1/p}, & b \neq 0, \end{cases} \qquad \nu = \begin{cases} 0, & b > 0, \\ -1, & b < 0. \end{cases} \tag{25.2}$$

These are the real branches with $v_0 \to +\infty$. For $b < 0$, the argument is negative and the lower branch selects the increasing large-v part of the core. The real branch requires the Lambert argument condition, positivity of the real-power base, and $b + cpv_0^p > 0$. In all cases the exact inverse in the original source is $\phi(y) = H(v_0(y))$.

The inverse in (25.2) must remain exact. Additional terms in the leading expression that would invalidate this identity belong to the perturbation R, provided they satisfy its stated scale assumptions.

25.2 Complete coefficients without expanding the core inverse

Put

$$D(v) = av^{b-1}(b + cpv^p), \qquad F_0'(v) = e^{cv^p} D(v), \qquad E(v) = e^{-cv^p}. \tag{25.3}$$

Introduce the marker λ in $F_0(v) + \lambda R(v) = y$. The exact-core Lagrange formula of Section 17 shows that its degree-n correction has the form $C_n(v_0)E(v_0)^n$. The coefficient C_n can be computed using only power–logarithmic and rational operations on v:

$$Q_{n,1}(v) = \frac{H'(v)R(v)^n}{D(v)}, \tag{25.4}$$

$$Q_{n,j+1}(v) = \frac{Q_{n,j}'(v) - jcpv^{p-1}Q_{n,j}(v)}{D(v)}, \qquad 1 \le j < n, \tag{25.5}$$

$$C_n(v) = \frac{(-1)^n}{n!} Q_{n,n}(v). \tag{25.6}$$

Indeed, differentiating $Q(v)E(v)^j$ gives $(Q' - jcpv^{p-1}Q)E^j$, and division by F_0' adds one factor of E. Thus the recurrence differentiates rational power–logarithmic coefficients rather than the Lambert inverse, and keeps the exponential scale separate from those coefficients.

The exact core identity gives a second, often more convenient expression for the small scale:

$$E_0 = e^{-cv_0^p} = \frac{v_0^b}{Y}. \tag{25.7}$$

Both expressions are exact and positive on the selected branch. Through inclusive sector degree N, the approximation is

$$\phi(y) + \sum_{n=1}^{N} C_n(v_0(y))E_0(y)^n. \tag{25.8}$$

Here one complete marker degree is also one exponential degree, because the forward perturbation contains no additional exponentials.

Example 25.1 (The inverse of $xe^x + x^2$)**.** Let $v_0 = W_0(y)$ and $E_0 = v_0/y$. The first coefficients are

$$C_1(v) = -\frac{v^2}{v+1}, \tag{25.9}$$

$$C_2(v) = \frac{v^3(-v^2+2v+4)}{2(v+1)^3}. \tag{25.10}$$

Thus the first correction to the exact core inverse is

$$(xe^x + x^2)^{-1}(y) = W_0(y) - \frac{W_0(y)^3}{(W_0(y)+1)y} + \cdots .$$

This is much smaller than the error of any fixed inverse–logarithmic truncation of $W_0(y)$. Retaining W_0 exactly makes that correction meaningful without requiring a second, deeper core approximation.

25.3 A uniform complete-tail theorem

For the finite perturbation in (25.1), choose

$$\begin{aligned} d &= \max(0, \max_\alpha(\alpha - b)), \qquad & k &= \max_\alpha \deg P_\alpha, \\ B(v) &= v^d(1 + |\log v|)^k, \qquad & \chi(v) &= B(v)e^{-cv^p}. \end{aligned} \tag{25.11}$$

The zero perturbation is treated separately and leaves the exact inverse unchanged.

Theorem 25.2 (Complete tail around a growing exact core)**.** *For fixed data in* (25.1)*, there exist positive constants C, K and a source threshold such that the selected real inverse has a convergent expansion* (25.8) *with its complete omitted tail bounded by*

$$|\,\mathrm{Tail}_N\,| \le C v_0^{r_*-p} \frac{(K\chi(v_0))^{N+1}}{1 - K\chi(v_0)}, \qquad K\chi(v_0) < 1. \tag{25.12}$$

In particular, for fixed N the remainder class is

$$\mathcal{O}\Big(E_0^{N+1} v_0^{r_*-p+(N+1)d}(1 + |\log v_0|)^{(N+1)k}\Big). \tag{25.13}$$

The bound covers the whole omitted series independently of cancellation in the first omitted coefficient.

Proof. Set
$$v = v_0(1+\epsilon V), \qquad \epsilon = v_0^{-p}.$$
The displacement scale v_0^{1-p} keeps the change in the exponential phase bounded. Dividing the forward equation by $av_0^b e^{cv_0^p}$ gives a core difference

$$(1+\epsilon V)^b \exp\left(c\frac{(1+\epsilon V)^p - 1}{\epsilon}\right) - 1. \tag{25.14}$$

It extends analytically to $\epsilon = 0$, and its derivative in V at $(\epsilon, V) = (0,0)$ is $cp \neq 0$.

The normalized perturbation is $E_0 R(v_0(1+\epsilon V))/(av_0^b)$. After extraction of its powers of v_0 and $\log v_0$, each of its finite coefficient factors is analytic and uniformly bounded near $\epsilon = 0, V = 0$, with $1/\log v_0$ as an additional small parameter. Its absolute envelope is a constant times $B(v_0)E_0$. Replace E_0 temporarily by a complex variable Z and rescale $\zeta = B(v_0)Z$. The remaining normalized coefficient parameters range over a bounded finite polydisc. Since the derivative cp at $Z = 0$ is independent of them, the analytic implicit function theorem gives a uniform solution near $V = 0$ for $|\zeta|$ smaller than a fixed radius. Equivalently, it is analytic on $|Z| < 1/(KB(v_0))$.

For the observable, subtract its exact core value. The nonconstant part is

$$H(v_0(1+\epsilon V)) - H(v_0) = \sigma v_0^{r_*-p}\frac{(1+\epsilon V)^{r_*} - 1}{\epsilon}.$$

The quotient is analytic near $(\epsilon, V) = (0,0)$ and uniformly bounded on a smaller polydisc. Cauchy estimates bound its Z^n coefficient by $C(KB)^n$, with the displayed factor $v_0^{r_*-p}$. The coefficient identity above identifies these coefficients with the exact-core Lagrange coefficients. Summing the geometric tail and setting $Z = E_0$ proves (25.12). Finally $\chi(v_0) \to 0$, and the eventual derivative of the real forward function has the core's nonzero sign, so this solution is the selected real inverse near the endpoint. □

For $xe^x + x^2$, one can take $d = 1, k = 0, p = 1, r_* = 1$. The complete tail after sector N is consequently $\mathcal{O}((v_0E_0)^{N+1})$. For $e^{1/x} + x$ at 0^+, the reconstruction has $r_* = -1$, and the first correction is
$$(e^{1/x} + x)^{-1}(y) = \frac{1}{\log y} + \frac{1}{y\log^3 y} + \mathcal{O}\left(\frac{1}{y^2\log^2 y}\right). \tag{25.15}$$
The last bound is conservative but follows directly from (25.13) with $d = 0$.

25.4 An unknown input error remains a separate first sector

Suppose the actual perturbation includes a real continuously differentiable unknown term $A(v)$ satisfying
$$A(v) = \mathcal{O}(v^{-\rho}(1+|\log v|)^\ell), \qquad A'(v) = \mathcal{O}(v^{-\rho-1}(1+|\log v|)^\ell). \tag{25.16}$$
Here ρ may be any fixed real number: the growing exponential core dominates every such finite power scale. Its derivative is asymptotic to $acp\,v^{b+p-1}e^{cv^p}$. The mean value transport argument, with the derivative of H, gives an additional inverse error

$$\mathcal{O}\Big(E_0 v_0^{r_*-b-p-\rho}(1+|\log v_0|)^\ell\Big). \tag{25.17}$$

The matching derivative assumption ensures eventual monotonicity and permits comparison on the same local inverse branch. The model inverse and the actual inverse are both asymptotic to the exact core coordinate, so the scale can be expressed in v_0.

Equation (25.17) has exponential degree one, even when the finite model has been expanded through a much larger sector depth. It must therefore be added as a separate first-sector error to the model-tail remainder. Additional coefficients of the exact model cannot improve the uncertainty inherited from the input data. The retained terms beyond this error still describe the explicitly specified finite model.

25.5 Scope of the single-phase construction

The complete-tail theorem provides existence of constants and a uniform threshold for fixed data. Explicit numerical values require a further quantitative analysis of the analytic neighborhoods in its proof.

The coefficient identities can also be checked by formal substitution. For the primary example, insert $X = v + \sum_{n=1}^{N} C_n(v) Z^n$ into

$$Xe^{X-v} - v + ZX^2 = 0.$$

Vanishing of the coefficients through degree N follows from the unique formal solution with constant term v. This verifies the finite algebraic coefficients independently of the complete-tail estimate.

The phase in this construction is a single growing monomial cv^p. A general phase polynomial needs an appropriate exact core inverse with its own branch analysis. A perturbation such as $e^{v/2}$ introduces a further exponential scale and is not a finite power–logarithmic perturbation. A finite inverse-logarithmic approximation to the core would not resolve this obstruction, since its error can dominate the very correction one intends to calculate.

26 Separate inner precision and calculus of flat sectors

A finite calculus for the scale of Section 21 uses a common positive monomial target coordinate

$$u = \left(\frac{y - y_0}{a}\right)^{1/q}, \qquad E = e^{-c/u^p}, \qquad p, c > 0,$$

for every operand. The phase and chart are part of the scale; combining different phases requires an additional change-of-scale argument.

The asymptotic representation distinguishes two errors:

$$S(y) = C_0(u) + \sum_{k=1}^{N} E^k \left(T_k(u) + \mathcal{O}(u^{\rho_k} \mathrm{M}(u)^{d_k})\right) + \mathcal{O}(E^{N+1} u^{\rho_*} \mathrm{M}(u)^{d_*}). \tag{26.1}$$

Here C_0 is exact, each T_k is a finite power–log expression, and $\rho_k = +\infty$ means that the coefficient is exact. The remainder is a sum with one term for each nonzero inner error and a separate term for the complete omitted exponential tail. These are asymptotic bounds for fixed data, whose constants and threshold are existential.

26.1 Inner truncation and multiplication

An inner truncation at h retains powers strictly below h in every positive sector. The zero sector remains exact, including any of its powers at or above h. If a coefficient loses its first term at power α, its inner error has power α and the degree of the corresponding logarithmic polynomial. An existing, less precise bound is preserved. Increasing the formal cutoff after truncation does not recover discarded coefficient information.

For example, truncating the inner coefficients of

$$S = y - y^2 E + (y^2 + 2y^3)E^2 + \mathcal{O}(y^2 E^3), \qquad E = e^{-1/y},$$

at $h = 3$ gives

$$S = y - y^2E + y^2E^2 + E^2\mathcal{O}(y^3) + \mathcal{O}(y^2E^3).$$

The error $E^2\mathcal{O}(y^3)$ cannot be put into the third sector: its ratio to any E^3 times a fixed power–log expression is unbounded as $y \to 0^+$.

Multiplication convolves exponential degrees and uses the ordinary precision-tracked power–log multiplication inside each degree. If $A = \sum_{i \le N_A} E^i A_i + R_A$ and $B = \sum_{j \le N_B} E^j B_j + R_B$, the result retains degrees through $N = \min(N_A, N_B)$. Its tail includes all three kinds of contributions:

$$\sum_{i+j>N} E^{i+j} A_i B_j, \qquad R_A \sum_j E^j B_j + R_B \sum_i E^i A_i, \qquad R_A R_B.$$

Each coefficient bound includes its own inner uncertainty. Since $0 < E < 1$, a term in degree $k > N$ can conservatively be bounded in degree $N + 1$ while retaining its power–log envelope. This can give a looser algebraic power in the final sector bound; it is always a valid bound. In particular, unknown tails are added by bounds and never cancelled because some displayed coefficients cancel.

Polynomial composition $H(S)$ follows from addition and multiplication. The coefficients of H may be exact finite real power–log expressions in the target chart. Horner's formula computes the composition through successive products and gives the exact zero sector $H(C_0)$. A constant observable has no dependence on the unknown input tail; the zero observable therefore has zero remainder. General nonpolynomial composition and changes of phase require additional analytic and scale-compatibility arguments.

26.2 Qualified differentiation

A magnitude bound alone does not justify its differentiated bound. For the exact flat inverse, the uniform holomorphic implicit-function construction supplies bounds for every fixed derivative order. The exact finite forward model is essential to this argument. Addition, multiplication, polynomial composition, and inner truncation preserve the required local holomorphic estimates. A general real-axis remainder has no such derivative bound without a separate regularity hypothesis.

For a retained monomial the target derivative is explicit:

$$\begin{aligned}\frac{d}{dy}\Big(E^k u^\alpha P(\log u)\Big) = \frac{E^k}{aq}\Big[&u^{\alpha-q}(\alpha P(\log u) + P'(\log u)) \\ &+ kcp\, u^{\alpha-p-q} P(\log u)\Big].\end{aligned} \tag{26.2}$$

Thus the exponential degree stays k. For $k > 0$, a qualified inner bound at (ρ_k, d_k) is transported to $(\rho_k - p - q, d_k)$. When known differentiated terms are then discarded at that same boundary power, their logarithmic degree can increase the final inner degree bound. The complete tail has the same power shift $\rho_* \mapsto \rho_* - p - q$. This formula applies also when $q < 0$; the shift is a signed exact number, not its absolute value.

For completeness, the analytic tail estimate supports this derivative claim. Fix a small positive real u and use a complex disk

$$|v - u| < \eta u^{p+1},$$

with fixed sufficiently small $\eta > 0$. On this disk the chosen local power and logarithm branches are holomorphic, their power–log envelopes are comparable to those at u, and

$$v^{-p} - u^{-p} = \mathcal{O}(1).$$

Consequently e^{-c/v^p} differs from e^{-c/u^p} by a bounded factor. The normalized implicit-function construction in Section 21 extends uniformly to these disks. Its geometric tail estimate therefore holds there with changed constants. Cauchy's estimate costs a factor u^{-p-1} for one source derivative. The chain-rule factor $du/dy = (aq)^{-1}u^{1-q}$ gives the asserted loss u^{-p-q} for one target derivative. Every fixed higher derivative is handled on nested disks, or by the corresponding target disk of radius comparable to u^{p+q}; constants may depend on the derivative order. No differentiation of an arbitrary real-axis Big-O bound is used.

The finitely many discarded inner terms are themselves analytic power–log functions, so the same bound, and the exact phase derivative, apply to their errors. After differentiation, terms are retained only within the transported precision. For the previous example, the derivative of the truncated representation has the valid form

$$S' = 1 - (1 + 2y)E + 2E^2 + E^2\mathcal{O}(y) + \mathcal{O}(E^3),$$

where the last bound is the transported $\mathcal{O}(y^2E^3)$. The full second-sector derivative contains further powers of y that the truncated data do not determine.

The two levels of precision remain distinct under all these operations. A power-sized uncertainty in sector k cannot be replaced by a tail in sector $k + 1$, even if many known coefficient terms cancel. The analytic regularity required for differentiation is likewise separate from the magnitude of the remainder. These distinctions prevent formal coefficient arithmetic from asserting precision not present in the original data.

27 Composition and differentiation in a reciprocal logarithm

The reciprocal-logarithmic unit of Section 20 has more regularity than an arbitrary function satisfying a magnitude Big-O estimate. For an exact rational logarithmic forward unit, its implicit inverse is holomorphic in the small reciprocal logarithm. This regularity supplies composition and differentiation rules without discarding the exact power carrier or asking for an unsupported derivative of an unknown remainder.

Consider exact rational logarithmic forward units with positive monomial inverse carriers and zero finite source and target offsets. Known positive source-power perturbations are not included in this particular analytic construction: although smaller than every reciprocal-logarithmic order, they require a separate differentiable-remainder argument. The composition and differentiation below preserve holomorphy of the coefficient function.

27.1 A common positive logarithmic coordinate

Write

$$t = \frac{\epsilon}{\log y} > 0, \qquad \epsilon = \begin{cases} -1, & y \to 0^+, \\ 1, & y \to +\infty. \end{cases} \qquad g(y) = Cy^\alpha A(t), \tag{27.1}$$

Here $C > 0$, α is a fixed real number, and A is holomorphic near $t = 0$. For an original inverse unit, $A(0) = 1$. Differentiation can produce a coefficient function whose constant vanishes or has either sign; these outcomes remain valid carriers for further differentiation and as outer functions in composition.

Suppose the original inverse has positive leading amplitude a and internal leading source power $p \neq 0$. Its small coordinate is $\tau = -1/\log((y/a)^{1/p})$, and therefore

$$\tau = \frac{|p|t}{1 - \epsilon(\log a)t}, \qquad \epsilon = -\operatorname{sign} p. \tag{27.2}$$

This is an analytic change of variable at zero with positive derivative. It transports an analytic $\mathcal{O}(\tau^{\beta})$ tail to $\mathcal{O}(t^{\beta})$. For an internal observable power r_*, the monomial carrier has $\alpha = r_*/p$ and $C = \sigma^r a^{-\alpha}$. Assume this C is positive. In addition to the original target domain, the canonical chart requires $y > 0$ and $\epsilon \log y > 0$. These are eventual branch conditions; they are not a computed radius of convergence.

The Taylor expansions have nonnegative integer powers of t and constant real coefficients. A finite exclusive cutoff h retains powers $n < h$; the error exponent is limited by the inherited Taylor remainder and any discarded coefficient. Cancellation can make the first omitted nonzero degree larger than the cutoff.

27.2 Composition of two monomial carriers

Let the outer function and inner function be

$$g(v) = C_o v^{\alpha} A\left(\frac{\epsilon_o}{\log v}\right), \qquad h(y) = C_i y^b V(t), \qquad t = \frac{\epsilon_i}{\log y}, \quad V(0) = 1.$$

A positive nonunit constant in the inner coefficient function is first absorbed into C_i. The composition requires $b \neq 0$ and

$$\epsilon_o b \epsilon_i > 0. \tag{27.3}$$

This condition says that the positive inner function approaches the outer function's selected endpoint. It permits a negative power to interchange zero and infinity when the two specified endpoints agree with that change.

Taking the real logarithm of the inner carrier gives the exact identities

$$T(t) = \frac{\epsilon_o t}{b\epsilon_i + t(\log C_i + \log V(t))}, \tag{27.4}$$

$$g(h(y)) = C_o C_i^{\alpha} y^{\alpha b} V(t)^{\alpha} A(T(t)). \tag{27.5}$$

The denominator at zero is nonzero, $T(0) = 0$, and $T'(0) = \epsilon_o/(b\epsilon_i) > 0$. The functions $\log V$ and V^{α} use their analytic branches with $V(0) = 1$. Consequently the new coefficient function is holomorphic near zero and the original monomial carrier remains exact.

If the outer and inner Taylor remainders are $\mathcal{O}(T^{\beta_o})$ and $\mathcal{O}(t^{\beta_i})$, respectively, a conservative relative remainder for (27.5) is

$$\mathcal{O}\left(t^{\min(\beta_o, \beta_i)}\right). \tag{27.6}$$

To see this, an inner perturbation of size $\mathcal{O}(t^{\beta_i})$ changes $\log V$ and V^{α} by that order, while it changes T by $\mathcal{O}(t^{\beta_i+2})$. The outer function has bounded derivatives near zero, and its original remainder is evaluated at a coordinate comparable to t. The Taylor algebra preserves these bounds and can improve them when a constant factor or derivative vanishes. Truncating the displayed polynomial adds its own first discarded order. A larger formal cutoff never removes either operand's unknown tail.

For example, if both functions have the form $yW(-1/\log y)$ near zero, the coefficient function of their composition is

$$W(t)\, W\left(\frac{t}{1 - t\log W(t)}\right).$$

At positive infinity the denominator is instead $1 + t\log W(t)$. These formulas determine the coefficients by ordinary Taylor expansion. For general powers and scales, (27.4) retains both the exponent factor b and the additive logarithm $\log C_i$; dropping either gives incorrect coefficients already at low reciprocal-logarithmic orders.

An inner coefficient function with a zero constant term is excluded from this composition rule. Its logarithm introduces an additional $\log t$ scale. A negative constant also fails the positive inner branch condition. Those failures describe a change in the required scale or branch, rather than a reason to take a logarithm of an unproved positive approximation.

27.3 Differentiating the analytic remainder

The coordinate derivative is

$$\frac{dt}{dy} = -\frac{\epsilon t^2}{y}.$$

Hence

$$\frac{d}{dy}(Cy^\alpha A(t)) = Cy^{\alpha-1}\left(\alpha A(t) - \epsilon t^2 A'(t)\right). \tag{27.7}$$

For the nth derivative, the coefficient function is obtained by applying the successive operators

$$\prod_{k=0}^{n-1}\left(\alpha - k - \epsilon t^2 \frac{d}{dt}\right) \quad \text{to } A(t), \tag{27.8}$$

and the carrier is $Cy^{\alpha-n}$. The factors differ only by constants and commute. Applying them successively displays the precision change and possible cancellation at each derivative order.

If a holomorphic remainder has a zero of order at least β at zero, then its jth derivative is $\mathcal{O}(t^{\beta-j})$ for every fixed j. This follows directly from its convergent Taylor series, or from Cauchy estimates on a smaller disk. Thus the t^2A' contribution in (27.7) has remainder $\mathcal{O}(t^{\beta+1})$, while a nonzero αA contribution has remainder $\mathcal{O}(t^\beta)$. If $\alpha = 0$, the latter contribution disappears and the relative logarithmic error improves by one power. This is a proved derivative contract for the exact rational-logarithmic implicit model; it is not a rule for differentiating an arbitrary magnitude Big-O.

For the inverse of $x + x/\log x$ near zero, the retained coefficient function begins

$$A(t) = 1 + t + t^2 + 2t^3 + \tfrac{7}{2}t^4 + \mathcal{O}(t^5).$$

The first derivative therefore has the expansion

$$1 + t + 2t^2 + 4t^3 + \tfrac{19}{2}t^4 + \mathcal{O}(t^5).$$

Its carrier is constant. Differentiating once more gives a carrier y^{-1} and a relative remainder $\mathcal{O}(t^6)$, provided the known terms generated through degree five are retained. This illustrates the precision improvement when a carrier derivative vanishes. At infinity the signs of the odd powers and the coordinate derivative change consistently.

27.4 Refinement and the analytic hypothesis

A stronger approximation to a composed or differentiated function requires additional coefficients of its underlying analytic functions. Merely truncating the same finite polynomial at a larger formal degree supplies no new information about its unknown tail. Once those extra coefficients are known, the exact substitution formulas and derivative operators above transport their precision to the result.

The remainder theorem depends on holomorphy in the reciprocal logarithm, rather than on finite coefficient identities alone. A source-power sector which is smaller than every power of t can be invisible to the Taylor series yet obstruct holomorphy at $t = 0$. Its differentiation therefore requires a separate sector estimate. The hypotheses identifying an exact rational logarithmic model are part of the mathematical statement and cannot be discarded after its finite Taylor coefficients have been found.

As with the underlying implicit-function theorem, the convergence radius and bounding constants exist for fixed data. The formulas in this section do not assign explicit numerical values to those constants.

28 Implicit inverse composition and real branch selection

An expression $F^{-1}(h(x))$ poses two distinct mathematical problems: selecting the inverse branch of F, and expanding its composition with h. Branch selection must precede the coefficient calculation. A real inverse germ consists of a defining equation, a source domain, a source endpoint, and an approach direction. These data distinguish solutions which may share the same limiting target.

28.1 Scalar implicit equations and parameters

For a function F of n real variables, inversion in its kth argument with the other arguments fixed means solving the scalar equation

$$F(a_1, \dots, a_{k-1}, t, a_{k+1}, \dots, a_n) = a_k. \tag{28.1}$$

This is a one-dimensional inverse problem with parameters. For example, $at + t^2 = y$, with fixed $a > 0$ and $t \geq 0$, has the positive local solution

$$t = \frac{\sqrt{a^2 + 4y} - a}{2} = \frac{y}{a} - \frac{y^2}{a^3} + \frac{2y^3}{a^5} + \mathcal{O}(y^4), \qquad y \to 0^+.$$

The parameter a is fixed in this statement. If it varies with y, the error bound and even the relevant leading balance can change.

An important family with varying coefficients admits the exact reduction

$$A(x)F(t) + B(x) = h(x) \quad \Longleftrightarrow \quad F(t) = \frac{h(x) - B(x)}{A(x)}.$$

Assume $A(x) \neq 0$ on the selected deleted neighborhood and that the source domain for t is independent of x. Then the variation of A and B belongs entirely to the target argument. Nonvanishing at the endpoint itself is unnecessary: $A(x) = x$ is permitted near 0^+. The quotient must nevertheless be expanded with its full uncertainty; replacing a varying coefficient by its limiting value is generally invalid.

The coupled family $a(x)t + t^2 = h(x)$ cannot in general be reduced to a fixed forward germ by this separation. Joint implicit asymptotics would require hypotheses on the relative rates and on uniform regularity in the varying parameters. Such an additional analysis is independent of the fixed-parameter inversion formulas.

28.2 Source, target, and parameter domains

Let $C(t)$ specify a source restriction. The admissible real source domain is its intersection with the real domain of the chosen forward expression $F(t)$, under the fixed parameter assumptions. Merely requiring t real does not imply $t > 0$, whereas $t + t^2(1 + \log t)$, with the real logarithm, is defined only for $t > 0$. Domain restrictions remain part of the inverse problem throughout the calculation.

A restriction on the expansion variable x instead specifies the approach in $F^{-1}(h(x))$. A condition on a fixed parameter specifies the family for which the conclusion holds. These three kinds of conditions have different roles: an approach condition must hold eventually along x, a source condition must hold on the selected inverse germ, and a parameter condition is fixed throughout the limiting argument.

28.3 Local existence and unambiguous selection

Let $D \subset \mathbb{R}$ denote the admitted source domain. A source endpoint t_* is approached through a positive coordinate $u \to 0^+$:

$$t = t_* + su \quad (t_* \in \mathbb{R}), \qquad t = \frac{s}{u} \quad (t_* = s\infty), \qquad s \in \{-1, 1\}.$$

The condition $t(u) \in D$ concerns a *deleted* neighborhood $0 < u < \delta$. It need not hold at $u = 0$, where the source expression may be undefined. Thus a restriction $0 < t < 1$ is compatible with $t \to 0^+$.

Proposition 28.1 (A selected real inverse germ)**.** *Suppose there is a deleted source neighborhood on which F is real, continuous, and differentiable, and F' has a fixed strict sign. Suppose also that $F(t(u)) \to \eta$, with the prescribed one-sided target approach when η is finite. Then a sufficiently small interval on that target side, or a sufficiently distant target tail when $\eta = \pm\infty$, has a unique inverse with values in this source neighborhood. It approaches t_* on the selected source side.*

Proof. The source chart parametrizes an interval. The derivative condition makes F strictly monotone there, and continuity makes its image an interval. Its endpoint limit and target side show that the image contains a sufficiently small target interval or tail of the stated kind. The inverse on that image is unique and continuous, and its endpoint behavior follows from strict monotonicity. □

A nonzero leading block of the derivative's real asymptotic expansion can prove its eventual sign. The leading coefficient and the parity of the leading logarithmic degree determine that sign because $\log u \to -\infty$. This is a qualitative eventual-sign proof; it does not compute a numerical radius on which the derivative bound holds. A neighborhood known to satisfy the source condition is therefore not automatically a quantitative neighborhood for the full asymptotic error bound.

Proposition 28.1 verifies a proposed branch. *Selection* among different source endpoints requires an additional argument. There are two useful sufficient routes.

1. **Complete real fiber and boundary information.** For a finite target η, all interior points satisfying $F(t) = \eta$ must be accounted for, together with all finite boundaries of D and both source infinities. Continuity in the interior forces any finite interior endpoint of an inverse germ to belong to this fiber. Boundary limits must be checked separately because F need not be defined there. Every admissible endpoint and source direction is then tested by Proposition 28.1. Exactly one successful choice gives an unambiguous branch. For an infinite target, the finite interior fiber is replaced by the observation that a continuous finite-valued function cannot diverge at an interior point.

2. **Strict monotonicity on a connected real domain.** If D is an interval and F is strictly monotone throughout D, at most one real inverse can have values in D. A separately verified local candidate therefore identifies that inverse even without an explicit enumeration of the limiting fiber. This route requires both connectedness and the global monotonicity proof; local monotonicity near one candidate alone does not exclude another branch elsewhere.

An arbitrary finite list of candidate endpoints does not establish that either route has succeeded. For example, the two source sides of t^2 give opposite inverses at $\eta = 0^+$ when only realness is required. Restricting $t > 0$ or $t < 0$ selects one of them. An explicit source endpoint and direction specifies a local problem even when the global inverse is ambiguous. The chosen candidate must still satisfy the domain, limit, target-side, and monotonicity hypotheses.

28.4 Transporting the target argument's uncertainty

After selecting the branch, composition must account for uncertainty in both the inverse and its argument. The following ordinary power–log lemma makes the two contributions explicit. Exact exponential carriers or other scales require their corresponding chart rules from Section 16.

Lemma 28.2 (Two independent composition errors)**.** *Let $w \to 0^+$ and $\mathrm{M}(w) = 1 + |\log w|$. Suppose the positive local target coordinate satisfies*

$$q(w) \sim cw^{\lambda}, \quad c, \lambda > 0, \qquad q(w) - \widehat{q}(w) = \mathcal{O}(w^{\rho}\mathrm{M}(w)^{d}), \quad \rho > \lambda.$$

Let G be the selected inverse, or a differentiable observable of it, on this positive target coordinate, and suppose

$$G(v) - T(v) = \mathcal{O}(v^{\beta}\mathrm{M}(v)^{k}), \qquad |G'(v)| \le Cv^{\nu-1}\mathrm{M}(v)^{\ell}$$

for sufficiently small $v > 0$. Here $\beta, \nu \in \mathbb{R}$ and $d, k, \ell \ge 0$. Then

$$G(q(w)) - T(\widehat{q}(w)) = \mathcal{O}(w^{\lambda\beta}\mathrm{M}(w)^{k}) + \mathcal{O}(w^{\rho+\lambda(\nu-1)}\mathrm{M}(w)^{d+\ell}). \tag{28.2}$$

Proof. The target uncertainty is $o(w^{\lambda})$, so q, $\widehat{q}$, and every point between them are positive and comparable to w^{λ}. Their logarithmic envelopes are comparable to $\mathrm{M}(w)$. Split the difference as

$$[G(q) - G(\widehat{q})] + [G(\widehat{q}) - T(\widehat{q})].$$

Substitution into the stated outer remainder gives the first term in (28.2). The mean value theorem and the derivative bound give the second. The derivative estimate is an independent hypothesis: no differentiation of a magnitude Big-O statement is used. □

Thus the two candidate power cutoffs are $\lambda\beta$ and $\rho + \lambda(\nu - 1)$. The smaller governs the final power order; at equal powers the logarithmic envelopes must also be combined. A vanishing derivative can improve transported input precision, while a singular derivative can lower it. For an inverse itself, a suitable derivative estimate can be obtained from the original equation and $G' = 1/(F' \circ G)$ on the selected branch. A larger formal cutoff cannot recover coefficients hidden by the input argument's uncertainty. Resonant contributions at a common weight are combined before complete blocks are retained or discarded.

28.5 Logarithmic and irrational-power examples

Consider $F(t) = t + t^2(1 + \log t)$ on $t > 0$. Its derivative satisfies

$$F'(t) = 1 + t(3 + 2\log t) \ge 1 - 2e^{-5/2} > 0,$$

because $t(3 + 2\log t)$ has its global minimum at $t = e^{-5/2}$. The strictly increasing positive real branch tends to 0 as $y \to 0^+$. Its inverse begins

$$G(y) = y - y^2(1 + \log y) + y^3(2\log^2 y + 5\log y + 3) + \mathcal{O}(y^4\mathrm{M}(y)^3).$$

This is reversion of a real source germ, although the inverse has no ordinary Taylor series at $y = 0$.

For $F(t) = t + t^{\sqrt{2}}$ on $t > 0$, $F'(t) = 1 + \sqrt{2}\,t^{\sqrt{2}-1} > 0$. At 0^+ its inverse has the expansion

$$G(y) = y - y^{\sqrt{2}} + \sqrt{2}\,y^{2\sqrt{2}-1} - \frac{6-\sqrt{2}}{2}\,y^{3\sqrt{2}-2} + \mathcal{O}(y^{4\sqrt{2}-3}).$$

At $+\infty$, a different normalization is required because $t^{\sqrt{2}}$ is dominant. Put $p = \sqrt{2}$, $z = y^{1/p}$, and $\varepsilon = z^{1-p}$. Writing $G(y) = zW(\varepsilon)$ gives

$$W^p + \varepsilon W = 1, \qquad W(0) = 1.$$

The implicit derivative in W at the origin is $p \neq 0$, so W has a convergent ordinary power series near $\varepsilon = 0$. Its first terms are

$$\begin{aligned} G(y) = y^{1/p}\Big(1 - \frac{\varepsilon}{p} + \frac{3-p}{2p^2}\varepsilon^2 \\ - \frac{(p-2)(p-4)}{3p^3}\varepsilon^3 + \frac{(5-p)(5-2p)(5-3p)}{24p^4}\varepsilon^4 + \mathcal{O}(\varepsilon^5)\Big), \end{aligned}$$

where $\varepsilon = y^{(1-p)/p}$. The same globally increasing real function thus has different natural asymptotic coordinates at its two endpoints.

An arithmetic expression such as $1 + G(x)$ or $G(x)^r$ is an observable of the inverse. Its defining identity is obtained by reconstructing the source value $G(x)$ before substituting into F. It is not itself an inverse of F. On compatible branches, inversion of G recovers F; composition with a general observable has no such interpretation.

28.6 Exact identities and the role of the selected root

A closed form does not remove the need to specify its real branch. For example, $W_0(y)$ inverts te^t on $t \geq -1$, while $W_{-1}(y)$ inverts it on $t \leq -1$. Their common threshold and their distinct large-argument or near-zero behavior require the corresponding source endpoint data. An exact algebraic radical has the same obligation to use the branch matching the source domain.

A residual evaluated at a reconstructed source root must verify both the forward equation and the source restriction. Checking a powered observable alone can lose the source sign. A quantitative interval argument must establish the defining conditions on its verification interval; an open source condition excludes an endpoint at equality. A numerical root satisfying the condition supplies evidence at that point, while a uniform inverse statement requires the neighborhood hypotheses proved above.

29 Exponential normalization for Gamma and Barnes functions

Power–logarithmic scales describe algebraic growth and its corrections. A function such as $\Gamma(x)^2$ grows faster than every power of x. Its logarithm, however, has a power–logarithmic expansion. The appropriate construction retains the nonvanishing part of that logarithm exactly and expands only the vanishing part. This yields a relative asymptotic expansion whose absolute error includes the extracted growth factor.

29.1 From absolute logarithmic error to relative error

Let $u \to 0^+$ and write $\mathrm{M}(u) = 1 + |\log u|$. Suppose a real function has fixed sign $\sigma \in \{1, -1\}$ sufficiently near the endpoint and

$$\log|f(u)| = A(u) + B(u) + R(u), \qquad R(u) = \mathcal{O}\big(u^\rho \mathrm{M}(u)^k\big), \quad \rho > 0, \tag{29.1}$$

where B is a finite sum of positive-power logarithmic blocks, so that $B(u) \to 0$. The finite expression A includes all retained nonvanishing blocks, including any constant. Put $C(u) = \sigma e^{A(u)}$.

Proposition 29.1 (Relative normalization)**.** *Under these hypotheses,*

$$f(u) = C(u) \left(e^{B(u)} + \mathcal{O}(u^{\rho} \mathrm{M}(u)^{k}) \right). \tag{29.2}$$

If $e^B = T + \mathcal{O}(u^{\eta} \mathrm{M}(u)^{\ell})$, the finite expression CT has absolute error

$$\mathcal{O}\left(|C(u)| \left(u^{\rho} \mathrm{M}(u)^{k} + u^{\eta} \mathrm{M}(u)^{\ell} \right) \right). \tag{29.3}$$

Proof. The identity $f = Ce^B e^R$ and the mean-value estimate $|e^R - 1| \leq |R| e^{|R|}$ apply for real R. Both B and R tend to zero, so e^B and $e^{|R|}$ are bounded. The first assertion follows, and the triangle inequality gives the second. □

The logarithmic remainder must be known to tend to zero; the hypothesis $\rho > 0$ ensures this. An error bound of order one in the logarithm does not imply a relative error tending to zero in the original function. Likewise, a relative expansion of the logarithm alone need not determine a relative expansion of its exponential.

29.2 Products, varying powers, and coefficient extraction

For positive real functions f_j and real functions r_j, multiplication and real exponentiation become addition before any truncation:

$$\log \left(\prod_{j=1}^{m} f_j(u)^{r_j(u)} \right) = \sum_{j=1}^{m} r_j(u) \log f_j(u). \tag{29.4}$$

Cancellation can remove leading growth terms. It is therefore useful to form the complete sum at the required precision before extracting A. If $r_j(u) = \mathcal{O}(u^{-s_j} \mathrm{M}(u)^{d_j})$, a logarithmic remainder $\mathcal{O}(u^h \mathrm{M}(u)^k)$ becomes $\mathcal{O}(u^{h - s_j} \mathrm{M}(u)^{k + d_j})$. To reach a desired vanishing absolute error in the logarithm, the expansion of $\log f_j$ must be correspondingly deeper. A real varying exponent cannot in general be treated as a constant when propagating precision.

For an ordinary correction series $B(t) = \sum_{n \geq 1} a_n t^n$, write $e^{B(t)} = \sum_{n \geq 0} c_n t^n$. Formal differentiation gives

$$c_0 = 1, \qquad c_n = \frac{1}{n} \sum_{j=1}^{n} j a_j c_{n-j} \quad (n \geq 1). \tag{29.5}$$

For real positive weights, the finite convolution algebra of Section 2 supplies the same construction. If the smallest positive weight is δ, powers B^n with $n\delta \geq h$ cannot contribute below the strict weight cutoff h. Polynomial logarithmic coefficients stay attached to their complete weight blocks.

29.3 Stirling's logarithmic expansion

On the positive real axis, Stirling's expansion is

$$\log \Gamma(x) = \left(x - \tfrac{1}{2} \right) \log x - x + \tfrac{1}{2} \log(2\pi) + \sum_{n=1}^{N} \frac{B_{2n}}{2n(2n-1)x^{2n-1}} + \mathcal{O}(x^{-2N-1}), \qquad x \to +\infty, \tag{29.6}$$

where B_{2n} are Bernoulli numbers; see [3, §5.11]. This is a Poincaré expansion: the assertion holds for every fixed truncation order. It is not a convergence assertion as $N \to \infty$. Multiplying, adding, and substituting into finitely many such expansions is justified at each fixed order with the corresponding error estimates.

For products $\prod_j \Gamma(a_j(x))^{r_j(x)}$, assume that every argument is eventually positive, tends to $+\infty$, and admits the substitutions and error bounds needed in (29.6). Formula (29.4) then determines the logarithmic model. Factors with arguments tending to positive finite points instead use the local analytic expansion of $\log\Gamma$. Exponential normalization combines the resulting growth and correction terms in either case.

Example 29.2 (A square and a ratio)**.** Squaring the Gamma function doubles its logarithm. Exponentiating the vanishing part gives

$$\Gamma(x)^2 = 2\pi e^{-2x} x^{2x-1}\left(1 + \frac{1}{6x} + \frac{1}{72x^2} - \frac{31}{6480x^3} - \frac{139}{155520x^4} + \mathcal{O}(x^{-5})\right). \tag{29.7}$$

For a ratio, subtract the logarithms first:

$$\log\frac{\Gamma(3x)}{\Gamma(x)} = (3x - \tfrac{1}{2})\log 3 + 2x\log x - 2x - \frac{1}{18x} + \frac{13}{4860x^3} + \mathcal{O}(x^{-5}). \tag{29.8}$$

$$\begin{aligned}\frac{\Gamma(3x)}{\Gamma(x)} = 3^{3x-1/2}x^{2x}e^{-2x}\Big(1 - \frac{1}{18x} + \frac{1}{648x^2}\\ + \frac{463}{174960x^3} - \frac{1867}{12597120x^4} + \mathcal{O}(x^{-5})\Big).\end{aligned} \tag{29.9}$$

The remainder inside each parenthesis is relative to the prefactor. The corresponding absolute error is that prefactor times $\mathcal{O}(x^{-5})$.

Example 29.3 (A varying exponent)**.** Multiplying (29.6) by x turns the first Stirling correction into the constant $1/12$. It therefore belongs to the exact prefactor:

$$\Gamma(x)^x = e^{1/12}\left(\sqrt{2\pi}\, e^{-x}x^{x-1/2}\right)^x\left(1 - \frac{1}{360x^2} + \frac{1447}{1814400x^4} + \mathcal{O}(x^{-6})\right). \tag{29.10}$$

The step from $\mathcal{O}(x^{-7})$ in $\log\Gamma(x)$ to $\mathcal{O}(x^{-6})$ in $x\log\Gamma(x)$ illustrates the required increase in working order.

29.4 Barnes' G-function and the argument shift

Barnes' G-function satisfies

$$G(z+1) = \Gamma(z)G(z), \qquad G(1) = 1. \tag{29.11}$$

It is positive on the positive real axis. Write $c_G = \zeta'(-1) = 1/12 - \log A$, where A is Glaisher's constant. For every fixed integer $N \ge 0$, its logarithmic expansion is

$$\begin{aligned}\log G(z+1) = \left(\frac{z^2}{2} - \frac{1}{12}\right)\log z - \frac{3z^2}{4} + \frac{z}{2}\log(2\pi) + c_G\\ + \sum_{k=1}^{N}\frac{B_{2k+2}}{2k(2k+2)z^{2k}} + \mathcal{O}(z^{-2N-2}), \qquad z \to +\infty.\end{aligned} \tag{29.12}$$

This form follows by inserting Stirling's expansion for $\log\Gamma(z+1)$ into [3, (5.17.5)]; the relation between A and $\zeta'(-1)$ is [3, (5.17.7)]. The factor z multiplying that logarithm requires one additional order of its absolute error. Expanding it through the Bernoulli term with index

$2N+2$ gives the remainder displayed in (29.12). As with Stirling's formula, each fixed truncation has its stated error, without asserting convergence of the infinite series.

The first vanishing logarithmic terms are

$$-\frac{1}{240z^2}+\frac{1}{1008z^4}-\frac{1}{1440z^6}+\frac{1}{1056z^8}+\mathcal{O}(z^{-10}). \tag{29.13}$$

Define the positive prefactor

$$P_G^+(z)=e^{c_G}(2\pi)^{z/2}e^{-3z^2/4}z^{z^2/2-1/12}. \tag{29.14}$$

Relative normalization and (29.5) give, for example,

$$G(z+1)=P_G^+(z)\left(1-\frac{1}{240z^2}+\frac{269}{268800z^4}+\mathcal{O}(z^{-6})\right). \tag{29.15}$$

Every correction power in this coordinate is even. A logarithmic remainder $\mathcal{O}(z^{-2N-2})$ becomes a relative remainder of that same order after exponentiation; the absolute remainder includes the positive factor $P_G^+(z)$.

For $G(x)$ the argument in (29.12) is $z=x-1$. Retaining powers of $1/(x-1)$ gives the shifted form above. Expressing the result in powers of $1/x$ changes both the exact nonvanishing logarithmic part and the correction coefficients. The recurrence (29.11) provides an equivalent direct derivation: subtract (29.6) from the expansion of $\log G(x+1)$. Put

$$P_G(x)=e^{c_G}(2\pi)^{(x-1)/2}e^{-3x^2/4+x}x^{x^2/2-x+5/12}. \tag{29.16}$$

Then

$$\begin{aligned}\log G(x)=\ &\log P_G(x)-\frac{1}{12x}-\frac{1}{240x^2}\\&+\frac{1}{360x^3}+\frac{1}{1008x^4}+\mathcal{O}(x^{-5}),\end{aligned} \tag{29.17}$$

and its first five relative blocks are

$$\begin{aligned}G(x)=P_G(x)\Big(1&-\frac{1}{12x}-\frac{1}{1440x^2}+\frac{157}{51840x^3}\\&+\frac{65911}{87091200x^4}+\mathcal{O}(x^{-5})\Big).\end{aligned} \tag{29.18}$$

The term $z=x-1$ is therefore essential when translating the logarithmic formula to an expansion of $G(x)$. A shift of its argument does not preserve the correction bracket in a fixed coordinate.

Products of positive Gamma and Barnes factors use (29.4), so their logarithmic expansions can be combined before choosing the prefactor. Real varying powers transport the logarithmic error with the precision loss described above. Positive finite limiting arguments instead use local analytic logarithms. Integer shifts can also produce exact simplifications; for example,

$$\frac{G(x+1)}{G(x)}=\Gamma(x),\qquad \frac{G(x+1)}{\Gamma(x)G(x)}=1\qquad (x>0).$$

These identities justify exact cancellation independently of a finite asymptotic coefficient calculation.

29.5 Exact identities and elementary growth

The identities $\Gamma(x+1) = x\Gamma(x)$ and $\mathrm{B}(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$ connect the same analysis to factorials, binomial coefficients, beta functions, and rising factorials. For example,

$$\binom{2x}{x} = \frac{4^x}{\sqrt{\pi x}}\left(1 - \frac{1}{8x} + \frac{1}{128x^2} + \frac{5}{1024x^3} + \mathcal{O}(x^{-4})\right) \qquad (x \to +\infty), \tag{29.19}$$

where the binomial coefficient is defined by its Gamma quotient. The recurrence also proves $\Gamma(x+1)/\Gamma(x) = x$ exactly. Vanishing coefficients in a finite asymptotic computation alone would not prove this identity: a function may have all algebraic coefficients zero and still have a nonzero flat remainder.

The construction applies equally to elementary growth. Two examples are

$$e^{x+1/x} = e^x\left(1 + \frac{1}{x} + \frac{1}{2x^2} + \mathcal{O}(x^{-3})\right), \tag{29.20}$$

$$x^{x+1/x} = x^x\left(1 + \frac{\log x}{x} + \frac{(\log x)^2}{2x^2} + \mathcal{O}\left(\frac{(\log x)^3}{x^3}\right)\right). \tag{29.21}$$

In each case it is the vanishing logarithmic correction, rather than the entire growing function, that is expanded in a small scale.

30 The increasing inverse of the Gamma function

The logarithmic expansion of the Gamma function also provides an inverse expansion, but its natural small coordinate is neither the reciprocal of the original target nor a bare reciprocal logarithm. Retaining one exact Lambert function isolates the dominant balance. The remaining coefficients are finite polynomials in the reciprocal logarithm of that balance.

30.1 The branch and its exact dominant balance

Let $g(z)$ denote the solution of $\Gamma(g(z)) = z$ on the source interval $g(z) > 2$. This is a unique real inverse for $z > 1$. Indeed, $\Gamma(2) = 1$, $\Gamma(x) \to +\infty$ as $x \to +\infty$, and the positive-axis digamma bound

$$\frac{\Gamma'(x)}{\Gamma(x)} = \psi(x) \geq \log x - \frac{1}{2x} - \frac{1}{12x^2} > 0 \qquad (x > 2)$$

proves strict monotonicity; see the Stirling bounds in Section 24 and [3, §5.11]. The source restriction matters. Since $\Gamma(x) \to +\infty$ also as $x \to 0^+$, a large positive target alone does not select the increasing source branch. The branch studied here satisfies $g(z) \to +\infty$.

Put

$$Y = \log z, \qquad X = \frac{Y}{W_0(Y/e)}, \qquad L = \log X, \qquad t = \frac{1}{X}, \qquad q = \frac{1}{L}, \qquad C = \log(2\pi). \tag{30.1}$$

For $z > 1$, $Y > 0$, $X > e$ and $L > 1$. The positive Lambert branch gives the exact identity

$$X(\log X - 1) = Y.$$

As $z \to +\infty$, both t and q tend to zero, with $q = -1/\log t$. Stirling's leading equivalence and monotonicity imply $g(z)/X \to 1$: for each fixed $a > 0$, $\log\Gamma(aX)/Y \to a$, which brackets the inverse between fixed multiples of X arbitrarily close to one.

30.2 A polynomial coefficient hierarchy

Write $g = X(1+U)$ and let

$$b_k = \frac{B_{2k}}{2k(2k-1)}.$$

Substitute the finite Stirling expansion (29.6) into $\log\Gamma(g) = Y$, subtract $X(\log X - 1)$, and divide by XL. The resulting finite equation is

$$\begin{aligned}\Psi_m(U,t,q) = U + q\big((1+U)\log(1+U) - U\big) - \frac{t}{2} + \frac{Ctq}{2} \\ - \frac{tq}{2}\log(1+U) + q\sum_{k=1}^{m} b_k t^{2k}(1+U)^{-(2k-1)}.\end{aligned} \tag{30.2}$$

For the actual inverse, the omitted part has order $\mathcal{O}(qt^{2m+2})$. This assertion concerns each fixed m, not convergence of Stirling's infinite series.

Temporarily regard t and q as independent variables. The formal solution has the form

$$U = \sum_{n\geq 0} a_n(q)t^{n+1}, \qquad a_n \in \mathbb{Q}[C,q]. \tag{30.3}$$

The coefficients are determined successively. If $U_{<n} = \sum_{j=0}^{n-1} a_j(q)t^{j+1}$, with $U_{<0} = 0$, then

$$a_n(q) = -[t^{n+1}]\,\Psi_m(U_{<n},t,q), \qquad m \geq \left\lfloor \frac{n+1}{2} \right\rfloor. \tag{30.4}$$

The new coefficient appears in Ψ_m with coefficient one; every other occurrence of U either has an additional positive power of t or is at least quadratic in U. This proves triangular uniqueness. Expansion of the finite powers and logarithms at $U = 0$ also proves by induction that every a_n is a polynomial in q. A larger m does not change a coefficient already determined at a lower power of t.

The asymptotic blocks are

$$X, \quad a_0(1/L), \quad X^{-1}a_1(1/L), \quad X^{-2}a_2(1/L), \quad \ldots.$$

Each block retains its entire polynomial in $1/L$. In particular, the two terms of a_0 belong to one block at power X^0. A nonzero polynomial in q has an eventual leading power of q; since every positive power of X^{-1} dominates every fixed power of L in decay, these complete blocks are ordered by increasing powers of X^{-1}. This is distinct from truncation by perturbation-marker degree around an exact core: one marker coefficient may contain several of these ordered power blocks.

30.3 Five complete blocks and their first omitted order

The first coefficients can be written compactly as

$$\begin{aligned} a_0(q) &= \frac{1-Cq}{2}, \\ a_1(q) &= \frac{q}{24} - \frac{C^2q^3}{8}, \\ a_2(q) &= \frac{Cq^2}{48}(1 + q - C^2q^2 - 3C^2q^3), \\ a_3(q) &= -q\Big[(a_0 - \tfrac{1}{2})a_2 + \tfrac{1}{2}a_1^2 + (-\tfrac{1}{2}a_0^2 + \tfrac{1}{2}a_0 - \tfrac{1}{12})a_1 \\ &\qquad + \tfrac{1}{12}a_0^2(a_0-1)^2 - \tfrac{1}{360}\Big], \end{aligned} \tag{30.5}$$

where the a_j on the last two lines are evaluated at q. Consequently the five-block approximation is

$$G_5(z) = X + a_0(1/L) + \frac{a_1(1/L)}{X} + \frac{a_2(1/L)}{X^2} + \frac{a_3(1/L)}{X^3}. \tag{30.6}$$

Their leading coefficient orders are

$$a_0(q) \sim \frac{1}{2}, \qquad a_1(q) \sim \frac{q}{24}, \qquad a_2(q) \sim \frac{Cq^2}{48}, \qquad a_3(q) \sim -\frac{7q}{2880}.$$

Thus all five blocks in (30.6) are nonzero eventually. The recurrence gives the complete next coefficient a_4, whose leading behavior is

$$a_4(q) = -\frac{7C}{1920}q^2 + \mathcal{O}(q^3). \tag{30.7}$$

Theorem 30.1 (Finite-order inverse Gamma expansion)**.** *For each fixed integer $N \geq 0$, the increasing inverse satisfies*

$$g(z) = X + \sum_{n=0}^{N} X^{-n} a_n(1/L) + \mathcal{O}\left(\frac{X^{-N-1}}{L}\right), \qquad z \to +\infty. \tag{30.8}$$

In particular,

$$g(z) - G_5(z) = \frac{a_4(1/L)}{X^4} + \mathcal{O}\left(\frac{1}{X^5 L}\right) = \mathcal{O}\left(\frac{1}{X^4 L^2}\right), \tag{30.9}$$

and its first omitted leading constant is

$$\lim_{z \to +\infty} X^4 L^2 (g(z) - G_5(z)) = -\frac{7\log(2\pi)}{1920}. \tag{30.10}$$

Proof. Fix N and choose $m \geq \lfloor (N+1)/2 \rfloor$. The function Ψ_m is analytic near $(U, t, q) = (0, 0, 0)$, and its derivative with respect to U at the origin is one. The analytic implicit function theorem gives a solution $U_m(t, q)$ in a common neighborhood of $(t, q) = (0, 0)$. Since $\Psi_m(U, t, 0) = U - t/2$, one has $U_m(t, 0) = t/2$. Hence $U_m - t/2$ is divisible by q as an analytic function. Taylor's theorem in t, uniformly for q in a smaller neighborhood, gives

$$U_m(t, q) = \sum_{n=0}^{N} a_n(q) t^{n+1} + \mathcal{O}(qt^{N+2}).$$

Now specialize $t = 1/X$ and $q = 1/\log X$. The Stirling remainder, uniform for $x = X(1+U)$ with U in a fixed sufficiently small real interval, contributes $\mathcal{O}(qt^{2m+2})$ to the normalized equation. Its true derivative with respect to U is $\psi(X(1+U))/L$, which tends uniformly to one. The actual inverse already satisfies $U \to 0$. The mean value theorem therefore bounds its difference from U_m by $\mathcal{O}(qt^{2m+2})$. Multiplying by $X = t^{-1}$ and using $2m + 1 \geq N + 1$ proves (30.8).

For the five-block approximation, use the result with $N = 4$ and $m = 2$, then separate the a_4 term. Equation (30.7) and $L/X \to 0$ yield both (30.9) and (30.10). □

The analyticity used in the proof belongs to each finite Stirling model in two independent small variables. The theorem makes no claim that the inverse expansion converges when $N \to \infty$. Its constants and the required target threshold can depend on N. A numerical error certificate at a fixed target needs explicit residual and derivative bounds in addition to the asymptotic statement.

30.4 Affine arguments, fixed powers and transformed targets

Let $H(Y)$ be the increasing solution of $\log\Gamma(H(Y)) = Y$, so $g(z) = H(\log z)$. The same coefficients and theorem give H directly by using Y as the target in (30.1). No new coefficient calculation is needed to pass between Gamma and log-Gamma inversion.

For fixed real constants $\alpha \neq 0$, $a \neq 0$, $r \neq 0$, β and d, consider

$$z = d + a\,\Gamma(\alpha x + \beta)^r, \qquad \alpha x + \beta > 2. \tag{30.11}$$

On the target branch

$$T = \frac{z-d}{a} > 0, \qquad Y = \frac{\log T}{r} > 0,$$

the inverse is exactly

$$x = \frac{H(Y) - \beta}{\alpha}. \tag{30.12}$$

The large-source regime is $Y \to +\infty$. For $r > 0$ it corresponds to $T \to +\infty$; for $r < 0$ it corresponds to $T \to 0^+$ and $z \to d$ from the side determined by a. The source tends to $+\infty$ when $\alpha > 0$ and to $-\infty$ when $\alpha < 0$. The shift and scaling in (30.12) transport the absolute error by the factor $1/|\alpha|$.

Likewise $z = d + a\log\Gamma(\alpha x+\beta)$ uses $Y = (z-d)/a \to +\infty$ and the same source reconstruction. These are exact real coordinate changes; their domain restrictions must precede logarithms and noninteger powers. All asymptotic assertions hold with the parameters fixed. No uniform claim as r, a or α tends to zero is implied.

30.5 Fixed powers of the reconstructed source

Let ν be a fixed real number and consider x^ν for the source in (30.12). If $\alpha > 0$, then x is eventually positive and the usual real power is defined. For $\alpha < 0$, restrict here to integer ν. Define the polynomial coefficients

$$P_j(q) = \alpha^{-\nu}[t^j]\left(1 - \beta t + \sum_{n=0}^{j-1} a_n(q) t^{n+1}\right)^{\nu}, \qquad j \geq 0, \tag{30.13}$$

with an empty sum when $j = 0$, so $P_0 = \alpha^{-\nu}$. For every fixed integer $J \geq 0$, the finite-order theorem and the analytic binomial expansion around one give

$$x^\nu = X^\nu \sum_{j=0}^{J} P_j(1/L) X^{-j} + \mathcal{O}(X^{\nu-J-1}). \tag{30.14}$$

The coefficients remain polynomials in q, with the fixed parameters absorbed into their coefficients. The remainder in this general statement need not contain a factor $1/L$: at $q = 0$ the normalized source is $1 + (1/2 - \beta)t$, whose fixed power can already have nonzero coefficients at arbitrarily high powers of t.

More precisely, suppose an approximation A retains every nonzero block before the first omitted index j_*, and $P_{j_*}(q) = \gamma q^h + \mathcal{O}(q^{h+1})$ with $\gamma \neq 0$ and an integer $h \geq 0$. Apply (30.14) through j_* to obtain

$$\begin{aligned} x^\nu - A &= X^{\nu-j_*} P_{j_*}(1/L) + \mathcal{O}(X^{\nu-j_*-1}), \\ \frac{x^\nu - A}{X^{\nu-j_*} L^{-h}} &\longrightarrow \gamma. \end{aligned} \tag{30.15}$$

Indeed, the contribution of every subsequent core power is smaller because $L^h/X \to 0$. Thus the first omitted polynomial determines both the bound $\mathcal{O}(X^{\nu-j_*}L^{-h})$ and its leading constant, even when later polynomials have a smaller order of vanishing at $q = 0$. This also covers cancellations of entire blocks: the index j_* is the next nonzero polynomial after those cancellations.

31 The increasing inverse of Barnes' G-function

The logarithmic expansion (29.12) permits inversion in a scale whose dominant term is expressed exactly by the Lambert function. Keeping that term exact separates the successive powers of the large source from the slower logarithmic coefficients.

31.1 The real branch above three

The Barnes function is strictly increasing on $(3, \infty)$ and maps this interval onto $(1, \infty)$. To establish monotonicity, put $v = x - 1$ and $C = \log(2\pi)$. Differentiating [3, (5.17.4)] gives the exact identity

$$\frac{d}{dv} \log G(v+1) = v(\psi(v+1) - 1) + \frac{C-1}{2}, \tag{31.1}$$

where ψ is the digamma function. Its defining series gives

$$\psi(v+1) = \lim_{n\to\infty} \left(\log n - \sum_{k=1}^{n} \frac{1}{v+k} \right).$$

The function $t \mapsto 1/(v+t)$ is strictly convex. Comparing each midpoint value with its integral over $[k-1/2, k+1/2]$ gives a positive gap; the gap from the first interval persists in the limit. Thus $\psi(v+1) > \log(v+1/2)$ for $v > 0$. Consequently the derivative in (31.1) exceeds

$$h(v) = v\big(\log(v+1/2) - 1\big) + \frac{C-1}{2}.$$

For $v \geq 2$, one has $h'(v) = \log(v+1/2) - 1/(2v+1) > 0$ and $h(2) > 0$. For an elementary strict bound, the inequality $\log u > 2(u-1)/(u+1)$ for $u > 1$, applied to the factors in $5/2 = (3/2)(5/3)$, gives

$$\log(5/2) = \log(3/2) + \log(5/3) > \frac{2}{5} + \frac{1}{2} = \frac{9}{10} > \frac{8}{9}.$$

Likewise, $2\pi > 6 = (3/2)2^2$ gives

$$C > \log(3/2) + 2\log 2 > \frac{2}{5} + \frac{4}{3} = \frac{26}{15} > \frac{5}{3}.$$

These bounds imply $h(2) > 1/9$. Thus the derivative is positive for $x > 3$. The recurrence gives $G(3) = 1$, and (29.12) gives $G(x) \to \infty$. Denote this increasing inverse by $g(y)$. All subsequent assertions concern $y \to +\infty$ on this branch.

31.2 An exact Lambert balance

Write $Y = \log y$ and define

$$X = \sqrt{\frac{4Y}{W_0(4Y/e^3)}}, \qquad L = \log X, \qquad t = \frac{1}{X}, \qquad q = \frac{1}{L-1}, \tag{31.2}$$

where W_0 is the principal real Lambert function. Then

$$\frac{X^2}{2}\left(\log X - \frac{3}{2}\right) = Y, \qquad L - 1 = \frac{1 + W_0(4Y/e^3)}{2} > 0. \tag{31.3}$$

As $y \to \infty$, both X and L tend to infinity. The leading balance and monotonicity imply $(g(y)-1)/X \to 1$. The coefficient coordinate q uses $L-1$ because the derivative of the dominant logarithmic phase is $X(L-1)$.

Let $A = \log \mathcal{A}$, where $\mathcal{A}$ is Glaisher's constant, so that $\zeta'(-1) = 1/12 - A$. Write $g(y) = 1 + X(1+U)$. After substituting the finite expansion (29.12) and dividing the logarithmic residual by $X^2(L-1)$, the normalized finite equation is $\Psi_m(U,t,q) = 0$, where

$$\begin{aligned}\Psi_m(U,t,q) = U + \frac{U^2}{2} + \frac{q}{2}\left((1+U)^2\log(1+U) - U - \frac{U^2}{2}\right)\\ + \frac{Cqt}{2}(1+U) - t^2\left(\frac{1}{12} + Aq\right) - \frac{qt^2}{12}\log(1+U)\\ + q\sum_{k=1}^{m} \frac{B_{2k+2}\, t^{2k+2}(1+U)^{-2k}}{2k(2k+2)}.\end{aligned} \tag{31.4}$$

The omitted part of the true normalized equation is $\mathcal{O}(qt^{2m+4})$, uniformly for U in a fixed sufficiently small real interval. The finite equation is analytic near the origin in the three independent variables U, t, q and has derivative one with respect to U there.

31.3 Polynomial coefficients and three complete blocks

Temporarily regard t and q as independent. Seek a formal solution

$$U = \sum_{n\ge 0} b_n(q) t^{n+1}, \qquad b_n \in \mathbb{Q}[A, C, q]. \tag{31.5}$$

With $U_{<n} = \sum_{j=0}^{n-1} b_j(q) t^{j+1}$, the triangular recurrence is

$$b_n(q) = -[t^{n+1}]\Psi_m(U_{<n}, t, q), \qquad m \ge \max\left(0, \left\lfloor \frac{n-1}{2} \right\rfloor\right). \tag{31.6}$$

After the linear terms inside the logarithmic bracket cancel, only the first term U contains the new coefficient at the requested power; every other occurrence has an additional factor t or U. This proves uniqueness and polynomiality by induction. A larger m does not change coefficients already determined.

The first coefficients are

$$\begin{aligned} b_0(q) &= -\frac{Cq}{2},\\ b_1(q) &= \frac{1}{12} + Aq + \frac{C^2q^2}{8} - \frac{C^2q^3}{8},\\ b_2(q) &= \frac{CAq^3}{2} + \frac{C^3q^4}{12} - \frac{C^3q^5}{16}. \end{aligned} \tag{31.7}$$

Thus the first three complete blocks of the original source are

$$G_3(y) = X + \left(1 - \frac{Cq}{2}\right) + \frac{1}{X}\left(\frac{1}{12} + Aq + \frac{C^2q^2}{8} - \frac{C^2q^3}{8}\right). \tag{31.8}$$

The source translation by one belongs to the block at power X^0. Each block retains its whole polynomial in q. Since $X^{-1}(L-1)^d \to 0$ for every fixed real d, distinct nonzero blocks are ordered by their powers of X^{-1} even when their coefficient polynomials have different orders of vanishing at $q = 0$.

Theorem 31.1 (Finite-order inverse Barnes expansion)**.** *For every fixed integer $N \ge 0$,*

$$g(y) = 1 + X + \sum_{n=0}^{N} X^{-n} b_n(q) + \mathcal{O}(X^{-N-1}). \tag{31.9}$$

In particular,

$$g(y) - G_3(y) = \frac{b_2(q)}{X^2} + \mathcal{O}(X^{-3}) = \mathcal{O}\left(\frac{1}{X^2(L-1)^3}\right), \tag{31.10}$$

and

$$\lim_{y\to\infty} X^2(L-1)^3\big(g(y) - G_3(y)\big) = \frac{\log(2\pi)\log\mathcal{A}}{2} > 0. \tag{31.11}$$

Proof. Fix N and choose $m \geq \max(0, \lfloor (N-1)/2 \rfloor)$. The analytic implicit function theorem gives a solution $U_m(t,q)$ of (31.4) in a common neighborhood of $(t,q) = (0,0)$. Taylor expansion in t, uniformly for q in a smaller neighborhood, gives

$$U_m(t,q) = \sum_{n=0}^{N} b_n(q) t^{n+1} + \mathcal{O}(t^{N+2}).$$

For the true normalized equation, differentiation with respect to U gives

$$\frac{\left.\dfrac{d}{dv}\log G(v+1)\right|_{v=X(1+U)}}{X(L-1)}.$$

By (31.1) and the digamma asymptotics, this tends uniformly to $1+U$ on any fixed sufficiently small interval about zero. It is therefore bounded away from zero there. The true inverse has $U \to 0$. The mean value theorem and the finite Barnes remainder bound its difference from U_m by $\mathcal{O}(qt^{2m+4})$. Multiplication by $X = t^{-1}$ and the inequality $2m+3 \geq N+1$ prove (31.9). For (31.10), use $N = 2$ and separate the b_2 term. Its leading coefficient is $CAq^3/2$, and $(L-1)^3/X \to 0$, which also proves (31.11). □

The generic finite-order remainder has no obligatory factor q. Indeed, at $q = 0$ the analytic finite-model solution is $U_m(t,0) = -1 + \sqrt{1+t^2/6}$, which already has nonzero coefficients at arbitrarily high even powers of t. The proof concerns each fixed finite model and makes no assertion that the inverse series converges as its truncation order tends to infinity.

31.4 Affine reconstruction and real powers

Let $H(Y)$ be the increasing solution of $\log G(H(Y)) = Y$, so $g(y) = H(\log y)$. For fixed real parameters $\alpha a r \neq 0$, the equation

$$y = d + a\,G(\alpha x + \beta)^r, \qquad \alpha x + \beta > 3,$$

has the exact source reconstruction

$$x = \frac{H(Y) - \beta}{\alpha}, \qquad Y = \frac{1}{r}\log\frac{y-d}{a} > 0, \qquad \frac{y-d}{a} > 0. \tag{31.12}$$

The regime $Y \to \infty$ corresponds to $(y-d)/a \to \infty$ for $r > 0$, and to $(y-d)/a \to 0^+$ for $r < 0$. The latter approaches the finite target d from the side determined by a. An affine logarithmic target $y = d + a\log G(\alpha x + \beta)$ instead uses $Y = (y-d)/a$. All these assertions keep the parameters fixed.

A fixed real power x^ν is defined eventually for $\alpha > 0$; for $\alpha < 0$, restrict here to integer ν. Its normalized expression is

$$x^\nu = \alpha^{-\nu} X^\nu \big(1 + (1-\beta)t + U\big)^\nu.$$

Finite analytic composition therefore gives polynomial coefficients in q at successive powers of X^{-1}, with the same argument as (30.14). If the first omitted nonzero coefficient at $X^{\nu-j}$ is $\gamma q^h + \mathcal{O}(q^{h+1})$, then the corresponding error is asymptotic to $\gamma X^{\nu-j}(L-1)^{-h}$: the next full core power has size at most $\mathcal{O}(X^{\nu-j-1})$, and $X^{-1}(L-1)^h \to 0$. Entire zero blocks are omitted before selecting this first coefficient.

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